

Deployment-Oriented Neural Network Methods For Battery Management Systems:

Ageing Estimation, Robustness,
Embedded Optimization, and Hardware
Integration

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Preface

This dissertation investigates deployment-oriented artificial intelligence for battery management systems and combines model development, robustness analysis, embedded implementation, and hardware integration within one common research framework.

Abstract

This dissertation develops a deployment-oriented perspective on artificial intelligence for battery management systems. The work begins with neural-network-based ageing estimation for stationary storage systems, continues with robustness benchmarking of embedded battery state estimation under disturbed measurements, and then addresses the transfer of SOC and SOH models to microcontrollers using pruning and quantization. A comparative analysis across model architectures extends the compression perspective, while a dedicated hardware chapter documents the custom battery management system platform and the practical route from algorithm development to embedded execution. The central result is that battery intelligence must be evaluated not only by nominal prediction accuracy, but against a combined set of battery-management requirements across performance, hardware, and deployment.

Executive Summary

This dissertation is structured as a continuous, application-oriented account of artificial intelligence in battery management. Its main focus lies on neural architectures for ageing estimation in stationary storage systems, the robustness of embedded state estimation under realistic measurement disturbances, the efficient execution of such models on microcontrollers through pruning and quantization, and their integration into a real hardware and BMS platform. The overarching idea is that AI-enabled BMS methods should be evaluated not only by accuracy, but along the three requirement dimensions of performance, hardware, and deployment.

Table of Contents

Preface	III
Abstract	V
Executive Summary	VII
Abbreviations	XIII
Symbols	XV
List of Figures	XVIII
List of Tables	XIX
1 Introduction and Motivation	1
1.1 Research Objective and Requirement Dimensions	1
1.2 Scientific Contributions	2
1.3 Specific Research Objectives	3
1.4 Structure of the Dissertation and Development Path	4
2 Battery-Management Requirement Framework	7
2.1 Performance Requirements	8
2.2 Hardware Requirements	9
2.3 Deployment Requirements	9
2.4 Coverage of the Framework in the Dissertation	9
3 State of the Art and Technical Background	11
3.1 Battery Management Systems	11
3.2 Neural Networks for Battery Ageing and State Estimation	11
3.3 Embedded AI in Battery Applications	11

3.4	Compression and Model-Hardware Co-Design	12
4	Experimental Basis and Hardware Ecosystem	13
4.1	Data Sources and Ageing Campaigns	13
4.1.1	NMC Dataset – Stationary-Storage Ageing Estimation (Chapter 5)	14
4.1.2	LFP Dataset – Robustness Benchmark and Embedded Deployment (Chapters 6 and 7)	16
4.2	Embedded Targets and Execution Chain	19
4.3	Custom BMS-STEP Platform	19
4.3.1	System Objective and Laboratory Scope	20
4.3.2	Hardware Architecture	21
4.3.3	Firmware and Execution Concept	22
4.3.4	Experimental Relevance for Embedded AI	23
5	Neural Network Architecture for Ageing Estimation in Stationary Storage Systems	25
5.1	Application Context and Related Work	25
5.2	Raw Cell Dataset, Parameters and Measured Quantities	27
5.3	Preprocessing Method and Feature Engineering	29
5.3.1	Cumulative Current	29
5.3.2	Estimation of the State of Health	30
5.3.3	Correlation Analysis	30
5.3.4	Formation of the Lag Sequence	31
5.3.5	Training, Validation, and Test Splits	33
5.3.6	Data Scaling	34
5.4	Multilayer Perceptron Neural Network Model Structure	35
5.5	Configuration and Hyperparameter Tuning of the Neural Network	37
5.6	Results and Performance Evaluation	38
5.7	Interpretation of the Ageing-Estimation Results	41
6	Robustness Benchmarking of Embedded Battery State Estimation	43
6.1	Motivation and practical relevance	43
6.2	State of the art in battery state estimation	44
6.3	Performance and embedded deployment gap	44
6.4	Contribution and study objectives	45

6.5	Methodology	46
6.5.1	Estimator classes under test	46
6.5.2	SOC estimation methods	47
6.5.3	SOH estimation method	50
6.5.4	Embedded-oriented model preparation	52
6.5.5	Performance benchmark design	54
6.5.6	Metrics and evaluation criteria	57
6.6	Experimental Setup	60
6.6.1	Dataset and data acquisition	60
6.6.2	Evaluation labels and online feature reconstruction	60
6.6.3	Test trajectory and benchmark scenarios	61
6.6.4	Implementation details	62
6.7	Results	63
6.7.1	Baseline performance	63
6.7.2	Current-bias sensitivity	64
6.7.3	Noise sensitivity	65
6.7.4	Initialization error and recovery	67
6.7.5	Missing samples, irregular sampling, and burst dropout	68
6.7.6	Spike sensitivity	71
6.7.7	Cross-scenario comparison	72
6.8	Discussion	73
6.8.1	Model-specific performance behavior	73
6.8.2	Accuracy–performance trade-offs	74
6.8.3	Implications for BMS deployment	74
6.8.4	Limitations and future extensions	77
6.9	Conclusion	78
7	Embedded Artificial Intelligence for SOC and SOH Estimation on Microcontrollers	79
7.1	Embedded Deployment Objective	79
7.2	Dataset Basis and Experimental Scope	80
7.3	Feature Engineering and Reference Labels	81
7.4	LSTM Architecture and Training Setup	82
7.5	Structured Pruning for MCU Deployment	86
7.6	INT8 Quantization and Manual STM32 Export	87
7.7	Benchmark Methodology and KPI Logic	89

7.8	Streaming Prediction Accuracy	91
7.9	Error Distribution and Local Transient Behaviour	93
7.10	Resource Efficiency, Latency, and Trade-Off Summary	96
7.11	Implications for MCU-Based BMS	99
8	Architecture-Dependent Effects of Pruning and Quantization	101
8.1	Problem Statement	101
8.2	Comparison Framework	101
8.3	Scientific Contribution	103
9	General Discussion and Cross-Paper Synthesis	105
9.1	Synthesis of the Main Findings	105
9.2	Synthesis Matrix	106
9.3	Scientific Overall Contribution	106
9.4	Methodological Strengths and Limitations Across the Studies	107
9.5	Positioning Within the State of Research	108
9.6	Technical Implications for AI-Based BMS	108
9.7	Integrated Design Guidelines	109
10	Conclusion and Outlook	111
10.1	Overall Conclusion	111
10.2	Boundary Conditions and Remaining Gaps	111
10.3	Outlook	112
	Literatur	113
A	Supplementary Material	129
A.1	Supplementary Robustness-Benchmark Result Tables	129

Abbreviations

AI	Artificial Intelligence
BMS	Battery Management System
ECM	Equivalent Circuit Model
EFC	Equivalent Full Cycles
LSTM	Long Short-Term Memory
MCU	Microcontroller Unit
MLP	Multilayer Perceptron
SOC	State of Charge
SOH	State of Health

Symbols

SOC	State of charge
SOH	State of health
Q	Cumulative charge or charge throughput
Q_c	Online cumulative charge state
t	Time
U	Voltage
I	Current

List of Figures

1.1	Overall work roadmap.	6
2.1	BMS requirement framework.	8
4.1	SOH progression across the 14 NMC cells.	15
4.2	LFP DoE operating-point design.	17
4.3	Representative LFP trajectory structure.	17
4.4	SOH progression across the LFP ageing campaign.	18
4.5	BMS-STEP board-level view.	21
4.6	BMS-STEP system block view.	22
5.1	Correlation heatmap for SOH-related input variables.	31
5.2	Lag-sequence construction for supervised MLP samples.	33
5.3	Workflow from preprocessing to SOH evaluation.	34
5.4	MLP architecture for SOH estimation.	36
5.5	MAE matrix for lag length and time resolution.	39
5.6	Predicted vs. measured SOH trajectories.	40
5.7	Predicted vs. measured SOH scatter.	41
6.1	Robustness-benchmark methodology.	46
6.2	Architectures of the recurrent estimators in the benchmark.	50
6.3	Robustness-benchmark disturbance taxonomy.	55
6.4	Baseline performance on the benchmark trajectory.	63
6.5	Current-bias sensitivity.	65
6.6	Additive current-noise sensitivity.	66
6.7	Recovery after initialization mismatch.	68
6.8	Continuity and timing disturbance summary.	69
6.9	Transition into burst dropout.	70
6.10	Recovery after burst dropout.	71

6.11	Voltage-spike sensitivity.	72
6.12	Cross-scenario error increase heatmap.	73
6.13	Decision-oriented robustness synthesis.	77
7.1	Stateful LSTM+MLP estimator structure.	82
7.2	Architectures of the embedded SOC and SOH estimators.	85
7.3	Embedded deployment pipeline.	86
7.4	Structured pruning concept.	87
7.5	Per-row INT8 quantization concept.	89
7.6	Streaming SOC dashboard.	92
7.7	Streaming SOH dashboard.	93
7.8	SOC error distribution.	94
7.9	SOH error distribution.	95
7.10	Pulse-region SOC analysis.	95
7.11	Check-up-region SOC analysis.	96
7.12	Resource landscape of embedded variants.	98
7.13	Latency distributions on STM32H753ZI.	99
8.1	Quantization workflow.	103
A.1	ADC-quantization robustness illustration.	132

List of Tables

1.1	Dissertation structure and chapter roles.	5
4.1	Overview of the two battery datasets.	14
5.1	Stationary-storage ageing scenarios.	29
5.2	Final tuned MLP layer configuration.	37
6.1	Reusable estimator-component footprints.	54
6.2	Robustness-benchmark scenario protocol.	56
6.3	Connection between raw metrics and result dimensions.	59
6.4	Decision-oriented recommendation map for estimator selection. . . .	76
7.1	STM32 KPIs for SOC deployments.	90
7.2	STM32 KPIs for SOH deployments.	90
7.3	Pruning and quantization summary.	99
8.1	Compression experiment matrix.	102
9.1	Synthesis matrix of contribution chapters.	106
A.1	Baseline metrics for the robustness benchmark.	129
A.2	Cross-scenario global robustness metrics.	129
A.3	Local robustness and recovery metrics.	130
A.4	Robustness scenario evidence map.	131
A.5	ADC-quantization robustness metrics.	132

Introduction and Motivation

Battery management systems require reliable internal-state information to operate lithium-ion batteries safely, efficiently, and over long service intervals. Estimates of state of charge and state of health influence charging control, power limitation, lifetime-aware operation, and higher-level energy-management decisions. At the same time, battery behaviour becomes increasingly nonlinear once ageing, operating-point variation, and real measurement imperfections are taken into account. This creates a strong motivation for AI-based estimation methods that can capture complex relations beyond compact analytical descriptions.

Practical BMS deployment, however, imposes requirements that go beyond nominal prediction accuracy under controlled evaluation settings. Estimators must remain useful under disturbed measurements, limited computation and memory, causal execution, and real firmware and sensing constraints. These requirements become especially relevant in decentralized stationary-storage systems and remote microgrid-like applications, where battery supervision must remain locally available over long maintenance intervals and without permanent cloud or internet connectivity. The central scientific gap addressed in this dissertation is therefore not the absence of accurate battery-AI models as such, but the limited availability of a consistent evaluation and development framework that brings model quality, robustness, embedded feasibility, and hardware-level realization together.

1.1 Research Objective and Requirement Dimensions

The research objective of the dissertation is to develop and validate deployment-oriented neural-network methods for battery management systems along three linked requirement dimensions:

- The first requirement dimension addresses estimator performance. This includes not only nominal accuracy under controlled conditions, but also robustness

against disturbed measurements, recovery behaviour after state inconsistency, and stable operation under realistic signal disturbances and imperfect online conditions.

- The second dimension concerns hardware requirements. Battery-AI methods intended for practical BMS use must be judged against the limits of the target platform, including memory demand, runtime latency, integration constraints, and restrictions imposed by the measurement chain itself.
- The third dimension focuses on deployment requirements from model concept to embedded realization. This includes feature engineering, architecture selection, compression, model-to-device transfer, adaptability, scalability, and the firmware-level steps required to turn a trained estimator into a deployable BMS component.

These three dimensions are treated as interdependent rather than separate evaluation layers. The central argument of the dissertation is that battery-AI methods become practically meaningful only when predictive quality, embedded realizability, and hardware-level integration are developed and assessed together.

1.2 Scientific Contributions

- At the conceptual level, the dissertation establishes a unified deployment-oriented framework for evaluating battery-AI methods across performance, hardware, and deployment requirements. This framework provides the common perspective through which the subsequent studies are interpreted rather than treating accuracy, robustness, and embedded realization as isolated concerns.
- For ageing estimation in stationary-storage cells, the work introduces cumulative-current features and temporal lag construction as a compact way of representing ageing-relevant information. In the first study, this representation is examined with an MLP-based baseline; within the broader dissertation context, however, it serves mainly as the starting point of a larger development path that later extends toward recurrent architectures and deployment-oriented estimator design.
- A further methodological contribution is the robustness benchmark for embedded battery state estimation. Here, direct-measurement, hybrid, and data-driven estimator classes are compared under controlled disturbances such as bias, noise,

missing samples, timing irregularities, and initialization mismatch, so that estimator quality is assessed under deployment-relevant rather than purely nominal conditions.

- On the implementation side, the dissertation analyses how SOC and SOH estimators can be transferred to microcontrollers through pruning, quantization, and deployment-oriented STM32 benchmarking. This part of the work links model design to concrete trade-offs in memory demand, runtime latency, and practical firmware integration.
- The experimental basis of the dissertation is provided by the Battery Management System State-Estimation Test Platform (BMS-STEP), a custom hardware environment for repeatable embedded-AI studies within one common BMS architecture. This platform creates the technical basis for comparing interchangeable estimation methods under controlled and reproducible hardware conditions.
- Taken together, these elements form a cumulative design perspective for deployment-oriented battery intelligence that connects representation, robustness, embedded optimization, and hardware integration. The overall contribution therefore lies not only in the individual studies, but also in their integration into one coherent development and evaluation logic.

1.3 Specific Research Objectives

From this overall framing, the dissertation pursues five specific research objectives:

- To identify how neural-network methods can represent battery ageing with sufficient practical utility for stationary-storage applications and long-term condition monitoring.
- To assess under which disturbance conditions embedded battery state estimators preserve acceptable accuracy, convergence behaviour, and recovery capability.
- To evaluate how AI-based SOC and SOH estimators can be transferred to microcontrollers through resource-aware pruning, quantization, and model-to-device adaptation.
- To analyse under which architectural conditions pruning and quantization remain effective without disproportionately degrading predictive performance.

- To define how a hardware platform and validation chain should be structured in order to support repeatable, hardware-level evaluation of AI-based BMS methods under realistic embedded conditions.

1.4 Structure of the Dissertation and Development Path

The dissertation follows a cumulative structure in which study-specific methods, results, and local interpretation remain within the contribution chapters, while the common technical basis and the cross-study implications are developed in the surrounding chapters.

Several contribution chapters build on previously published or manuscript-based work. In particular, the ageing-estimation study in Chapter 5 is adapted from the published neural-network architecture paper on SOH estimation for stationary storage systems.[1]

The progression of the work is intentionally staged. The ageing-estimation chapter first addresses how battery-relevant information can be represented in a neural estimator. The robustness benchmark then shifts the focus from nominal performance to disturbance-aware evaluation under realistic measurement disturbances. The embedded-AI chapter extends this perspective toward microcontroller deployment, where compression, memory, and latency become decisive design variables. The subsequent compression chapter generalizes this question across architectures, while the hardware chapter and the BMS-STEP platform provide the common laboratory basis on which these deployment questions can be tested under realistic embedded conditions.

The shared laboratory hardware basis of the dissertation is the Battery Management System State-Estimation Test Platform (BMS-STEP), a custom 4s research BMS developed to compare and exchange embedded state-estimation methods on one reproducible system basis. The final synthesis chapter integrates the findings across all studies and derives the cumulative contribution of the work to deployment-oriented battery intelligence.

Tabelle 1.1: Cumulative structure of the dissertation and function of the core contribution chapters within the overall argument.

Order	Contribution chapter or basis	Primary dimension	Function within the overall argument
1	Experimental and hardware basis from the BMS-STEP platform and the associated reports	System context and validation basis	Establishes the shared measurement, firmware, and hardware environment in which the embedded-AI contributions of the dissertation are technically grounded.
2	Neural-network-based ageing estimation for stationary storage systems	Model design and ageing estimation	Establishes the representation-oriented starting point and shows how ageing-relevant information can be encoded through feature engineering and temporal input structure.
3	Robustness benchmarking of embedded battery state estimation	Robustness and deployment-facing performance	Defines the common deployment-oriented evaluation logic and shows why clean-condition error alone is insufficient for estimator selection.
4	Embedded artificial intelligence for battery management systems	Embedded deployment and optimization	Connects estimator design to practical microcontroller execution through pruning, quantization, latency, and memory measurements.
5	Architecture-dependent compression analysis	Compression sensitivity and co-design	Extends the embedded perspective toward broader model-hardware design rules across estimator families.

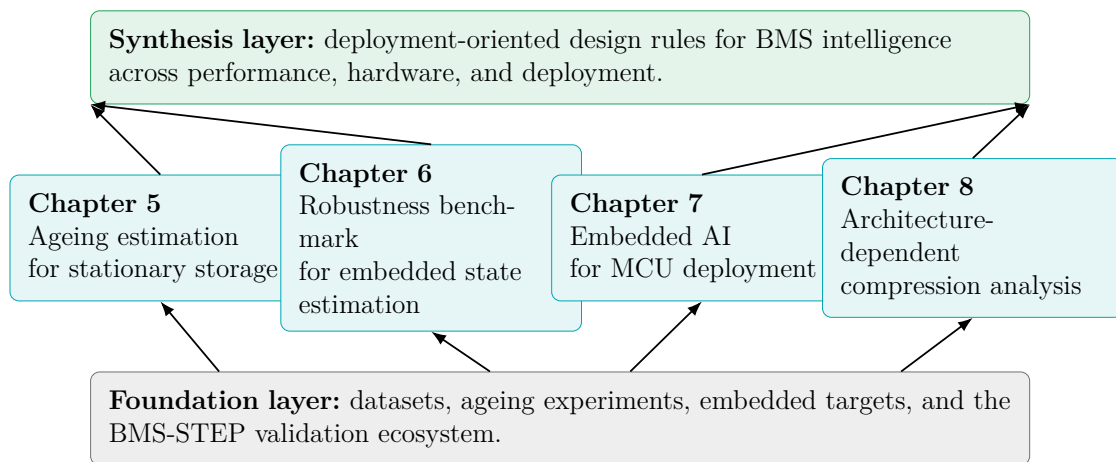


Abbildung 1.1: Roadmap of the overall work. The study chapters are embedded in a shared technical basis and are connected through a final synthesis toward deployment-oriented BMS intelligence.

Battery-Management Requirement Framework

Battery-management systems require estimation methods that satisfy several conditions simultaneously. In practical operation, a model must provide accurate and stable state information, remain robust under realistic measurement imperfections, fit within the available embedded resources, and be transferable into an executable BMS implementation. For this reason, the present work organizes the relevant battery-management requirements into three connected areas: performance requirements, hardware requirements, and deployment requirements. This framework provides the common basis on which the individual studies of the dissertation are interpreted. Each study emphasizes different requirement groups, but all of them contribute to the same overarching question of how neural-network methods can be developed for deployment-oriented battery management.

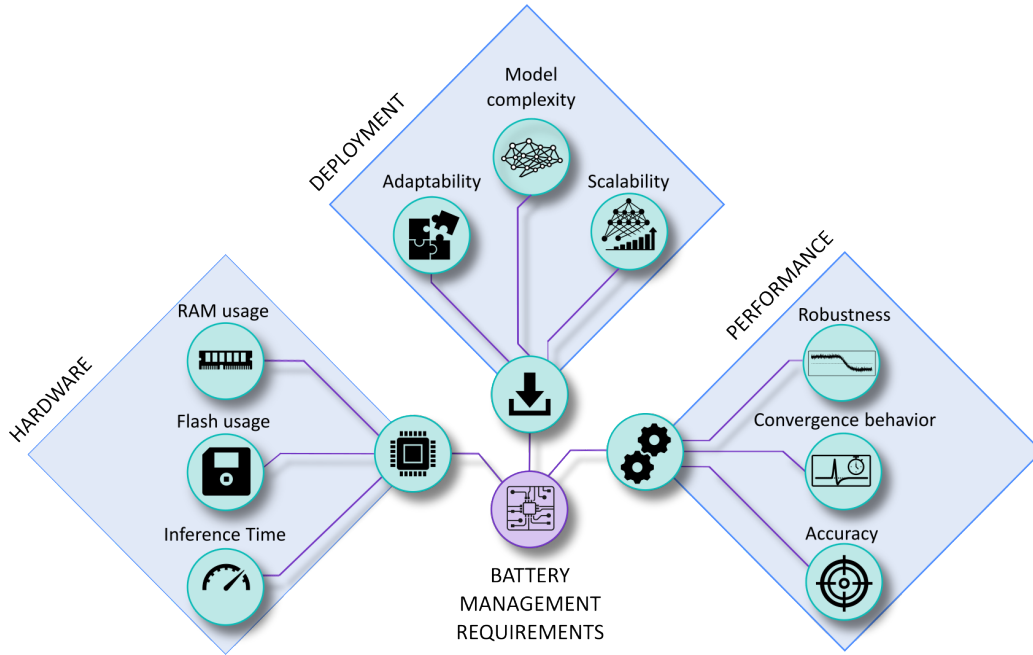


Abbildung 2.1: Battery-management requirement framework used throughout the dissertation. The framework groups deployment-oriented BMS requirements into performance, hardware, and deployment aspects and relates estimator quality to embedded feasibility and system realization.

2.1 Performance Requirements

Performance requirements describe the estimation quality that a BMS must obtain from an AI-based method during operation. In this dissertation, the central criteria are accuracy, robustness, and convergence behaviour. Accuracy is required because the estimator must provide reliable state information under nominal conditions. Robustness becomes essential once the input signals are affected by bias, noise, missing samples, or timing irregularities. Convergence behaviour determines whether the estimator can recover in a controlled manner after initialization mismatch, signal disturbances, or temporary state inconsistency. These criteria are examined most directly in the robustness benchmark, but they also remain relevant when the results of the ageing-estimation and embedded-AI studies are interpreted in a broader deployment context.

2.2 Hardware Requirements

Hardware requirements arise because BMS estimators are executed on resource-limited embedded platforms rather than in unrestricted desktop environments. The most important quantities considered in this work are RAM usage, Flash usage, and inference time. These metrics determine whether a model can be executed on the target controller with sufficient timing margin and without exhausting the available memory. In addition, hardware-related conditions such as measurement quality, controller structure, and system integration influence which estimator concepts remain feasible in practice. These questions become particularly visible in the embedded-AI chapter, where memory footprint and latency are quantified directly, and in the hardware chapter, which provides the BMS-oriented environment in which such constraints can be assessed experimentally.

2.3 Deployment Requirements

Deployment requirements describe how readily a model can be transferred from research development into practical BMS use. In this dissertation, the most relevant aspects are model complexity, adaptability, and scalability. Model complexity influences both the effort of implementation and the feasibility of later compression and hardware transfer. Adaptability refers to the extent to which modelling concepts, feature constructions, and estimator structures can be reused or modified for related battery-state estimation tasks. Scalability concerns whether a method and its optimization strategy remain viable across different model classes and target platforms. This requirement area therefore connects feature design, architecture choice, pruning, quantization, and firmware transfer within one common perspective.

2.4 Coverage of the Framework in the Dissertation

- The ageing-estimation study addresses primarily the performance and deployment sides of the framework. Its focus lies on how ageing-relevant battery information can be represented in a neural estimator and how this representation can be translated into useful prediction quality for stationary-storage data.
- The robustness benchmark concentrates most strongly on performance requirements. It shows that estimator quality in battery management cannot be reduced

to nominal accuracy alone, but must also include robustness against disturbed measurements and convergence behaviour after inconsistencies.

- The embedded-AI study connects deployment requirements with hardware requirements. In this part of the dissertation, model transfer, pruning, quantization, RAM usage, Flash usage, and inference latency become central because they determine whether a method can actually be executed on a microcontroller target.
- The hardware chapter provides the experimental and system-level basis for this framework. It establishes a BMS-oriented platform on which different estimation methods can be integrated, exchanged, and tested under reproducible embedded conditions.
- The architecture-dependent compression study extends the deployment perspective toward more general design rules. It examines how strongly pruning and quantization depend on model class and how scalable such optimization strategies remain across different architectures.

Taken together, these studies cover complementary parts of one common battery-management requirement framework. The framework therefore functions as the conceptual link between the chapters and as the basis for the final synthesis of the dissertation.

State of the Art and Technical Background

3.1 Battery Management Systems

Battery management systems supervise cell voltages, current, temperature, balancing, protection paths, and communication with higher-level control systems. Within this functional structure, SOC and SOH are not isolated diagnostic variables, but central quantities for power limitation, charging control, lifetime-aware operation, and system-level energy management.

3.2 Neural Networks for Battery Ageing and State Estimation

Neural methods are attractive in battery applications because they can represent nonlinear signal relations, exploit temporal context, and combine measured and derived variables in a data-driven manner. At the same time, their practical value depends not only on predictive accuracy, but also on interpretability, training stability, robustness, and deployment effort.

3.3 Embedded AI in Battery Applications

Embedded AI in battery applications shifts the focus from model quality alone toward deployment quality. Relevant criteria therefore include not only prediction error, but also Flash demand, RAM demand, latency, numerical stability, reproducibility, and the effort required to transfer a model from high-level research code to microcontroller inference.

3.4 Compression and Model-Hardware Co-Design

Compression and model-hardware co-design become essential as soon as AI is treated as a practical BMS component. Pruning and quantization are therefore not secondary optimizations, but central design choices that influence memory footprint, runtime behaviour, numerical precision, and ultimately the feasibility of embedded deployment.

Experimental Basis and Hardware Ecosystem

- The experimental basis of this work combines laboratory battery datasets, embedded execution targets, and the custom BMS-STEP hardware platform for battery management.
- This combined setup makes it possible to study battery-related artificial intelligence not only as an offline modelling problem, but also as an embedded systems problem shaped by sensing, communication, firmware partitioning, and hardware constraints.
- The resulting experimental ecosystem forms the common basis for the ageing-estimation work, the robustness benchmark, and the embedded implementation studies.

4.1 Data Sources and Ageing Campaigns

This dissertation draws on two distinct battery datasets with different cell chemistries, ageing protocols, and application contexts. The first dataset uses NMC cells and supports the ageing-estimation study in Chapter 5. The second uses LFP cells and forms the experimental basis for both the robustness benchmark in Chapter 6 and the embedded SOC/SOH study in Chapter 7. Across both datasets, the directly measured quantities are voltage, current, temperature, and time; dataset-side processing then derives labels for training and evaluation, such as capacity, SOC, and SOH, as well as engineered quantities such as transferred charge and cumulative charge state. Table 4.1 provides a compact side-by-side comparison of the two campaigns.

Tabelle 4.1: Key properties of the two battery datasets used in this dissertation. The NMC dataset supports the SOH ageing-estimation study in Chapter 5, while the LFP dataset provides the experimental basis for the robustness benchmark in Chapter 6 and the embedded deployment study in Chapter 7.

Property	NMC Dataset	LFP Dataset
Cell type	INR18650 NMC	JGCFR18650 LFP
Nominal capacity	2,6 A h	1,8 A h
Nominal voltage	3,7 V	3,2 V
Number of cells	14	15
Campaign type	Cyclic ageing	DoE cyclic ageing
Varied parameters	SOC _{mean} , DOD, C-rate, temperature	C_{chg} , C_{dis} , DOD
Temperature	35 °C / 45 °C	25 °C (constant)
Data source	TU Berlin	TU Berlin / MG-FARM[2]
Used in chapters	Chapter 5	Chapters 6 and 7
Based on publication	MDPI paper	JES paper and EAAI paper

4.1.1 NMC Dataset – Stationary-Storage Ageing Estimation (Chapter 5)

The first dataset consists of cyclic ageing tests on Li-ion INR18650 NMC cells carried out at Technische Universität Berlin. The cells have a nominal capacity of 2,6 A h, a nominal voltage of 3,7 V, a maximum charging voltage of 4,2 V, and a discharge cut-off at 2,75 V. In total, 14 cells are investigated across seven ageing scenarios, with two cells per scenario. The scenarios systematically vary four operating parameters: mean state of charge, depth of discharge, discharge C-rate, and operating temperature. This parametric design is intended to reproduce representative operating windows of stationary-storage systems rather than highly dynamic mobile profiles. Charging follows a CC-CV procedure; discharging uses constant current. Periodic capacity tests inserted at defined intervals track the SOH progression over time and supply the degradation labels used for model training.

The state of health is defined as the ratio of the current cell capacity to the

nominal capacity,[3, 4]

$$\text{SOH}(t) = \frac{C(t)}{C_N}, \quad (4.1)$$

where $C(t)$ is determined from periodic capacity tests in which the cell is fully charged and then discharged under controlled conditions. Because capacity tests exist only at discrete time points, the SOH label is linearly interpolated onto the continuous measurement timeline. Figure 4.1 shows the resulting SOH progression for all 14 cells over both equivalent full cycles and absolute time in days. Temperature has a pronounced impact on degradation, whereas the interactions between the remaining parameters are less obvious from direct visual inspection alone.

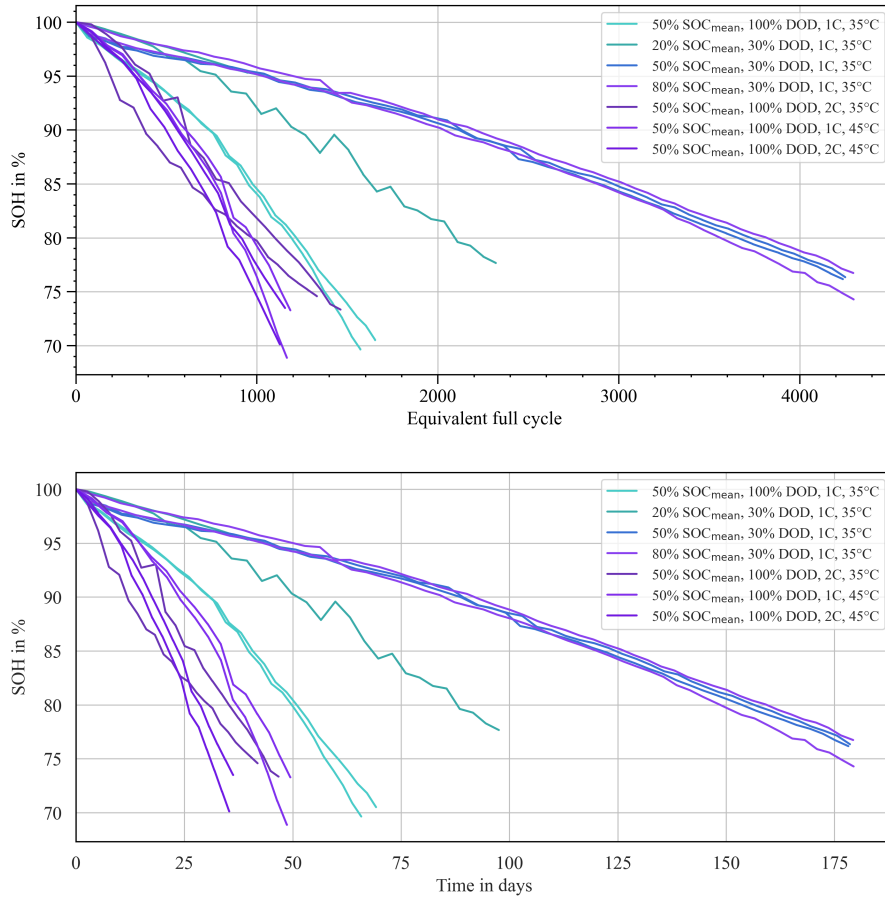


Abbildung 4.1: SOH progression of the 14 investigated NMC cells: top, over equivalent full cycles; bottom, over absolute time in days. Colors denote the seven ageing scenarios, with two cells per scenario.

This dataset is used exclusively in Chapter 5, where the ageing scenarios and chapter-specific feature engineering are described in detail.

4.1.2 LFP Dataset – Robustness Benchmark and Embedded Deployment (Chapters 6 and 7)

The second dataset is the published LFP DoE Cycle Ageing Dataset recorded at Technische Universität Berlin within the MG-FARM project.[2] It uses commercially available JGCFR18650-1800mAh-3.2V LFP cells with a nominal capacity of 1,8 Ah and a nominal voltage of 3,2V. The ageing campaign follows a three-factor design-of-experiments scheme conducted at a constant ambient temperature of 25 °C. The three varied factors are the charge rate $C_{\text{chg}} \in [0.5 \text{ C}, 0.9 \text{ C}]$, the discharge rate $C_{\text{dis}} \in [1.0 \text{ C}, 2.5 \text{ C}]$, and the depth of discharge DoD $\in [45\%, 65\%]$. A mixed full/fractional factorial plan assigns 15 test cells across eight operating points. Between ageing loops, dedicated check-up phases with capacity tests, open-circuit-voltage sweeps, and hybrid pulse-power characterisation are inserted. Stationary-storage load profiles derived from residential PV home storage are additionally applied so that the data contain both tightly controlled diagnostics and application-oriented dynamic behaviour. Raw measurements are available at 1 Hz and include voltage, current, terminal temperature, and auxiliary channels.

Figure 4.2 visualizes the three-factor operating-point design. Figure 4.3 shows a representative segment of the resulting trajectory structure, combining cyclic ageing blocks, check-up phases, and stationary-storage load-profile segments. Figure 4.4 shows the SOH spread across cells and operating points over the full campaign duration.

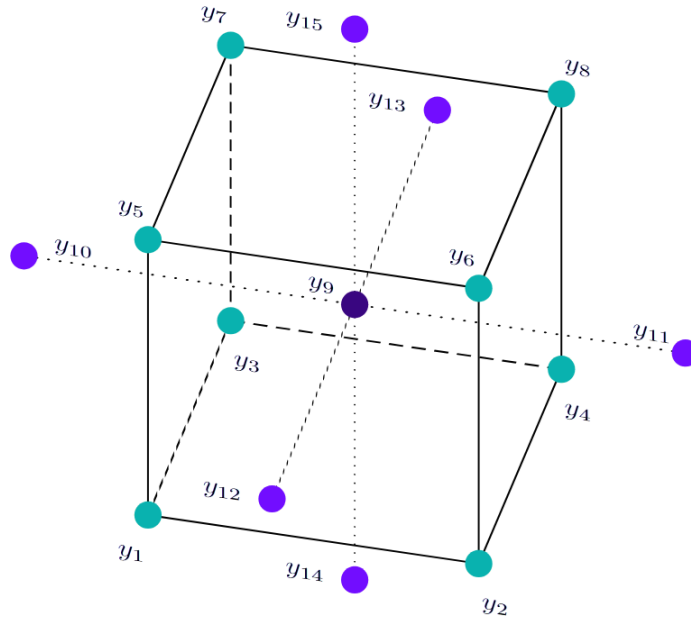


Abbildung 4.2: Design-of-experiments cube of the LFP ageing dataset. The three axes correspond to charge C-rate, discharge C-rate, and depth of discharge; the highlighted points define the factorial test matrix assigned to the 15 ageing cells.

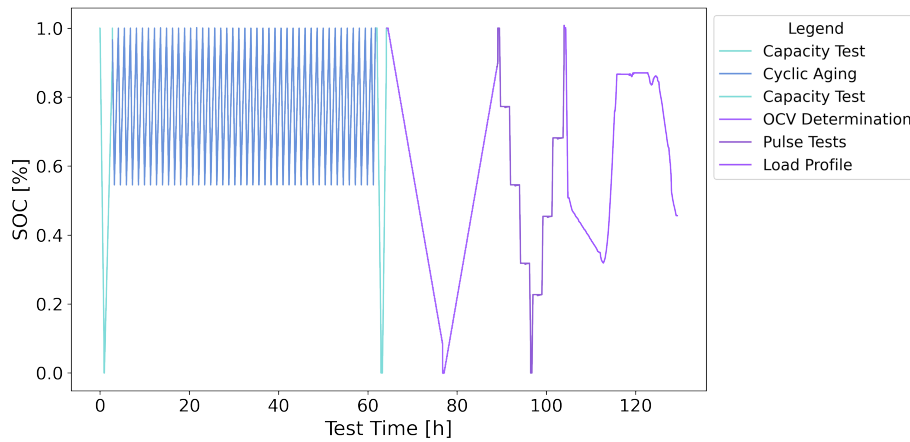


Abbildung 4.3: Representative SOC trajectory illustrating the experimental structure of the LFP campaign. The sequence combines cyclic ageing operation, diagnostic check-up blocks, and stationary-storage load-profile segments, capturing both laboratory reference phases and application-oriented dynamic behaviour.

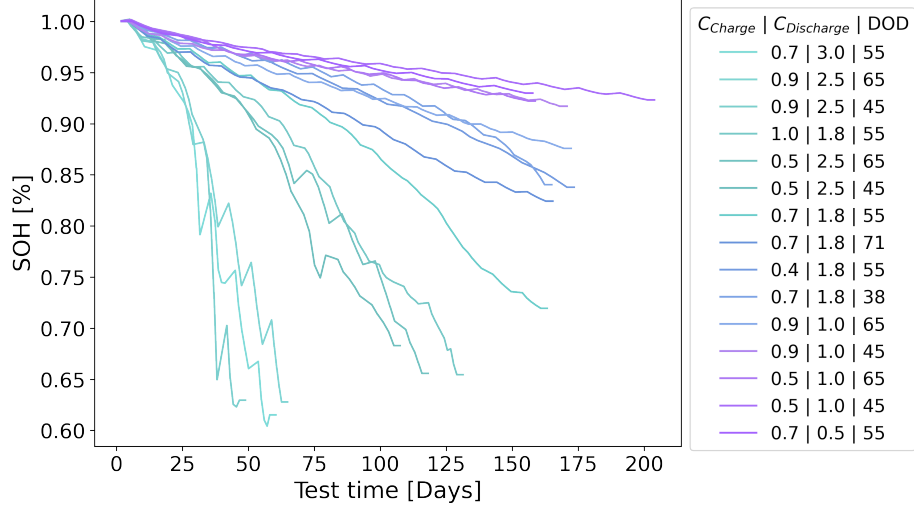


Abbildung 4.4: SOH progression across the 15 LFP cells over the full ageing campaign. The spread between trajectories reflects the different operating points in the factorial plan and motivates the use of data-driven SOH estimators that generalise across diverse loading histories.

Across both Chapters 6 and 7, the SOC and SOH labels are constructed from dataset-side processing. The SOC label is derived from Coulomb counting with drift compensation,

$$\text{SOC}(t) = \text{SOC}(t_0) - \frac{1}{C_{\text{app}}} \int_{t_0}^t I(\tau) d\tau, \quad (4.2)$$

with resets to $\text{SOC} = 1$ after sustained constant-voltage full-charge phases to reduce long-horizon drift. The SOH label is obtained from periodic capacity check-ups,

$$\text{SOH}_k = \frac{C_k}{C_{\text{nom}}}, \quad (4.3)$$

where C_k is the measured discharge capacity and C_{nom} is the nominal capacity.[5, 6] Both labels rely on dataset-side information – full-capacity discharge tests and sustained CV-based full-charge resets – that is not available during embedded deployment. They therefore serve as reference labels for model training, estimator calibration or validation, and evaluation in Chapters 6 and 7.

This dataset is used in Chapter 6 for the robustness benchmark and in Chapter 7 for the embedded SOC and SOH estimation study. Chapter-specific feature engineering and normalisation are described in the respective chapters.

4.2 Embedded Targets and Execution Chain

- The embedded work relies on two complementary targets: the final BMS-STEP hardware and an STM32H755-based evaluation-board setup for rapid firmware iteration and controlled implementation tests.
- The integrated BMS hardware is used whenever sensing, balancing, protection, communication, telemetry, and on-board state estimation must be validated as one coupled system rather than as isolated software modules.
- The evaluation-board setup is used whenever estimator code, toolchain details, memory demand, and runtime behaviour have to be checked quickly before transfer to the full BMS-STEP platform.
- In both cases, the deployment path follows the same sequence described in the reports: state-estimation methods are first prototyped in Python or MATLAB, then rewritten as plain C modules, compiled into the STM32 firmware, and executed locally on the target hardware instead of off-board.
- During validation, the firmware records voltage, current, temperature, and relevant intermediate estimator variables. After each run, these traces are aligned with higher-fidelity host-side SOC and SOH reference labels. D7.11 explicitly states that no hardware-in-the-loop setup is used in this workflow.
- This split between evaluation-board work and integrated BMS operation is important for the dissertation because it separates pure implementation questions from questions of sensing, protection, and full system integration.

4.3 Custom BMS-STEP Platform

- The custom laboratory hardware developed in this work is referred to as the Battery Management System State-Estimation Test Platform, abbreviated as BMS-STEP.
- BMS-STEP is a physically realised 4s laboratory BMS designed to monitor and protect the cells, execute on-board SoC and SoH estimation, perform balancing, and provide the technical basis for controlled experiments with embedded battery intelligence.

- A central reason for the in-house design is adaptability. In contrast to a commercial off-the-shelf BMS, the BMS-STEP platform gives direct access to the sensing chain, firmware structure, interfaces, and estimator integration, so that state-estimation methods can be exchanged, modified, and benchmarked repeatedly on the same hardware basis.
- The board should therefore be understood less as a finished product and more as a configurable research platform that was built deliberately to support rapid iteration, customisation, and comparative testing.

4.3.1 System Objective and Laboratory Scope

- The present implementation is a compact and reproducible 4s laboratory BMS for battery ageing studies and embedded state-estimation experiments at TU Berlin EET.
- The platform is intended for ageing tests, cyclic and calendar experiments, and representative charging and discharging profiles rather than unrestricted field deployment in large battery racks.
- D7.11 makes clear that exact field loads are not reproduced one-to-one in the laboratory. Instead, they are approximated within a controlled setup that allows calibration, diagnostics, repeated validation, and structured dataset generation.
- This laboratory orientation is deliberate because it enables benchmark-style comparison of different state-estimation approaches under the same measurement chain, including classical model-based observers and modern data-driven or neural methods executed directly on the STM32 controller.
- The platform concept is explicitly geared toward experimentation: one hardware base should support multiple estimator variants without rebuilding the full BMS around each method.
- Validation campaigns reported in D7.11 include duty cycles derived from decentralized stationary-storage use cases with limited maintenance access. These tests were used to record voltage drift, current ripple, temperature gradients, and embedded estimator outputs under realistic operating conditions.

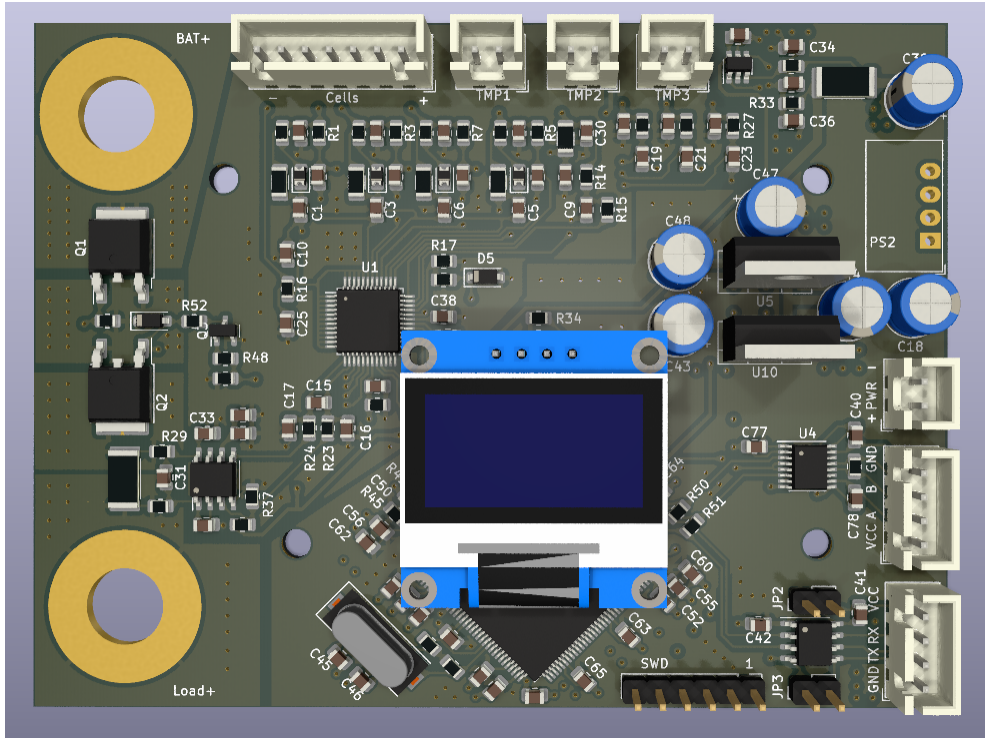


Abbildung 4.5: Board-level view of the custom BMS-STEP platform with integrated sensing, controller, communication, and auxiliary interfaces.

4.3.2 Hardware Architecture

- The BMS-STEP platform can be understood as three tightly coupled hardware domains: the TLE9012DQU-based sensing and balancing stage, the current-measurement and protection interface, and the controller/communication island around the dual-core STM32H755.
- The board layout keeps the high-current zone, shunt, and relay path physically separated from the low-voltage controller and communication domain. This separation matters not only for safety, but also for electromagnetic compatibility and stable measurement quality.
- The Infineon TLE9012DQU forms the measurement backbone of the pack. In the laboratory build it supervises four series-connected cells with buffered RC filtering and provides cell-voltage and temperature acquisition together with passive balancing capability.
- The active front-end includes balancing circuitry, while unused monitor inputs

are terminated according to the 12-to-4 cell adaptation of the reference design so that measurement integrity is preserved on the active channels.

- Temperature instrumentation is extended beyond the standard TLE inputs by an NTC harness that can connect additional probes at cell terminals, busbars, or ambient reference points. This provides a finer spatial thermal picture during ageing and state-estimation tests.
- The controller domain uses an STM32H755ZITx dual-core microcontroller. This design choice is central because it creates a clean architectural separation between conventional BMS functionality and the estimator layer that is investigated in the dissertation.

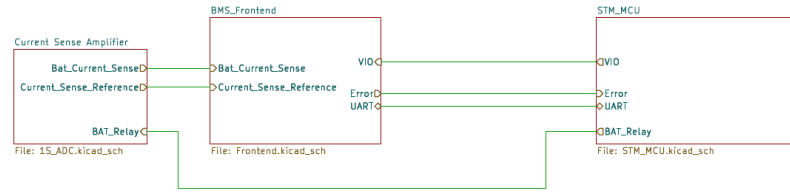


Abbildung 4.6: Reduced system-level block view of the BMS-STEP platform showing the interaction between current sensing, BMS front-end, and microcontroller domain.

4.3.3 Firmware and Execution Concept

- The firmware concept is intentionally centred on the dual-core STM32 architecture.
- The Cortex-M4 core is used for conventional BMS functionality such as measurement handling, supervision, and general system management, whereas the Cortex-M7 core is reserved for the computationally heavier state-estimation routines.

- This split is important for the dissertation because it makes the estimator layer exchangeable. Different SoC and SoH methods can be integrated, tested, and replaced on the M7 side without redesigning the complete BMS firmware structure.
- The platform was built exactly for this purpose: one stable hardware and BMS basis should support repeated experimentation with different embedded-AI methods, different estimator classes, and different implementation variants.
- From the perspective of the dissertation, the central contribution of the firmware architecture is therefore not the low-level implementation detail itself, but the fact that it creates a reusable and customisable execution environment for embedded battery state estimation.

4.3.4 Experimental Relevance for Embedded AI

- The BMS-STEP platform is not only a monitoring board, but an experimental platform for comparing estimator classes under realistic embedded conditions.
- Its main value for the dissertation lies in controllability and openness: the same measurement hardware can be reused while state-estimation methods are changed, refined, compressed, or replaced.
- This makes it possible to compare classical model-based observers and data-driven or neural approaches on one common embedded basis instead of distributing them across unrelated hardware environments.
- The BMS-STEP platform therefore closes the gap between algorithm development and embedded validation. It was chosen precisely because it can be modified and extended in ways that would be difficult with a purchased closed BMS.
- At the same time, the 4s implementation remains deliberately compact. It serves as a reproducible laboratory baseline from which later transfer to larger stacks or a 16s/48 V variant can be reasoned without changing the overall control and estimator architecture.

Neural Network Architecture for Ageing Estimation in Stationary Storage Systems

5.1 Application Context and Related Work

- State-of-health estimation is a central prerequisite for reliable stationary storage operation, especially in smart-grid and microgrid settings in which storage units must remain available over long maintenance intervals and under fluctuating renewable-power conditions.[7, 8]
- This application context is more demanding than conventional laboratory estimation tasks because smart-grid storage operates under varying load, charging, and renewable-infeed conditions and therefore requires methods that remain informative outside narrowly standardized test windows. The ageing tests considered in this chapter are designed to reflect these operating characteristics in a controlled laboratory setting so that the later findings remain relevant for practical stationary-storage deployment rather than only for idealized benchmark cases.
- Within this field, the literature usually distinguishes between model-based and data-driven SOH estimation. Model-based methods exploit explicit physical battery descriptions, whereas data-driven methods learn the relation between measured operating variables and the degradation state directly from data. The data-driven route is chosen here because it is better suited to strongly nonlinear ageing behaviour and does not require the same level of electrochemical model knowledge for practical deployment.[9, 10]

- Representative model-based alternatives include Kalman-filter-based joint SOC/SOH estimation, particle-filter-based ageing tracking, and fractional-order ageing models. Reported errors in this group range from about 1 % for dual adaptive Kalman-filter approaches to around 3 % for capacity-fade estimation and 4 % for power-fade estimation, while fractional-order-model studies report remaining-capacity errors near 3,1 %. These approaches can therefore achieve strong nominal accuracy, but they depend on accurate parameter identification, carefully tuned filters, and, in some cases, computationally demanding physical formulations. Their robustness may further deteriorate when the initial SOC is uncertain, when empirical ageing relations do not generalize cleanly across operating regimes, or when offline parameter-identification steps are required before deployment.[11, 12, 13]
- On the measurement-driven and machine-learning side, the discussion covers pressure-based abnormality detection, partial-voltage and incremental-capacity methods, hybrid deep-learning–Kalman combinations, autoregressive prediction, support-vector-machine approaches, fuzzy-entropy features, and SOC-domain preprocessing. Direct-measurement approaches based on reduced charging-voltage windows report average errors below 2,5 %, and hybrid deep-learning–Kalman combinations reduce estimation errors to about 2 %. At the same time, these studies reveal typical limitations such as sensitivity to selected operating windows and charging conditions, growing estimation errors at higher degradation levels, dependence on feature validity, susceptibility to abnormal data points, and limited transferability to varying operating conditions.[14, 15, 16, 17, 18, 19, 20, 21]
- More recent hybrid and sequence-based approaches process raw voltage, current, and temperature histories directly by neural networks or combine data-driven inference with physics-informed components. Sequence-to-sequence hybrid degradation prediction has, for example, reported mean absolute percentage errors below 2,5 % for the capacity trajectory and 6,5 % for the remaining usable cycles when only 20 % of the data are used, with further reductions to below 1,3 % and 5 % when 50 % of the data are available. These approaches are methodologically close to the later development path of this dissertation. The present ageing-estimation study nevertheless starts from a deliberately simpler MLP baseline in order to test how far feature engineering and lag-sequence construction can already push a comparatively lightweight model.[22, 23, 24, 25]

- The intention is therefore not to claim that MLPs are universally superior to recurrent or hybrid models. The central assumption is instead that a suitable representation of voltage, temperature, current, and energy history can make a simple feedforward network surprisingly effective while keeping the model easier to train, easier to interpret structurally, and closer to later deployment-oriented questions. This choice also reflects a practical concern: for continuously operated storage systems, classical SOH assessment from clearly delimited capacity tests is often unavailable online, so the estimator must infer degradation from operational trajectories observed in less structured states.[26, 27, 28]
- The methodological contribution of the chapter lies less in proposing an exotic neural-network architecture than in combining a compact MLP with a preprocessing pipeline tailored to stationary-storage data. In particular, lag-sequence construction allows the network to learn from segmented operating histories instead of relying on explicitly repeated capacity-test signatures, which improves data efficiency and keeps storage and training effort manageable. The additional inclusion of energy as an input variable broadens the information basis beyond the standard trio of current, voltage, and temperature and is intended to improve robustness under heterogeneous operating conditions.
- Under this framing, the approach is meant as a resilient compromise between accuracy, interpretability, and deployability. It targets imperfect and partially irregular measurement streams, avoids dependence on detailed electrochemical cell models, and is designed to remain transferable in principle across different stationary-storage configurations when sufficient training data are available. The chapter is organized accordingly: it first introduces the dataset and preprocessing logic, then the MLP structure and evaluation setup, and finally the experimental results. The overall workflow discussed in these sections is summarized later in Figure 5.3.

5.2 Raw Cell Dataset, Parameters and Measured Quantities

- The NMC dataset used in this chapter is described in Section 4.1. In brief, it consists of cyclic ageing tests on 14 INR18650 NMC cells at Technische Universität Berlin, grouped into seven scenarios with two cells each. All cells start from

an initial SOH of 100 %.

- Table 5.1 lists the seven scenarios and their parametric variation. The four varied parameters – mean state of charge, depth of discharge, discharge C-rate, and operating temperature – are selected to reproduce representative operating windows of stationary storage systems rather than highly dynamic mobile-load profiles.[29] The chosen mean-SOC levels are particularly intended to reflect stationary-storage operation around recurring operating points instead of idealized full-charge/full-discharge behaviour. Their systematic variation makes it possible to examine how the main operating conditions shape both degradation progression and the later estimation problem.
- After a defined number of ageing cycles, capacity tests are inserted to determine the degradation state of the cells, consistent with established battery ageing and calendar-testing practice.[30] Because the sampling intervals depend on the respective process step, all measurements are resampled to 1 s before feature engineering is applied.
- One scenario with 20 % mean state of charge, 30 % depth of discharge, 1C discharge rate, and 35 °C contains premature test termination and irregular measurements for cells 3 and 4. These data are nevertheless retained in the training pool in order to test whether meaningful ageing relations can still be learned from imperfect measurements. This is a deliberate robustness test: real stationary-storage data may also contain atypical interruptions, measurement irregularities, or operating deviations caused by changing demand, grid disturbances, or physical boundary effects. The model is therefore not only exposed to ideal trajectories but also to non-ideal data in order to assess whether the learned ageing relation remains usable under realistic conditions.

Tabelle 5.1: Ageing scenarios of the stationary-storage dataset, showing the seven investigated test cases and the systematic variation of SOC_{mean} , DOD, discharge C-rate, and operating temperature.

Scenario	Cell No.	SOC_{mean}	DOD	C-Rate _{dis}	T (°C)
1	1 and 2	50%	100%	1C	35
2	3 and 4	20%	30%	1C	35
3	5 and 6	50%	30%	1C	35
4	7 and 8	80%	30%	1C	35
5	9 and 10	50%	100%	2C	35
6	11 and 12	50%	100%	1C	45
7	13 and 14	50%	100%	2C	45

5.3 Preprocessing Method and Feature Engineering

5.3.1 Cumulative Current

- The preprocessing pipeline is built around the argument that instantaneous current alone is not sufficient as an explanatory variable for ageing estimation.
- Cumulative current is introduced as the central preprocessing feature because it captures the two quantities that are most relevant at this stage: the time dependence of the measurements and the cumulative throughput stress experienced by the cell.
- In its basic form, integrating the absolute current over time encodes not only the present loading level but also the prior electrochemical exposure and, therefore, a compact proxy for the cyclic age that is not visible from an instantaneous sample alone.
- The formulation is also physically motivated by the interaction between current and voltage over a sequence: their joint evolution carries ageing-relevant information that is closer to an impedance-like gradient than a single current reading. The preprocessing is therefore extended by distinguishing charging and discharging contributions, because both directions can induce different voltage gradients and different effective ageing signatures.

- The corresponding error reduction can be observed empirically, but because these effects are strongly collinear, the improvement cannot be attributed cleanly to one isolated mechanism alone.
- In this formulation, cumulative current becomes the feature that allows a relatively simple feedforward network to still access ageing-relevant temporal context without explicitly using a recurrent structure.
- Voltage and temperature are retained as further direct inputs, so the final feature set combines operating-state information with a compact representation of the prior load exposure.

$$Q(t) = I_{\text{cumulative}}(t) = \int_0^{\Delta t} |I(t)| dt \quad (5.1)$$

5.3.2 Estimation of the State of Health

The SOH definition, the construction of SOH labels from periodic capacity tests, and the resulting progression across all 14 NMC cells are described in Section 4.1 (Equation 4.1, Figure 4.1). Temperature has a pronounced impact on degradation, which motivates the correlation analysis below as a more structured view of the parameter–SOH relation.

5.3.3 Correlation Analysis

- To better understand which measured quantities should be retained for learning, the SOH trajectories are supplemented with the correlation analysis summarized in Figure 5.1.
- This matrix provides a more structured view of the parameter–SOH relation because the influence of several operating variables and their mutual interactions is not sufficiently clear from the SOH trajectories alone.
- The resulting heatmap indicates that temperature and cumulative current, both in positive and negative direction, show the clearest direct relation to SOH.
- In contrast, the time derivative of the voltage, although itself strongly temperature-dependent, does not show a comparably direct influence in the linear correlation view.

- The correlation analysis can only reveal linear dependencies and therefore serves only as a starting point for feature selection rather than as a substitute for the later model-based evaluation. Non-linear effects may still be relevant and can subsequently be captured by the neural network.
- Even with this limitation, the analysis provides a concrete justification for retaining temperature and cumulative-current descriptors as core inputs for the final model.

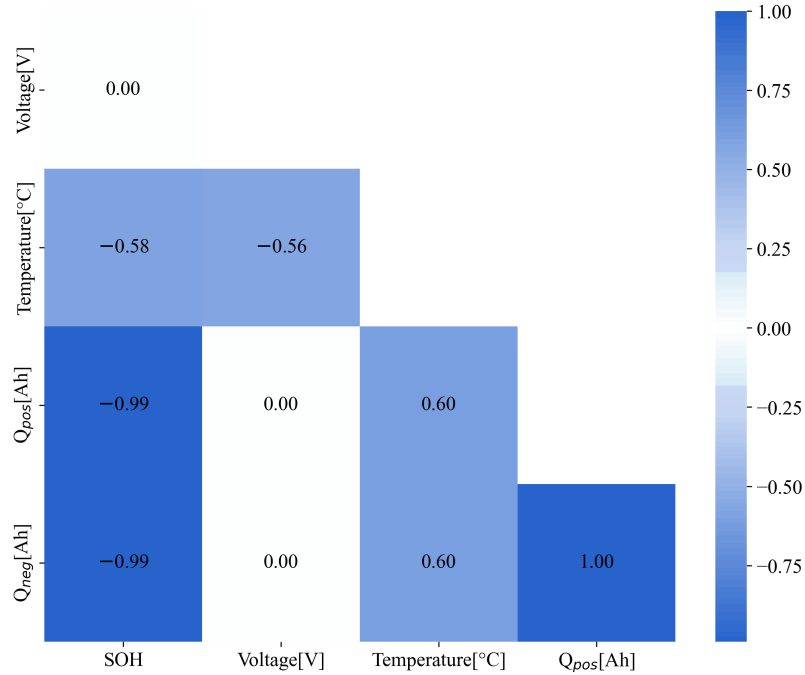


Abbildung 5.1: Correlation heatmap illustrating the linear relation between SOH and candidate input variables, including temperature and cumulative current Q in positive and negative direction. Correlation coefficients near $+1$ or -1 indicate strong linear dependence, whereas values near 0 indicate little or no direct linear relation.

5.3.4 Formation of the Lag Sequence

- Each learning sample is constructed as a lag sequence of voltage, temperature, and cumulative current, with the SOH at the current point in time used as the corresponding label, as illustrated in Figure 5.2.
- This transforms the continuous measurement stream into structured sliding-window samples that can be processed by a feedforward network without pas-

sing the complete life history of an individual cell directly into the model. The inclusion of past input values is intended to improve prediction quality, while the sliding window defines the temporal duration of the usable history.

- The initial parameterisation uses a time resolution of 30 min and 12 lag steps, which corresponds to a history length of 6 h. This starting choice depends on the expected usage pattern, the charging and discharging durations, and the presence of short high-current phases in the data.
- The time resolution has to match the actual charging and discharging behaviour. If the spacing is too coarse, relevant dynamics disappear. If it is too fine, the model sees an unnecessarily long and noisy representation that can make training less effective. If short pulses fall within one interval, they are represented only through the averaged feature values of that interval.
- This design can also be linked to the sampling theorem: the spacing between data points has to remain compatible with the temporal structure of the charging and discharging cycles so that the network can still recognise the relevant periodic behaviour.[31]
- Because the raw measurements are resampled before sequence generation, the preprocessing step also acts like a low-pass filter and strongly attenuates high-frequency noise components. Noise is therefore not treated as a dominant issue for the present estimation setup.
- A further benefit of the lag-sequence approach is that the training samples are no longer tied to one specific cell trajectory. Instead of memorising individual cells, the model is trained on many local windows drawn from the full dataset.

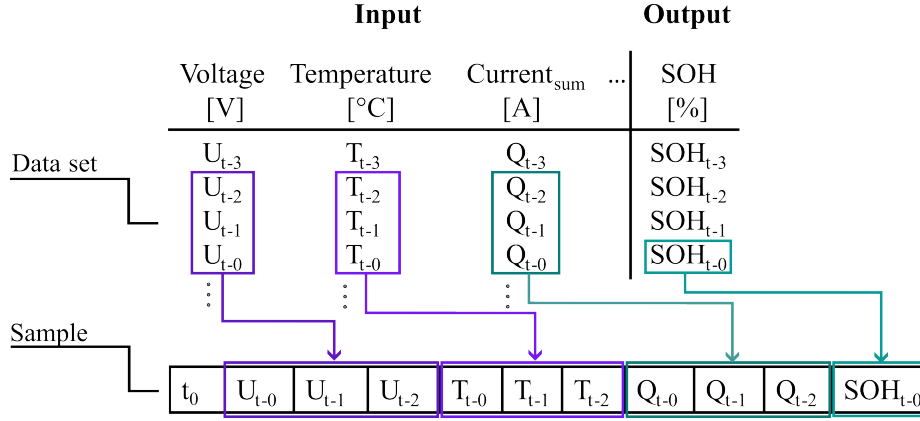


Abbildung 5.2: Illustration of lag-sequence construction from raw cell measurements into supervised MLP samples. Voltage, temperature, and cumulative current form the structured input history, while the SOH at the current time step serves as the corresponding label.

5.3.5 Training, Validation, and Test Splits

- The available data are separated into training, validation, and test subsets, as summarised in Figure 5.3.
- The training split is used to adapt the network weights, while 20% of the available training sequences are reserved for validation.
- The validation split is used during training to monitor overfitting and generalisation behaviour and to support hyperparameter decisions without exposing the model to the final test cells.
- At the same time, several complete cells are withheld from the training process for the final performance assessment. In total, four out of fourteen cells are retained for this independent accuracy evaluation.
- This choice is important because the method is intended to generalise to previously unseen cells and not only to previously unseen windows extracted from known cells. The explicit separation of training, validation, and test data therefore makes the resulting performance assessment more robust and more informative.

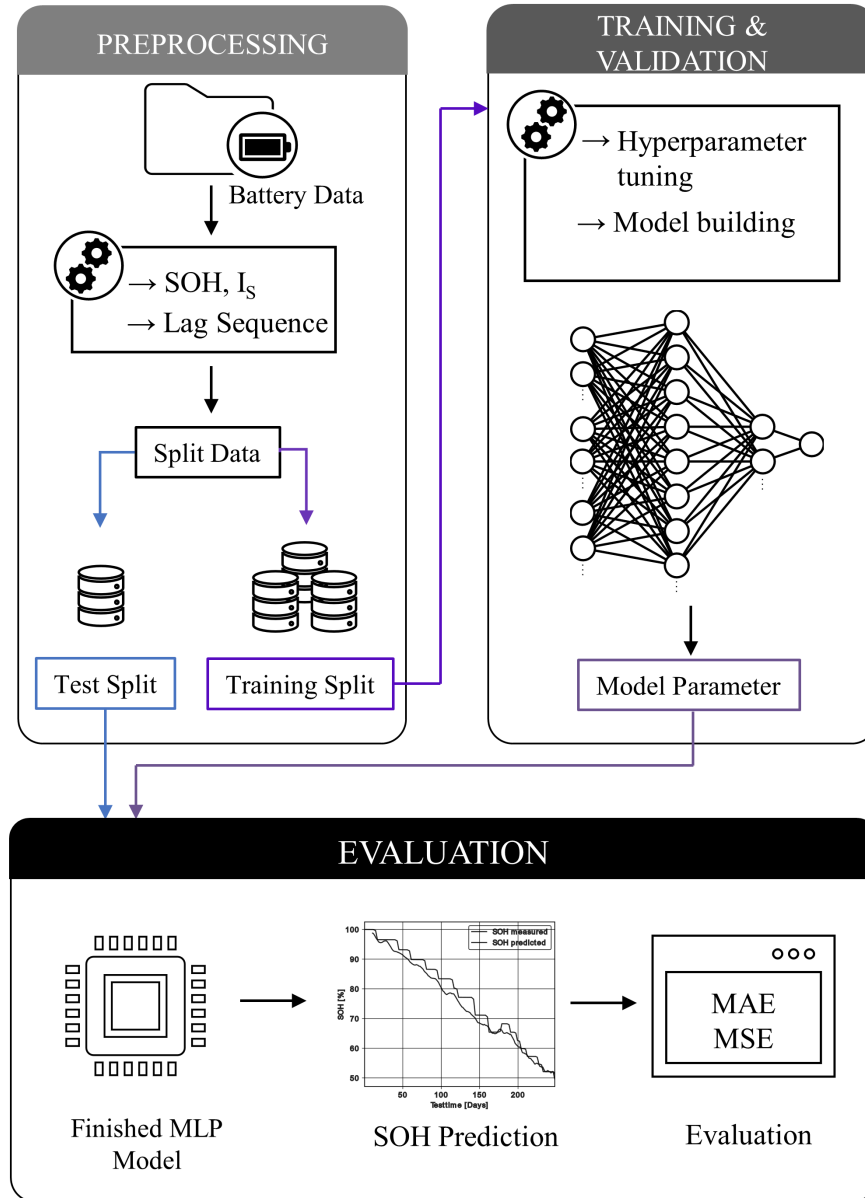


Abbildung 5.3: Workflow of the proposed MLP-based SOH estimation approach, from raw-cell preprocessing and data splitting through training and hyperparameter tuning to prediction and final evaluation.

5.3.6 Data Scaling

- All input features are scaled to the interval between 0 and 1 before training.[32]
- This step is necessary because the numerical ranges of the features differ strongly. Voltage remains in a narrow interval between 2,75 V and 4,2 V, whereas

cumulative current can rise to several thousand ampere-hours across the full ageing experiment.

- Uniform scaling improves numerical stability, supports convergence during optimisation, and avoids a situation in which large-magnitude inputs dominate the network weights while smaller but still relevant signals are effectively ignored.[32]
- The same scaling convention must be applied consistently to the training, validation, and test data so that the learned mapping is evaluated under comparable feature ranges and the resulting performance metrics remain interpretable across all splits.

5.4 Multilayer Perceptron Neural Network Model Structure

- The selected model type is a multilayer perceptron with fully connected layers.
- A convolutional interpretation is not pursued because the data do not exhibit a spatial image-like structure. Recurrent architectures such as LSTMs are also not used at this stage, because the temporal dependence is already encoded through the lag-sequence construction and because the recurrent alternatives are more resource-intensive and more difficult to tune in the chosen setting.[33]
- This architecture choice is also connected to the later hyperparameter-search setup: the chapter keeps the estimator architecture feedforward and shifts the main optimization effort to lag-window design, network depth, node count, and training parameters, which are then searched efficiently with Hyperband.[34]
- As illustrated in Figure 5.4, the MLP consists of an input layer, several dense hidden layers with ReLU activation, and a single linear output node for the SOH estimate.
- The original starting point is intentionally small and uses one hidden layer with eight nodes. The final network depth and width are then determined in the hyperparameter search.
- In the tuned configuration, summarised in Table 5.2, the model grows into a deep dense architecture with nine hidden/output stages and a final linear regression

output. The resulting structure remains conceptually simple from an architectural point of view, but it is substantially larger than the initial baseline.

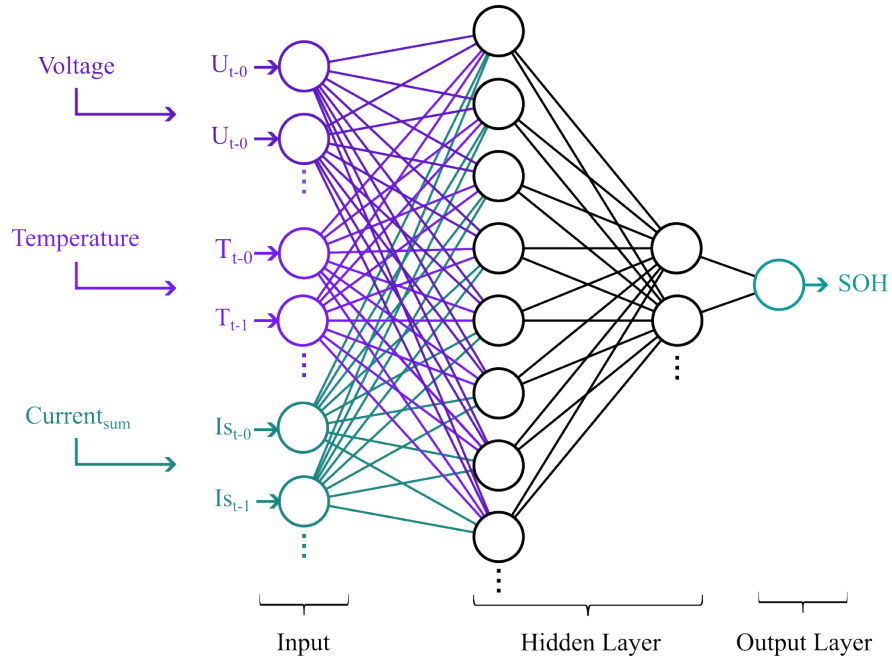


Abbildung 5.4: Illustration of the multilayer perceptron (MLP) used for SOH estimation. The network receives voltage, temperature, and cumulative current as inputs, with the number of input nodes scaling with the number of lag-sequence elements in a sample. Each input node is fully connected to the hidden layers, which is characteristic of an MLP, and a single output node provides the SOH estimate. The figure highlights the feedforward flow of information from the input features through the hidden representation to the final prediction.

Tabelle 5.2: Final tuned configuration of the SOH-estimation MLP after hyperparameter optimisation, listing the number of nodes and the activation function used in each layer of the network.

Layer	Nodes	Activation Function
0	128	ReLU
1	256	ReLU
2	256	ReLU
3	256	ReLU
4	128	ReLU
5	512	ReLU
6	128	ReLU
7	256	ReLU
8	256	ReLU
9	1	Linear

5.5 Configuration and Hyperparameter Tuning of the Neural Network

- This section describes the transition from the initial small MLP baseline to the tuned final network, with the resulting architecture summarised in Table 5.2.
- Model quality is evaluated by means of mean squared error and mean absolute error.
- Mean squared error is used as the central optimisation-oriented metric during training and validation, whereas both MSE and MAE are used for the final interpretation on the held-out test cells. The original ageing-estimation study reports errors in percentage points of SOH; for consistency with the later chapters, the corresponding normalized SOH error is obtained by dividing MAE values by 100 and MSE values by 10,000.
- The hyperparameter search covers the number of lag steps, the temporal resolution of the lag sequence, network depth, number of nodes, batch size, kernel regularisation, optimizer type, and activation function. Within this set, the number of lag steps and the temporal resolution are the most method-specific

parameters, because they define how much history is provided to the model and at which temporal granularity it is represented.

- The number of nodes controls the representational capacity of the network, whereas the network depth influences how strongly the model can build more abstract internal feature combinations across layers. Both parameters therefore affect the balance between flexibility, generalisation, and model complexity.
- Hyperband is used as the main search strategy because it distributes training resources adaptively, evaluates many candidate settings quickly, and allocates more epochs only to the most promising configurations.[34]
- The study reports that ADAM outperforms RMSProp, Adadelata, and Adagrad for this task. Larger batch sizes reduce overfitting tendencies, whereas explicit kernel regularisation does not lead to a meaningful improvement.
- Batch size also affects the training trade-off directly: larger batches are computationally efficient and converge reliably, whereas smaller batches can improve generalisation at the cost of longer or less stable optimisation.
- The optimizer analysis is complemented by an activation-function comparison. ReLU yields the lowest loss in the tested setup, while sigmoid, tanh, and softmax perform worse.
- The models are trained on approximately 15,000 sequences with varying batch-size and epoch combinations, and larger batches converge to a comparable minimum while reducing overfitting tendencies.

5.6 Results and Performance Evaluation

- This section evaluates how the two most influential sequence-design parameters, namely the number of lag steps and the temporal resolution, affect the final SOH-estimation performance.
- For the detailed analysis, Figure 5.5 compares three exemplary cells aged under different conditions. Four time resolutions are considered, namely 1 min, 10 min, 30 min, and 60 min, while the number of lag steps is varied between 8, 12, 16, and 20. All MAE values in this section are shown in percentage points of SOH to

match the source study; the dissertation-wide normalized MAE on a 0–1 SOH scale is the same value divided by 100.

- Across the exemplary cells, the most favourable time-resolution range lies between 10 min and 30 min. This matches the characteristic charge and discharge durations of the investigated scenarios, which typically fall in the range of approximately 20 min to 1 h.
- If the time resolution is too large, relevant behaviour is averaged out and important information is lost. If it is too small, the training process becomes less effective, the effective histories become unnecessarily long, and the resulting representation does not translate automatically into better prediction quality.
- The MAE matrix shows the smallest errors for 16 lag steps at a time resolution of 10 min. This corresponds to a history length of 160 min.
- Extending the sequence length beyond 20 steps is reported as disadvantageous because the resulting history becomes excessively long and does not improve prediction quality further.

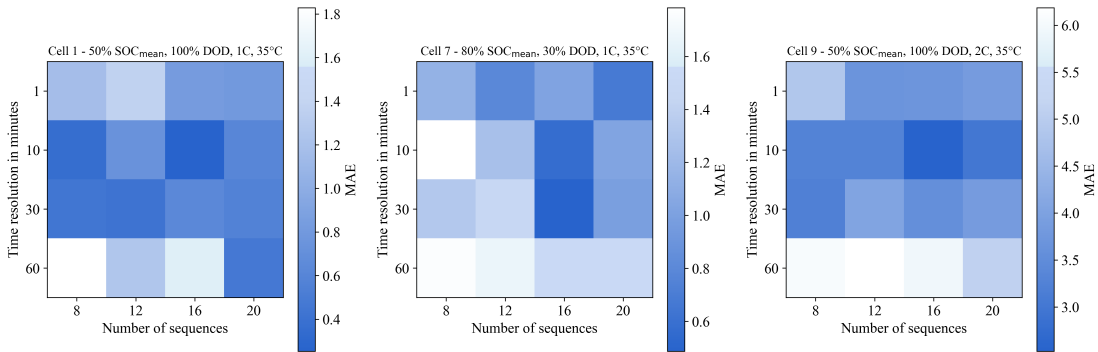


Abbildung 5.5: Matrix of MAE values shown in percentage points of SOH, following the source study. The corresponding normalized MAE on a 0–1 SOH scale is obtained by dividing the displayed values by 100. The colour scale indicates the magnitude of the prediction error and highlights the most favourable operating region, which lies in the range of approximately 10 min to 30 min for the time resolution.

- Figure 5.6 compares the predicted and measured SOH trajectories. Cells 7 and 1 represent typical ageing behaviour and are reproduced well, whereas a cell with a

pronounced U-shaped SOH progression leads to a substantially larger prediction error.

- Cells 7 and 1 are used as representative examples of typical ageing behaviour, whereas cell 9 serves as a deliberately selected atypical case.
- Figure 5.7 complements this trajectory-based view with a direct comparison between predicted and measured SOH values. For the favourable individual cells, the reported best errors are MSE values of $0.410\%^2$ and $0.118\%^2$ and MAE values of 0.487% and 0.253% ; these correspond to normalized MAE values of 0.00487 and 0.00253 on a 0–1 SOH scale.
- Across the evaluated cases, the average MAE is 1.10% , corresponding to a normalized MAE of 0.0110 , and the average MSE is $2.95\%^2$, corresponding to a normalized MSE of $2.95 \cdot 10^{-4}$.
- The prediction plots additionally show that a post-rolling filter is applied to reduce visual noise and to make the comparison between measured and estimated SOH easier to interpret.

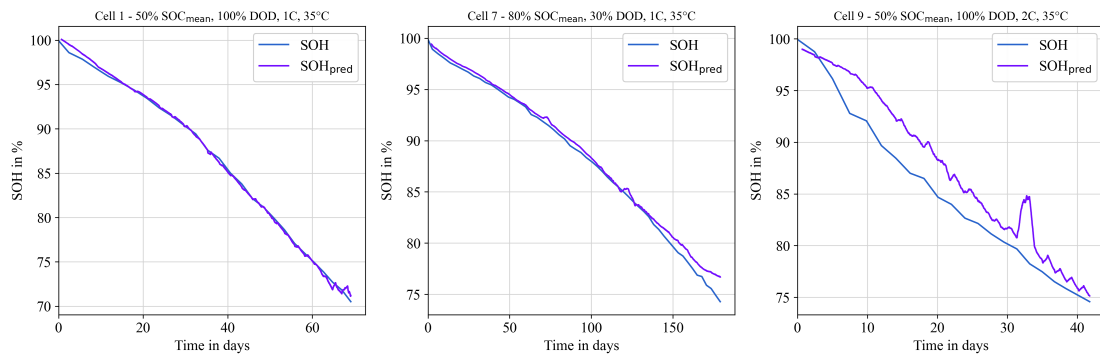


Abbildung 5.6: Comparison of predicted and measured SOH trajectories over time for three representative test cells. Cells 7 and 1 show the typical ageing progression most frequently observed in the dataset and are estimated accurately by the network, whereas Cell 9 was intentionally chosen as an atypical example with a pronounced U-shaped SOH curve. A post-rolling filter is applied to reduce visual noise and to make the agreement and deviation between measured and predicted trajectories easier to interpret.

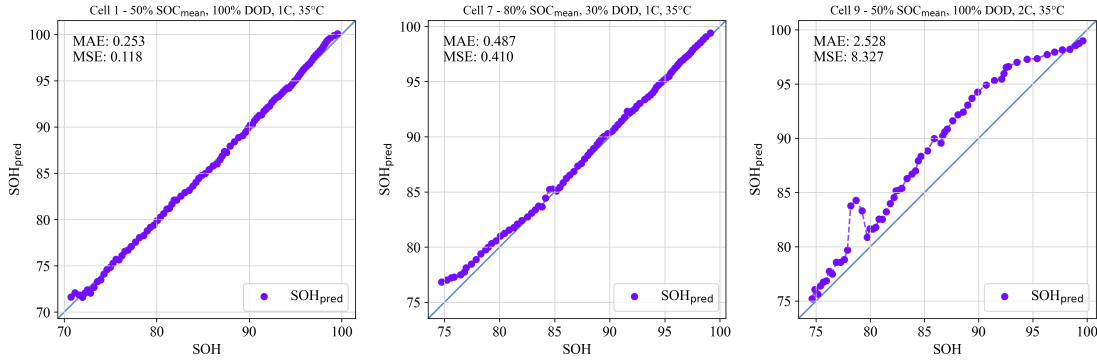


Abbildung 5.7: Scatter comparison between predicted and measured SOH values for the three representative test cells. Cells 7 and 1 exhibit only small dispersion and low absolute error, indicating accurate prediction, whereas Cell 9 shows substantially larger deviation in both MSE and MAE because of its atypical ageing behaviour. Error values are shown in percentage points of SOH (MAE in %, MSE in $\%^2$); normalized MAE and MSE values on a 0–1 SOH scale are obtained by dividing by 100 and 10,000, respectively.

- The effect of removing the problematic cells 3 and 4 from the training basis is also examined. In that variant, the aggregate error improves only marginally to an MAE of 1.07 % (normalized MAE 0.0107) and an MSE of 2.52 $\%^2$ (normalized MSE $2.52 \cdot 10^{-4}$).
- At the same time, the behaviour on the atypical cell 9 becomes more unstable, which indicates that cleaner data alone do not replace broader data diversity.
- The result section therefore already makes a broader methodological point: the exact optimum for sequence length and time resolution depends strongly on the available dataset, and direct comparison with other literature remains difficult if the underlying cell data differ substantially.

5.7 Interpretation of the Ageing-Estimation Results

- A meaningful SOH estimate can be obtained with a comparatively simple feed-forward model when the preprocessing and feature engineering are aligned with the underlying ageing problem.
- One of the central insights is that the input representation is at least as important as the model class itself. Cumulative-current features, lag-sequence construction,

and the explicit reconstruction of the SOH target are decisive parts of the overall method and form the actual methodological core of this chapter.

- The results simultaneously show the dependence of neural SOH estimation on the available dataset. Atypical ageing trajectories, interrupted tests, and limited data diversity remain the dominant constraints on generalisation.
- Even under these limitations, the approach reaches a level of accuracy that remains competitive with more complex alternatives. Within the dissertation, this chapter should therefore not be read as a claim that feedforward MLPs are universally preferable to recurrent models. Its role is to establish a first data-driven baseline in which ageing-relevant temporal information is represented explicitly while the estimator architecture itself is kept deliberately simple.
- The method is not restricted to one narrowly defined technology, because it relies only on directly measured quantities such as voltage, temperature, and current-derived throughput features. In principle, the concept can therefore be transferred to other cells and chemistries if sufficiently rich data are available.
- This baseline role is important for the broader argument of the dissertation. Once a workable representation for ageing information has been identified, the next question is no longer only how well the estimator performs under nominal conditions, but how different estimator classes behave under deployment-relevant disturbances such as bias, noise, missing samples, and initialization errors. This motivates the shift toward the robustness benchmark in the following chapter.
- Building on this robustness perspective, the later chapters extend the analysis to recurrent LSTM–MLP models for SOC and SOH estimation. There, the aim is no longer only to compare estimator concepts at a general level, but to examine whether stateful recurrent models offer advantages when temporal information has to be processed causally over long streaming sequences and ultimately transferred to embedded microcontroller execution.

Robustness Benchmarking of Embedded Battery State Estimation

6.1 Motivation and practical relevance

This chapter investigates deployment-facing battery-state estimation under realistic Battery Management System conditions rather than under clean laboratory input signals alone. Reliable SOC and SOH estimation is a core BMS function because it directly shapes safety margins, usable energy, operating limits, charge control, and degradation-aware management in both mobile and stationary battery systems.[35, 36, 37] Practical deployment quality is not defined by clean-condition accuracy alone, however, because real BMS operation is exposed to sensor bias, additive noise, missing samples, irregular timing, restart-related initialization mismatch, and transient signal spikes caused by the measurement chain or by communication faults.[38, 39, 40]

In the requirement framework introduced in Figure 2.1, this chapter therefore occupies the performance branch of the dissertation. The focus is not a full production-BMS validation campaign, but the question of how estimator classes behave when realistic measurement-side disturbances are applied under otherwise matched conditions. Estimator quality is accordingly interpreted as a three-part performance problem consisting of nominal baseline accuracy, convergence behavior after state inconsistency, and robustness under disturbed measurements and measurement-continuity faults.

6.2 State of the art in battery state estimation

The relevant literature usually distinguishes between direct-measurement or integration-based estimators, typically centered on Coulomb counting, model-based estimators built around equivalent-circuit, observer-based, or electrochemical structure, hybrid approaches that combine physical structure with learned adaptation, and purely data-driven recurrent or sequence models.[41, 35, 36, 37] Direct-measurement approaches remain attractive because they are simple, computationally inexpensive, and easy to deploy, but they are structurally exposed to current bias, missing data, and initialization errors because they do not contain an internal correction mechanism.[41, 38]

Equivalent-circuit and observer-based estimators offer physical interpretability and voltage-based correction capability, but their practical performance depends strongly on parameter quality, operating-point observability, and measurement quality.[42, 36, 38] Hybrid approaches combine physical state propagation with learned adaptation of capacity, parameters, or auxiliary states and are therefore attractive for joint SOC/SOH estimation;[43, 44, 45] such combinations can improve baseline calibration, but they do not automatically remove structural sensitivity to measurement disturbances. Data-driven recurrent estimators can exploit nonlinear relations between voltage, current, temperature, and derived online variables, and more recent work has also shown that such models can be prepared for embedded deployment through TinyML-oriented implementations, pruning, and quantization.[46, 47, 48, 49, 37]

6.3 Performance and embedded deployment gap

The main benchmark gap is not only that much of the literature still emphasizes clean-data accuracy, average error metrics, or computational efficiency, but also that comparatively fewer studies analyze estimator behavior under realistic measurement faults and timing disturbances.[35, 47, 48, 49] Where robustness is studied, it often remains confined to a single estimator family, so cross-family comparisons under the same disturbance protocol and under comparable embedded-ready execution conditions are still limited.[38, 39]

Broader BMS benchmarking studies have highlighted the importance of reproducible scenario design and deployment-relevant validation, but they do not by themselves close the gap between clean-data estimator comparison and measurement-

level performance benchmarking across fundamentally different estimator classes.[40] This gap is especially relevant in embedded BMS design, where estimator selection must jointly consider nominal accuracy, convergence after inconsistency, robustness under disturbed measurements, and computational constraints.

6.4 Contribution and study objectives

The central objective is a measurement-only robustness benchmark in which estimator performance is isolated from broader system-level effects such as hardware-resource optimization or implementation-effort comparison. The benchmark is deliberately restricted to disturbances applied only to measurable inputs and their timing, including quantization effects, bias, additive noise, missing samples, irregular sampling, burst dropout, voltage spikes, and initialization mismatch. This design keeps the scenarios physically interpretable and allows the resulting failure modes to be compared under reproducible conditions.

The chapter compares four embedded-ready estimator classes on the same LFP ageing trajectory and under the same disturbance protocol: a direct-measurement model (DM), a hybrid direct-measurement model with learned SOH support (HDM), a hybrid equivalent-circuit model with voltage feedback (HECM), and a data-driven recurrent model with GRU-based SOC estimation and LSTM-based online SOH context (DD). Under clean conditions, HDM provides the lowest baseline error with $\text{MAE} = 0.0024$ (0.24 percentage points), followed by HECM with 0.0090, DD with 0.0100, and DM with 0.0311. Under disturbed measurements, however, the ranking changes substantially, which is the core reason for the benchmark. The benchmark should therefore be read as a controlled cross-class comparison that isolates estimator-side failure modes under matched data, matched disturbance definitions, and matched online execution rules. Its purpose is not to identify one universal winner, but to expose complementary strengths and weaknesses across estimator families and to translate them into decision-oriented guidance for embedded BMS design.

6.5 Methodology

6.5.1 Estimator classes under test

Figure 6.1 summarizes the benchmark chain from measured signals to online feature reconstruction, disturbance injection, estimator execution, and global or local performance evaluation. The comparison covers four estimator classes with deliberately different correction mechanisms: DM is the fixed-capacity Coulomb-counting reference, HDM adds capacity adaptation through the shared online SOH estimate, HECM adds voltage-residual feedback through a SOH-aware two-RC observer,[42, 36] and DD represents the compact recurrent neural estimator with stateful GRU-based SOC prediction.[46, 37]

HDM, HECM, and DD all use the same LSTM-based online SOH estimator. The benchmark therefore compares how the SOC estimators use a common health context instead of comparing different SOH backbones. The causal online quantities required by several estimators, especially cumulative charge Q_c , equivalent full cycles EFC, derivatives, and hourly SOH aggregates, are reconstructed from the measured stream during benchmark execution. This prevents precomputed dataset-side features from entering only one estimator class.

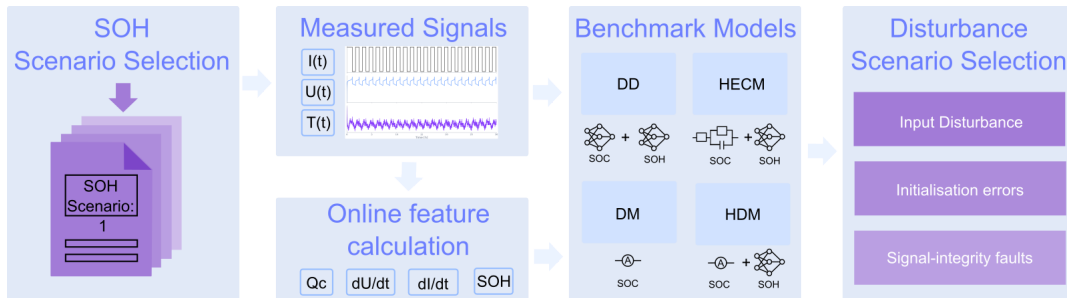


Abbildung 6.1: Methodology overview of the robustness benchmark. Measured current, voltage, temperature, and time signals are first converted into causal online auxiliary quantities such as cumulative charge Q_c , derivative channels, equivalent full cycles EFC, and shared SOH context. The resulting online stream is then processed by the four estimator classes under matched execution conditions, exposed to measurement-only disturbance scenarios, and finally evaluated through global and local performance metrics that separate baseline accuracy, convergence behavior, and disturbed-operation robustness.

6.5.2 SOC estimation methods

Direct-measurement model (DM)

The overview in Figure 6.1 is implemented through four estimator variants that differ in their SOC correction mechanism, but share the same causal measurement stream. The direct measurement branch first constructs the cumulative charge state

$$Q_c(k) = Q_c(k-1) + \frac{I_k \Delta t_k}{3600 \text{ s h}^{-1}}, \quad (6.1)$$

where I_k is the measured current and Δt_k is the elapsed time between two samples. The division by 3600 s h^{-1} converts the current-time product into ampere-hours. In the DM estimator this charge state is mapped to SOC with a fixed nominal capacity,

$$\widehat{\text{SOC}}_{\text{DM}}(k) = \text{clip}\left(1 + \frac{Q_c(k)}{C_{\text{nom}}}, 0, 1\right). \quad (6.2)$$

Hybrid direct-measurement model (HDM)

The same integration core is also used in the HDM estimator. The difference is that the denominator is updated through the shared online SOH estimate,

$$C_{\text{eff}}(k) = C_{\text{nom}} \cdot \widehat{\text{SOH}}(k), \quad (6.3)$$

which gives the hybrid direct-measurement SOC estimate

$$\widehat{\text{SOC}}_{\text{HDM}}(k) = \text{clip}\left(1 + \frac{Q_c(k)}{C_{\text{eff}}(k)}, 0, 1\right). \quad (6.4)$$

DM and HDM therefore differ only in the assumed usable capacity. Both variants keep the same constant-voltage full-charge reset logic: if the upper-voltage condition is sustained for the configured duration, the internal charge state is reset to $Q_c = 0$ and the SOC estimate is anchored at full charge.

Hybrid equivalent-circuit model (HECM)

The HECM extends the current-driven charge balance with voltage-residual correction. It is implemented as a two-RC observer with state vector

$$\mathbf{x}_k = \left[\widehat{\text{SOC}}_k \quad U_{1,k} \quad U_{2,k} \right]^\top, \quad (6.5)$$

where $U_{1,k}$ and $U_{2,k}$ denote the polarization voltages of the two RC branches. The prediction step uses the previous current sample and propagates the state according to

$$\mathbf{x}_k^- = \mathbf{A}_k \mathbf{x}_{k-1} + \mathbf{b}_k I_{k-1}, \quad (6.6)$$

with

$$\mathbf{A}_k = \text{diag}\left(1, e^{-\Delta t_k/\tau_1}, e^{-\Delta t_k/\tau_2}\right), \quad \mathbf{b}_k = \begin{bmatrix} \eta \Delta t_k / C_b \\ R_1 \left(1 - e^{-\Delta t_k/\tau_1}\right) \\ R_2 \left(1 - e^{-\Delta t_k/\tau_2}\right) \end{bmatrix}. \quad (6.7)$$

Here $C_b = C_0 \widehat{\text{SOH}}$ is the SOH-scaled capacity used by the observer. The voltage prediction used for the correction step is

$$\hat{U}_k = \text{OCV}\left(\widehat{\text{SOC}}_k, \widehat{\text{SOH}}_k, I_k\right) + U_{1,k} + U_{2,k} + R_i I_k. \quad (6.8)$$

The Kalman correction then uses the residual between this predicted terminal voltage and the measured voltage to update the internal state. The parameters R_i , R_1 , R_2 , τ_1 , τ_2 , OCV, and dOCV/dSOC are not treated as constants; they are interpolated from pre-identified SOC/SOH lookup surfaces with separate charge and discharge parameterizations. This is the main structural reason why HECM can re-anchor the integrated SOC estimate through voltage information while still using the same online SOH context as HDM and DD.

Data-driven SOC model (DD)

The DD estimator uses a stateful GRU instead of an explicit observer equation. At each online time step it receives the feature vector

$$\mathbf{z}_k = \left[U_k \quad I_k \quad T_k \quad \widehat{\text{SOH}}_k \quad Q_c(k) \quad \frac{dU}{dt}(k) \quad \frac{dI}{dt}(k) \quad \Delta t_k \right]^\top. \quad (6.9)$$

Figure 6.2(a) shows the block structure of this model. In its floating-point base configuration, a single GRU layer with hidden size 96 processes the eight-channel feature vector; an MLP head with one hidden layer of size 96, ReLU activation, and a sigmoid output then maps the recurrent state to a SOC estimate constrained to $[0, 1]$. The benchmark itself uses the deployment-prepared variant of this model, in which structured pruning reduces the hidden size from 96 to 67 and the weights are stored as INT8 values (Section 6.5.4). During training, the recurrent core is

exposed to causal sequence segments of

$$L_{\text{SOC}} = 2024 \quad (6.10)$$

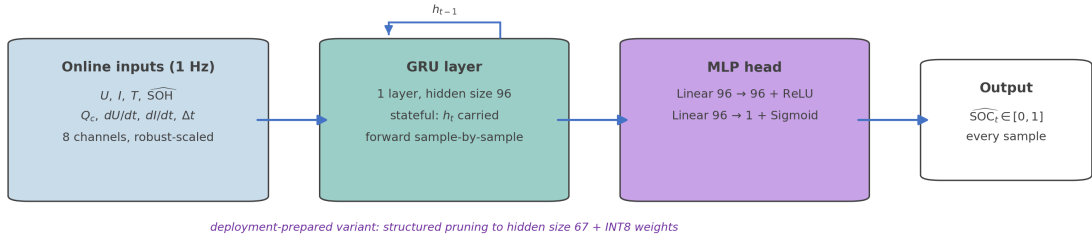
consecutive time steps. During benchmark execution the same model is run statefully and causally, so the hidden state is carried forward sample by sample and the MLP head produces a continuous SOC trajectory without access to future samples.

The derivative channels in the DD input are rebuilt online from the disturbed measurement stream,

$$\frac{dI}{dt}(k) = \frac{I_k - I_{k-1}}{\Delta t_k}, \quad \frac{dU}{dt}(k) = \frac{U_k - U_{k-1}}{\Delta t_k}. \quad (6.11)$$

Together with the explicit Δt_k channel, these quantities allow the neural model to distinguish local voltage or current dynamics from timing changes under irregular sampling. The inclusion of $Q_c(k)$ also ensures that the DD model receives the same charge-throughput information that drives DM and HDM.[46, 37]

(a) Data-driven SOC estimator (DD): stateful GRU + MLP



(b) Shared SOH estimator: hourly LSTM sequence-to-sequence model

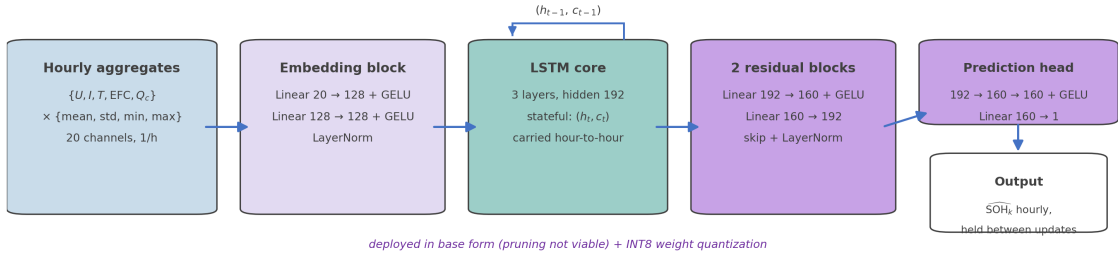


Abbildung 6.2: Block-level architectures of the data-driven estimator components used in the benchmark. (a) The DD SOC estimator processes the eight online input channels with a single stateful GRU layer of hidden size 96, followed by an MLP head with one ReLU-activated hidden layer of size 96 and a sigmoid output. (b) The shared SOH estimator aggregates the five online channels into a 20-dimensional hourly feature vector, projects it through a two-layer feature projection block of width 128, and processes the projected sequence with a three-layer stateful LSTM of hidden size 192, two residual MLP blocks with an internal width of 160, and a prediction head that maps 192 to 160 to 1. The model is deployed with INT8 weight quantization; see Section 6.5.4.

6.5.3 SOH estimation method

HDM, HECM, and DD all use the same LSTM-based SOH estimator. Its input is an hourly feature vector obtained from the online channels $\{U, I, T, \text{EFC}, Q_c\}$ by computing mean, standard deviation, minimum, and maximum for each channel. The five base channels therefore form a 20-dimensional hourly input vector.[50, 51, 52, 1]

Figure 6.2(b) summarizes the architecture of this network. It consists of four sequential stages.

The first stage is a **feature projection block**. The term *projection* is used deliberately here to avoid confusion with hardware *embedding* discussed elsewhere in

the dissertation; the function of this block is a learned transformation of the 20-dimensional statistical input into a compact, task-adapted representation before the recurrent processing begins. This is necessary because the 20 input values are heterogeneous statistics – means, standard deviations, minima, and maxima – drawn from physically different channels with different units and scales. The projection block consists of two successive linear layers of width 128 with layer normalisation and GELU activation between and after them. Each linear layer is a full matrix multiplication with its own trainable weights, so the first layer learns which combinations of the 20 input statistics are jointly relevant for SOH, and the second layer refines these combinations further. The GELU activation function is used instead of ReLU because it is smooth and differentiable everywhere, which supports gradient flow in deeper networks without the dying-neuron effect of hard-clipping negative values to zero; this choice is consistent with the residual-style architecture that follows.[53]

The second stage is a **three-layer unidirectional LSTM** with hidden size 192. Three stacked LSTM layers are used because SOH evolves over long time horizons – weeks to months of hourly aggregates – and a hierarchical recurrent structure allows the first layer to capture short-term variation patterns, the second layer to aggregate these into medium-term trends, and the third layer to distil the slowest degradation dynamics that unfold over weeks to months. A single layer would conflate all these timescales in one set of weights. The network is strictly unidirectional: the hidden state at each hourly step accumulates information only from past observations, with no access to future measurements, which is a hard requirement for deployment in a continuously running BMS.

The third stage consists of **two residual MLP blocks**. Each block receives the 192-dimensional LSTM output as input $\mathbf{x} \in \mathbb{R}^{192}$ and computes

$$\mathbf{y} = \text{LayerNorm}(\mathbf{x} + \text{Dropout}(W_2 \text{GELU}(W_1 \mathbf{x}))), \quad (6.12)$$

where $W_1 \in \mathbb{R}^{160 \times 192}$ contracts the input to an internal width of 160 and $W_2 \in \mathbb{R}^{192 \times 160}$ expands it back to 192. The key property is the *skip connection*: the original LSTM output $\mathbf{x}$ is added directly to the transformed output before normalisation. As a result, the block does not have to learn the full output from scratch; instead it learns only the *residual correction* – the difference between what the LSTM already computed and a better representation. This protects the useful temporal information already contained in the LSTM output: in the worst case, the block

weights converge toward zero and the signal passes through unchanged. Without the skip connection, the block could overwrite the LSTM output entirely, which would waste the recurrent computation. The input and output dimension stays at 192 throughout both blocks; they refine the representation without compressing it.

The fourth stage is the **prediction head**, a three-layer MLP that maps the 192-dimensional LSTM output to a single scalar SOH estimate: a first layer reduces 192 to 160, a second hidden layer stays at width 160, both with GELU activation and dropout, followed by a final linear layer with a single output neuron. There is no skip connection here because the purpose of this stage is precisely to compress the high-dimensional recurrent state into a scalar; the final linear output is unconstrained so that the network can represent the full SOH range without saturation effects.

Three width values appear in this section with distinct roles: 128 is the feature-projection width, 192 is the LSTM hidden size, and 160 is the shared internal width of the residual blocks and the prediction head. The base model has approximately 1 041 000 trainable parameters and is deployed with INT8 weight quantization (Section 6.5.4). In online execution, only one hourly aggregate is processed at a time; the hidden and cell states are carried forward step by step using only past observations, the first output is anchored with the configured initial SOH value, and later estimates are held piecewise constant between hourly updates. This common SOH context is consumed differently by the SOC estimators: HDM uses it through C_{eff} , HECM uses it in the observer capacity and lookup scheduling, and DD receives it as part of its recurrent feature vector.

6.5.4 Embedded-oriented model preparation

All compared estimators are constrained to a common STM32H755ZI-class reference platform with 2 MB dual-bank flash (1 MB per bank) and 1 MB RAM. This shared feasibility criterion keeps the comparison deployment-grounded: it ensures that all performance differences are interpretable within a common hardware class and prevents an asymmetric comparison between compact analytical estimators and neural models that would not be feasible on the same platform.

DM requires no compression: its four persistent scalar states occupy 4.2 KiB of flash. HECM also requires no neural-weight compression, but its flash footprint is dominated by the embedded ECM parameter lookup tables – two SOH×SOC grids of seven electrical parameters (internal resistance, RC time constants, open-circuit voltage, and its derivative) for the charge and discharge directions, stored as float32

constants – bringing its flash total to 442.5 KiB and making it the largest single component. The recurrent components used by HDM and DD are deployment-prepared variants. For the SOC-GRU, structured width reduction and finetuning reduce the hidden size from 96 to 67, followed by INT8 weight quantization. For the SOH-LSTM, structured pruning at the applied intensity collapses model accuracy entirely, so this component is deployed in its base form with INT8 weight quantization only, retaining all 192 hidden units across three layers. The resulting deployment-ready hidden sizes are therefore 67 for the SOC-GRU and 192 for the SOH-LSTM. The SOH-LSTM in INT8 carries approximately 1 034 KiB of quantized weight data plus a further 29 KiB of float32 bias and normalisation parameters, for a total of approximately 1 038 KiB – occupying essentially one full 1 MB flash bank on its own. The remaining components (SOC-GRU at 27.9 KiB, DM at 4.2 KiB, HECM at 442.5 KiB) fit together in the second bank. Table 6.1 lists the reusable building blocks; when these components are combined to form DM, HDM, HECM, and DD, all four estimators remain within the 2 MB dual-bank flash capacity of the STM32H755ZI reference platform.

Tabelle 6.1: Estimated persistent flash and RAM footprints of the reusable estimator components used in the deployment-prepared benchmark variants. For the SOC-GRU, the flash value refers to the structured-pruned, finetuned, and INT8-quantized version. RAM* counts only architecture-intrinsic persistent numerical states; temporary buffers, framework overhead, stack usage, and system-level runtime memory are excluded. The SOH-LSTM in INT8 occupies approximately one full 1 MB flash bank; the remaining components fit in the second bank of the STM32H755ZI dual-bank flash (2 MB total).

Component	Flash [KiB]	RAM* [B]	Basis
DM core (Coulomb counting)	4.2	16	Four persistent scalar states in the online update logic
SOC-GRU core	27.9	268	One recurrent hidden state with size 67 stored as float32
SOH-LSTM core	1038	4608	INT8 weight matrices (1034 KiB) plus float32 biases and LayerNorm parameters (29 KiB); three-layer LSTM h192 in base form without pruning; hidden and cell states stored as float32
HECM core (2RC observer)	442.5	52	Flash dominated by two SOH×SOC parameter tables (40×100 grid, 7 ECM parameters, charge/discharge, float64); RAM: observer state $x \in \mathbb{R}^3$, covariance $P \in \mathbb{R}^{3 \times 3}$, previous current

6.5.5 Performance benchmark design

The benchmark perturbs only signals that a BMS controller would actually receive from its sensors: current, voltage, temperature, sample availability, and sampling timing. Model-internal quantities – such as LSTM hidden states or the model weights themselves – are never modified directly. This restriction keeps all disturbance scenarios physically interpretable: every failure mode tested corresponds to something that can go wrong at the sensor, wiring, or communication layer of a real embedded system, rather than to artificial tampering with the estimator internals.[40]

Figure 6.3 separates the resulting cases into input disturbances, initialization mismatch, and measurement-continuity faults, meaning disturbances of sample availability, sample timing, or frozen data updates. The numerical levels in Table 6.2 combine three roles: realistic acquisition limits, adverse but plausible measurement

disturbances, and deliberate stress cases for recovery analysis.[54, 55, 56, 57, 58, 59] During every run, all four estimators receive the identical disturbed sensor time series – the same modified sequence of voltage, current, temperature, and timing samples – so that any difference in output is attributable to estimator structure rather than to differences in the input. Each estimator then reconstructs its auxiliary quantities (cumulative charge Q_c , equivalent full cycles EFC, current derivative, hourly SOH aggregates) causally and online from this disturbed time series, not from precomputed clean dataset columns. A sensor bias therefore propagates into the derived features exactly as it would in a real BMS, which keeps the comparison physically consistent across all estimator classes.

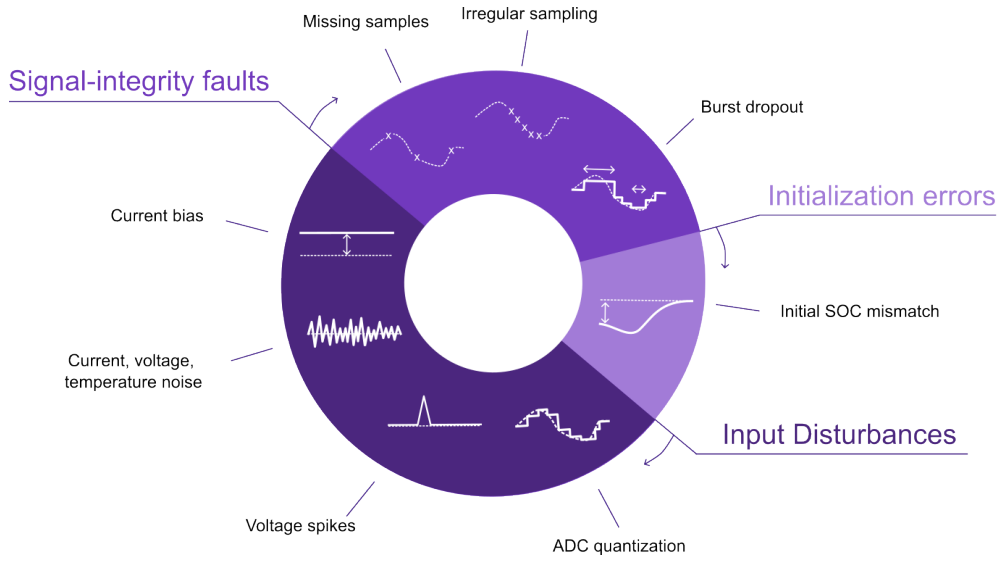


Abbildung 6.3: Disturbance taxonomy used in the robustness benchmark. Input disturbances, initialization mismatch, and measurement-continuity faults are separated so that drift, loss of correction, delayed recovery, and transient instability can be interpreted as different estimator-side effects.

Tabelle 6.2: Scenario protocol used in the robustness benchmark. The entries define the fixed disturbance settings and the practical motivation behind each measurement or continuity/timing case.

Scenario	Benchmark setup	Practical rationale
ADC quantization	$\Delta I = 0,01 \text{ A}$, $\Delta U = 0,005 \text{ V}$, $\Delta T = 0,5 \text{ }^\circ\text{C}$	Deployment-level discretization, sensor scaling, and rounding effects in embedded acquisition chains.[54, 55, 60]
Current bias	Constant $+0,05 \text{ A}$ in the compact matrix; additional sweep of 0.5%, 1.5%, and 3.0% of $ I _{\max}$	Sensor offset, gain error, miscalibration, and long-term drift stress on integration-driven estimators.[55, 56, 57]
Current noise	Gaussian noise with $\sigma_I = 0,10 \text{ A}$ in the compact matrix; local-detail sweep up to $\sigma_I = 0,20 \text{ A}$	Adverse current-measurement noise and noise-driven disturbance propagation.[38, 39]
Voltage noise	Additive Gaussian noise with $\sigma_U = 0,01 \text{ V}$	Wiring noise, front-end noise, insufficient filtering, and EMI-sensitive correction channels.[38, 54, 58]
Temperature noise	Additive Gaussian noise with $\sigma_T = 1,0 \text{ }^\circ\text{C}$	Sensor tolerance, conversion noise, thermal gradients, and sensor-to-cell mismatch.[60, 45]
Initial SOC mismatch	Additive initial mismatch of -0.10 SOC , corresponding to a 10-percentage-point offset; in DD realized through perturbed online Q_c	Restart conditions, missing recent history, or weak initial anchoring under practical deployment.[38, 42]
Missing samples	Every 50th sample frozen on the nominal 1 s grid	Packet loss, blocked updates, logger dropouts, or periodic frozen-data events.[39, 59]
Irregular sampling	Uniform timing jitter of $\pm 0,1 \text{ s}$ around the nominal 1 s grid	Scheduling jitter, bus latency, and asynchronous acquisition timing.[40, 39, 59]
Burst dropout	One central frozen gap of 3600 s	Communication interruption, frozen tasks, or severe delayed-information recovery stress.[40, 39]
Voltage spikes	One single-sample spike of $\pm 0,20 \text{ V}$ every 1000 samples (1000 s)	Transient voltage-sample corruption caused by EMI or switching disturbances.[58]

The compact scenario matrix gives one common protocol for the main result figures. Current bias and current noise are additionally examined through local sweeps, while ADC quantization remains included as a deployment-level sensing case even though it is not expected to dominate the ranking. In this form, the benchmark is a controlled estimator comparison under matched disturbances, not a complete validation of a production BMS.

6.5.6 Metrics and evaluation criteria

The benchmark evaluates three performance dimensions: nominal accuracy, convergence behavior after state inconsistency, and robustness under both longer-lasting and short-term measurement disturbances. These three dimensions are not independent post-hoc labels, but the aggregation logic used later in the result interpretation and in the decision-oriented synthesis. Nominal accuracy is read from the clean baseline run, robustness is read from the additional error caused by each disturbed scenario relative to that estimator's own baseline, and convergence is read from the time needed to return to a baseline-related error band after an imposed inconsistency or interrupted signal stream.

For each estimator m and disturbed scenario s , the main global robustness penalty is therefore defined as

$$\Delta\text{MAE}_{m,s} = \text{MAE}_{m,s} - \text{MAE}_{m,\text{base}}. \quad (6.13)$$

Positive values indicate that the disturbance degrades the estimator relative to its own clean-operation reference; values close to zero indicate that the disturbance does not materially change the full-trajectory average error. Global disturbed-operation performance is quantified through MAE, RMSE, bias, maximum error, and the 95th percentile absolute error. MAE and RMSE describe the average error level, bias describes systematic over- or underestimation, the maximum error reports the single largest deviation, and P95 is the absolute-error value that 95 % of all evaluated samples do not exceed. P95 therefore describes the larger-error range of the trajectory without being determined by the single worst sample.

Local disturbance performance is additionally analysed through scenario-specific quantities such as jump counts, output variance, absolute-error variance, excess rolling MAE, recovery time, and aligned post-disturbance error, because several scenarios do not primarily manifest as a collapse of full-run MAE but as short-term

output fluctuations, delayed drift, or slow re-synchronization. For scenarios in which the affected samples or time interval are explicitly known, the analysis separates the trajectory into pre-disturbance, disturbance, and post-disturbance phases. This makes it possible to quantify error growth during the disturbance, recovery time after the event, and residual error after the input stream has returned to normal. In the current benchmark phase, ADC quantization, current bias, voltage noise, temperature noise, missing samples, and irregular sampling are interpreted mainly through global metrics, whereas burst dropout, voltage spikes, current-noise detail, and initialization mismatch additionally require local evidence because their most relevant effects are episodic, recovery-related, or strongly localized in time.

Recovery after initialization errors is interpreted through two baseline-relative operating bands rather than through one universal fixed threshold. This keeps the recovery criterion tied to the normal clean-condition performance level of each estimator and avoids judging models with very different baseline accuracy against the same absolute tolerance. The strict band is defined as

$$B_{\text{strict}} = \max(\text{P95}_{\text{base}}, 1.5 \cdot \text{MAE}_{\text{base}}). \quad (6.14)$$

In the strict criterion, P95_{base} ensures that the recovery band includes the larger errors that already occur during clean baseline operation, while $1.5 \cdot \text{MAE}_{\text{base}}$ provides a minimum margin around the average baseline error. Taking the maximum of both terms means that recovery is judged against the more tolerant of these two baseline-derived limits, so recovery under this band means re-entry close to the normal baseline regime without imposing unrealistically narrow limits. The fair band is

$$B_{\text{fair}} = \max(2 \cdot \text{P95}_{\text{base}}, 3 \cdot \text{MAE}_{\text{base}}, 0.02), \quad (6.15)$$

and acts as a more deployment-tolerant criterion for estimators that re-enter a practically acceptable operating range without immediately returning to their strict near-baseline regime. The lower bound of 0.02 avoids unrealistically tight fair-band thresholds for very accurate baseline models. Both recovery bands require sustained re-entry over a fixed horizon so that the reported recovery times reflect stable resynchronization rather than an accidental short crossing of the threshold.

Reporting both bands avoids making the recovery conclusion depend on one manually chosen limit. If an estimator returns only to the fair band, it has reached a practically acceptable range but not its near-baseline regime; if it also returns to the

strict band, the recovery is stronger. This multi-scale metric design therefore evaluates performance as a structured combination of nominal accuracy, convergence behavior, global degradation, local disturbance response, and long-term drift.

Tabelle 6.3: Connection between raw benchmark metrics, result figures, and the three performance dimensions used in the robustness interpretation. The table makes explicit how the later statements on accuracy, robustness, and recovery/convergence are derived from the same metric set rather than from separate evaluation criteria.

Dimension	Primary metric evidence	Role in the result interpretation	Typical result sections
Nominal accuracy	Baseline MAE, RMSE, P95 absolute error, maximum error, and bias from the undisturbed run	Quantifies clean-sensing estimator quality before any robustness penalty is applied. This is why an estimator can lead in accuracy even if it is not the most robust under disturbances.	Baseline performance; decision-oriented accuracy score
Disturbed-scenario robustness	$\Delta\text{MAE}_{m,s}$ relative to the estimator's own baseline, complemented by disturbed-window MAE, P95 error, maximum error, bias, jump counts, and local excess rolling MAE where needed	Quantifies how strongly each disturbance changes the estimator's behavior. This supports scenario-specific robustness claims instead of ranking models only by absolute disturbed MAE.	Current bias, noise sensitivity, continuity/timing faults, voltage spikes, cross-scenario comparison
Convergence and recovery	First sustained re-entry into the strict or fair baseline-related error band after initialization mismatch or signal interruption; non-recovery inside the inspected window is treated as weakest evidence	Quantifies whether an estimator can return to a useful operating regime after state inconsistency. This explains why recovery behavior is evaluated separately from both clean accuracy and global disturbed-operation MAE.	Initialization mismatch, burst-dropout recovery, decision-oriented recovery score

For the decision-oriented synthesis in Section 6.8.3, the same logic is condensed into relative lower-is-better scores across the four estimator classes. The accuracy score is based on the clean baseline errors, the robustness score is based on ΔMAE across the disturbed scenarios, and the recovery score is based on recovery time after initialization mismatch. This aggregation is used only as a compact decision aid; the individual scenario figures and local recovery analyses remain the primary evidence for the detailed conclusions.

6.6 Experimental Setup

6.6.1 Dataset and data acquisition

The benchmark uses the LFP DoE Cycle Ageing Dataset described in Section 4.1.[2] It provides high-resolution ageing-test records from repeated cycling and check-up phases that extend from early-life operation into low-SOH operation, including sections below the common 80 % end-of-life reference level. This makes the dataset suitable for reproducible disturbance injection and evaluation-label construction over a broad degradation range. In the current benchmark phase, all estimator classes are evaluated on the same representative ageing trajectory from one LFP cell and under the same disturbance protocol so that performance differences can be attributed to estimator structure under matched data and label definitions before extending the study to additional cells and cross-cell validation.

The available raw channels are current, voltage, temperature, and time. Feature engineering converts these measurements into two separate data types: evaluation labels and causal online features. The labels are derived from capacity-test intervals and physically motivated SOC reset criteria and contain the capacity, SOH, and SOC traces used for benchmark scoring. The online features are reconstructed during each benchmark run from the same input stream that each estimator receives; they include the cumulative charge state Q_c , equivalent full cycles EFC, derivative channels such as dU/dt and dI/dt , and hourly SOH-context aggregates. For these hourly aggregates, the online channels $\{U, I, T, \text{EFC}, Q_c\}$ are summarized by mean, standard deviation, minimum, and maximum.

6.6.2 Evaluation labels and online feature reconstruction

For label construction, the signed transferred charge is computed sample-wise as

$$Q_k = \frac{I_k \Delta t_k}{3600}, \quad (6.16)$$

and the capacity label used as evaluation reference is obtained from the absolute transferred charge during dedicated capacity-test intervals,

$$C_{\text{ref}} = \sum_{k \in \mathcal{T}_{\text{cap}}} |Q_k|. \quad (6.17)$$

The SOH label used as evaluation reference is then defined as

$$\text{SOH}_{\text{ref}} = \frac{C_{\text{ref}}}{C_{\text{ref},0}}, \quad (6.18)$$

where $C_{\text{ref},0}$ denotes the first measured reference capacity of the trajectory, and the corresponding capacity and SOH columns are interpolated between the dedicated capacity-test intervals. The SOC label used as evaluation reference follows from Coulomb counting,

$$\text{SOC}_{\text{ref}} = 1 + \frac{Q_c}{C_{\text{ref},0}}. \quad (6.19)$$

During preprocessing, the cumulative charge state is reset to $Q_c = 0$ only after the voltage has remained close to the upper voltage limit for the configured CV duration; in the benchmark implementation this duration is 300 s. This prevents a short voltage excursion from being interpreted as a full-charge point. At the lower-voltage limit, Q_c is reset to $-C_{\text{ref},0}$.

In the benchmark implementation, the online EFC feature is reconstructed from absolute charge throughput,

$$Q_{\text{thr}}(k) = \sum_{j=1}^k \frac{|I_j| \Delta t_j}{3600}, \quad \text{EFC}(k) = \frac{Q_{\text{thr}}(k)}{C_{\text{nom}}}. \quad (6.20)$$

Here C_{nom} is the nominal cell capacity used in the benchmark feature pipeline. Unlike the SOC and SOH labels, EFC is an online feature and is therefore recomputed from the current and timing stream of each benchmark run.

This distinction between labels and online features is central to the benchmark. The labels remain fixed reference values for scoring the online estimates, whereas Q_c , EFC, dU/dt , dI/dt , and the hourly aggregates are rebuilt from the same clean or disturbed inputs that are passed to the estimators. Consequently, disturbed measurements affect the estimator inputs and derived features, but not the reference labels used for scoring.

6.6.3 Test trajectory and benchmark scenarios

All estimator classes are run on the same benchmark trajectory and the same scenario protocol from Table 6.2. This ensures that differences in the results are attributable to estimator structure and disturbance propagation rather than to different data splits or label definitions. The reported scenarios cover the nominal

baseline, current bias, additive current noise, voltage noise, temperature noise, initial SOC mismatch, missing samples, irregular sampling, burst dropout, voltage spikes, and ADC quantization. Together they probe long-term integration drift, voltage-correction sensitivity, disturbance propagation through derived features, frozen updates, timing uncertainty, and local post-event recovery. Input disturbances are applied directly to measured channels, initialization errors are injected only where structurally meaningful, and continuity/timing cases manipulate sample availability or timing while preserving online execution logic.

6.6.4 Implementation details

All compared estimators are executed under online conditions. Disturbed inputs are processed sample by sample, and the required auxiliary quantities are rebuilt directly from the disturbed signals rather than copied from precomputed dataset columns. This applies in particular to Q_c , EFC, dU/dt , dI/dt , and the hourly SOH aggregates, so that no hidden dataset-side variables leak into the benchmark runs.

For the recurrent neural estimators, the benchmark uses the intended stateful embedded execution mode. The SOH-LSTM carries its hidden and cell states from one hourly update to the next, while the SOC-GRU preserves its recurrent state along the sample-wise measurement stream. The neural variants used in the benchmark correspond to the compact pruned-and-quantized architectures introduced above. In freeze and dropout scenarios, disturbance propagation is handled consistently by freezing or distorting the derived online features together with the corresponding measured inputs, which is necessary for a physically meaningful comparison between model classes.

The direct-measurement estimators include the same sustained-CV full-charge reset criterion that is also used to define the full-charge reference point in preprocessing, whereas the data-driven initial-SOC test is implemented through an initial offset of the online Q_c feature because the neural estimator does not expose an explicit SOC state variable. The simulation environment therefore keeps the base nominal capacity used for online feature reconstruction fixed across all estimator classes, while all models receive the same disturbed input streams and are evaluated against the same reference labels and global and local metrics. This is one of the main reasons why the benchmark remains close to later embedded deployment: the estimators are not evaluated as offline regressors, but as causal state estimators under matched measurement-side constraints.

6.7 Results

Unless stated otherwise, all SOC error metrics in this chapter are reported as fractions on the normalized 0–1 SOC scale. A value of $\text{MAE} = 0.0024$, for example, corresponds to 0.24 percentage points.

6.7.1 Baseline performance

Under clean measurements, the hybrid direct-measurement model is the nominal-accuracy leader on the selected ageing benchmark trajectory. It reaches $\text{MAE} = 0.0024$, followed by HECM with 0.0090, DD with 0.0100, and DM with 0.0311. This baseline ranking is the reference point for the later ΔMAE comparisons and recovery bands, but it is not yet a deployment recommendation by itself. The complete baseline metric set, including RMSE, P95 error, maximum error, and signed bias, is listed in Appendix Table A.1; Figure 6.4 visualizes the same clean-condition comparison.

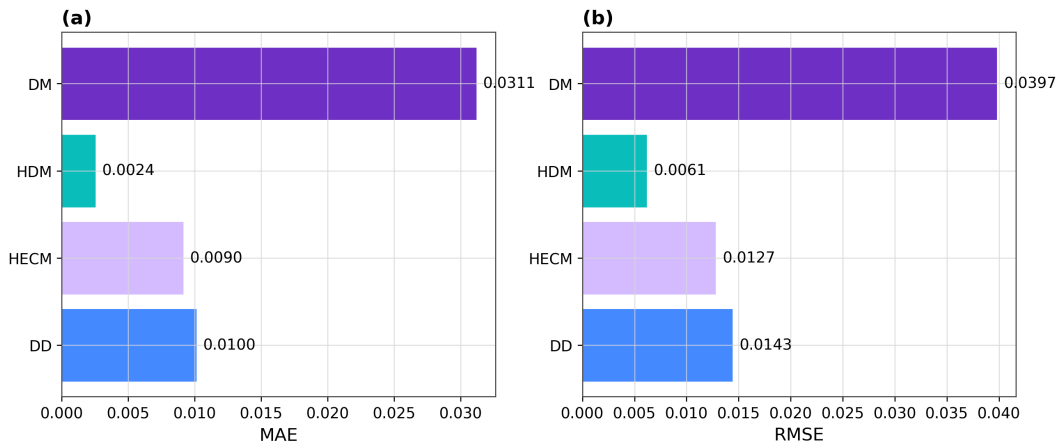


Abbildung 6.4: Baseline performance on the selected ageing benchmark trajectory. The summary compares the nominal MAE, RMSE, 95th-percentile error, and signed bias of the four estimator classes and therefore provides the clean-condition reference point for all later disturbed-scenario comparisons. The hybrid direct-measurement model provides the lowest baseline error, whereas the direct-measurement model shows the largest nominal deviation. Error values shown in the figure are SOC fractions on a 0–1 scale; multiply by 100 to obtain percentage points.

6.7.2 Current-bias sensitivity

Persistent current bias is the most severe integration stress in the benchmark. The dedicated sweep applies offsets of 0.5%, 1.5%, and 3.0% of the maximum absolute current of the benchmark trajectory, so the percentages describe the bias magnitude relative to $|I|_{\max}$ rather than an error percentage of SOC. Across this sweep, HECM remains the most robust estimator, DD is intermediate, and DM and HDM degrade strongly because the biased current accumulates directly through their integration path. For the strongly biased case reported in the compact scenario matrix, HECM reaches $\text{MAE} = 0.0932$, whereas DD, HDM, and DM increase to 0.2859, 0.3068, and 0.3143, respectively. The full MAE, RMSE, and P95 values are collected in Appendix Table A.2; Figure 6.5 shows the corresponding local bias example, sweep trend, and trajectory divergence.

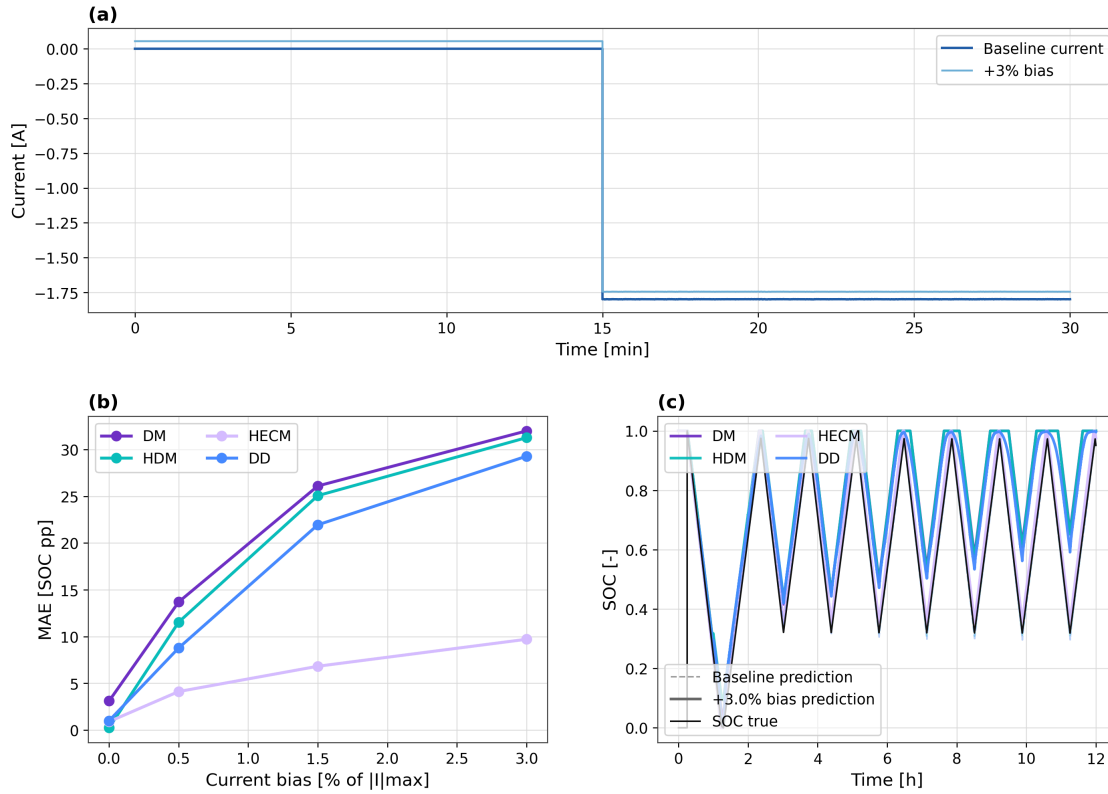


Abbildung 6.5: Sensitivity to current-measurement bias. The figure combines a representative local example of the applied bias, the global degradation of SOC accuracy across the matched bias sweep, and the resulting SOC divergence over time for a strongly biased case. Together, these views show why persistent current bias clearly separates the integration-based estimators from the voltage-corrected observer variant.

6.7.3 Noise sensitivity

The noise block is interpreted separately from the current-bias scenario because the injected noise terms are zero-mean disturbances rather than persistent offsets. The relevant question is therefore not only whether SOC drifts over the full trajectory, but also how a corrupted measurement channel propagates into observer correction, neural input features, and local output variability. For current noise, this propagation occurs through the measured current itself and through current-derived quantities such as Q_c , dI/dt , EFC, and the hourly SOH aggregation. In the compact benchmark matrix, the high-current-noise case uses $\sigma_I = 0,10$ A; the additional sweep in Figure 6.6 extends beyond this value to make local output-variability trends visible. Voltage noise with $\sigma_U = 0,01$ V mainly tests the voltage-correction

path in HECM and the voltage-dependent input path in the learned estimators; in this scenario HECM remains nearly unchanged, whereas DM, HDM, and DD show larger penalties. Temperature noise with $\sigma_T = 1,0^\circ\text{C}$ is weaker in the global metrics, but it still affects the SOH-aware paths because temperature enters the learned health context. The numerical global and local noise metrics are given in Appendix Tables A.2 and A.3.

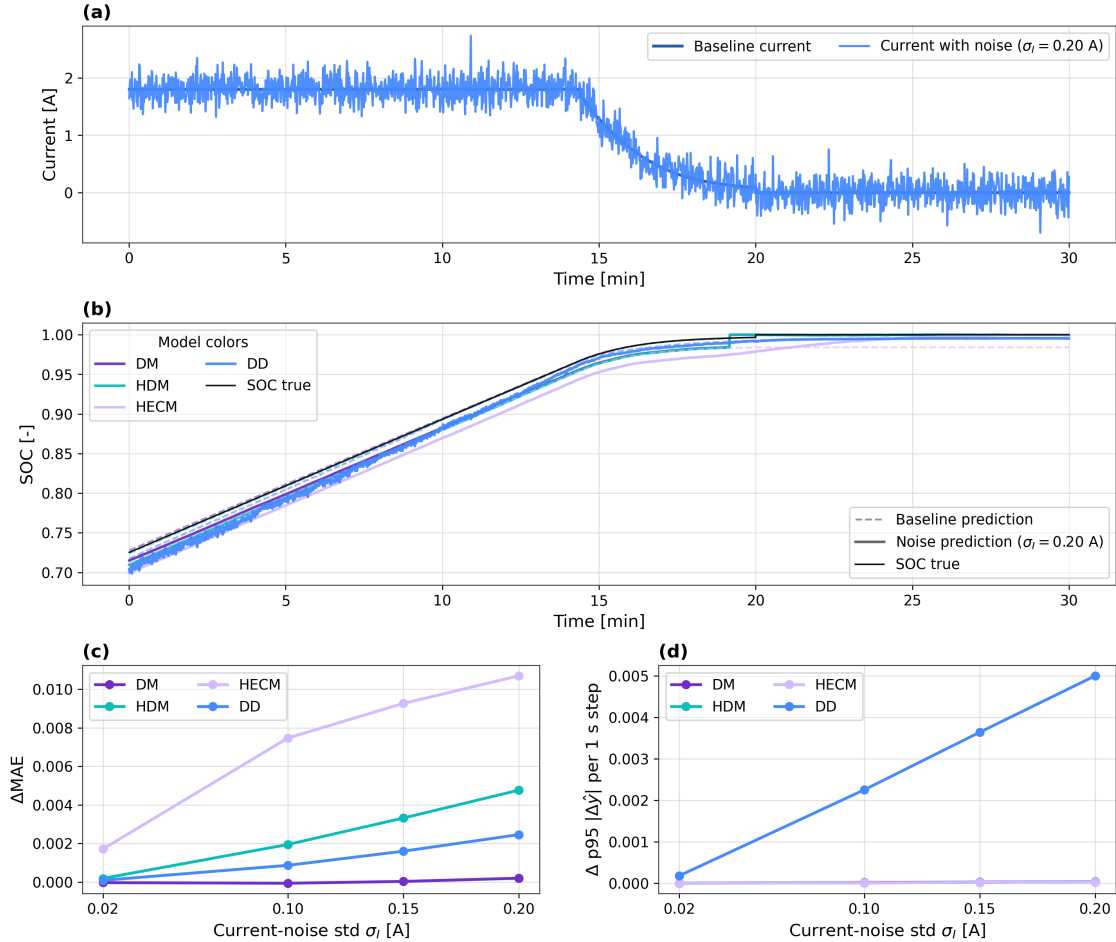


Abbildung 6.6: Noise sensitivity under additive current noise. The figure combines a representative disturbed window, a matched SOC comparison in the same interval, the global MAE penalty in the compact benchmark matrix, and the local output-variability trend across an extended current-noise sweep. This makes visible that zero-mean current noise is mainly a local disturbance-propagation test rather than the same type of long-term drift test as persistent current bias.

6.7.4 Initialization error and recovery

Initialization mismatch is a recovery problem rather than a simple full-run MAE problem, because the relevant question is whether the estimator can re-synchronize after starting from a wrong SOC value. The benchmark therefore uses a reset-free evaluation window and injects an initial mismatch of -0.10 on the normalized SOC scale, meaning that the online estimate starts 10 percentage points too low. This disturbance represents a deliberately visible restart inconsistency, for example after missing recent history or weak anchoring of the initial SOC. For DD, the mismatch is implemented as an offset of the online cumulative-charge feature instead of an artificial reset of an inaccessible latent state, which keeps the test aligned with deployment-time observables.

Recovery is evaluated with the strict and fair baseline-related bands defined in Section 6.5.6. These thresholds are not times and not fixed SOC setpoints; they are admissible absolute SOC errors on the normalized 0–1 scale. They differ between estimator classes because they are computed from each estimator’s own clean-condition MAE and P95 error. The strict band marks return to the near-baseline regime, while the fair band is a more tolerant operating range. In the inspected window, DD returns to the strict band after about 1.05 h and to the fair band after about 1.00 h; DM returns after about 1.17 h and 0.25 h, respectively; HECM returns after about 2.37 h and 1.16 h; and HDM does not re-enter either band. The exact threshold values and recovery times are listed in Appendix Table A.3; Figure 6.7 shows the corresponding trajectory-level recovery behavior.

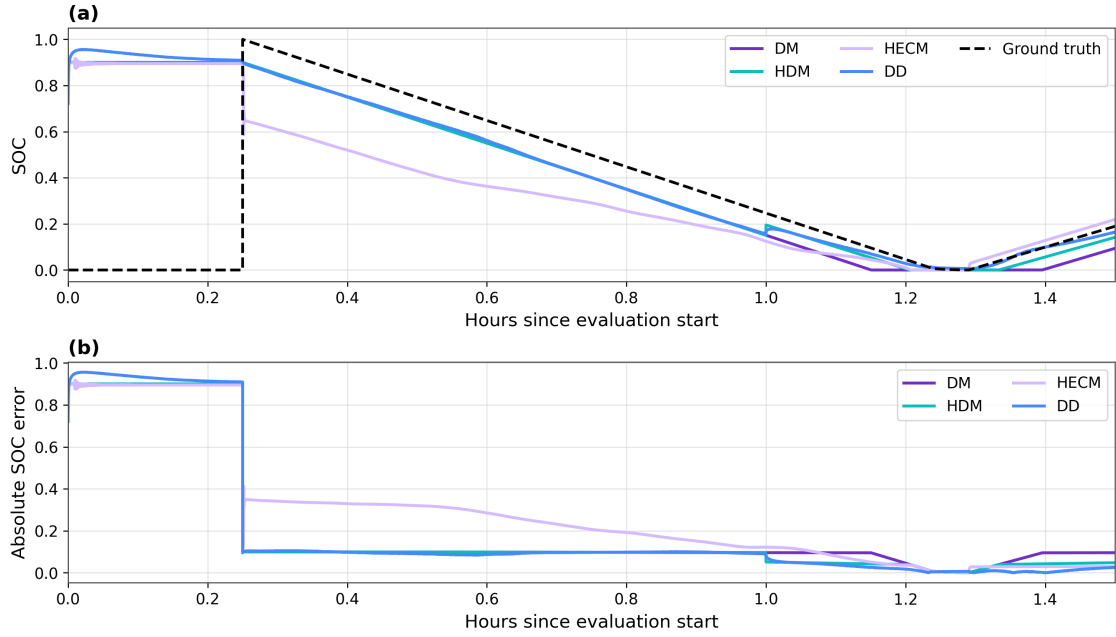


Abbildung 6.7: Recovery after initialization mismatch in a reset-free evaluation window. The figure shows the disturbed SOC trajectories after the imposed initial-state error together with the corresponding absolute SOC error over time, so that re-synchronization to the baseline-relative recovery bands becomes directly visible. This local view is required because full-run averages alone do not capture restart behavior adequately.

6.7.5 Missing samples, irregular sampling, and burst dropout

This block covers disturbances that affect the continuity and timing of the measurement time series rather than the amplitude of one sensor channel alone. In this context, signal integrity means that the sampled input stream remains complete, time-consistent, and updated; it does not refer only to electrical signal quality. The three cases therefore belong together: individual samples can be frozen, the sampling interval can jitter, or the input stream can be held constant over a longer dropout window.

Missing samples are the strongest global case in this block when the effect is measured by ΔMAE relative to each estimator's own baseline. Under the every-50th-sample freeze, DD changes least with $\Delta\text{MAE} = 0.0016$, whereas DM, HDM, and HECM show larger increases of 0.0076, 0.0079, and 0.0029, respectively. Irregular sampling with $\pm 0.1\text{s}$ timing jitter remains nearly neutral in the present implementation, with ΔMAE values close to zero for all estimator classes. Burst

dropout is different: the ΔMAE values remain moderate, but one central 3600 s frozen-data gap causes long post-gap re-synchronization times for all estimators. Figures 6.8–6.10 therefore combine the global ΔMAE summary with transition and recovery views. The underlying metrics are listed in Appendix Tables A.2 and A.3, and the scenario-to-evidence mapping is summarized in Appendix Table A.4.

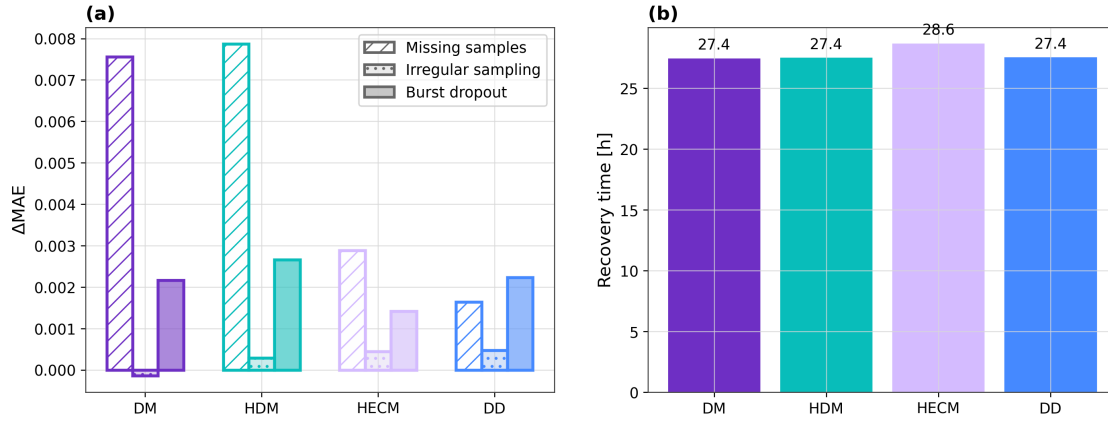


Abbildung 6.8: Continuity and timing disturbance summary. The figure compares the global ΔMAE penalties for missing samples, irregular sampling, and burst dropout, and it complements these full-run penalties with the burst-dropout recovery times. This compact summary shows why missing samples, timing irregularity, and longer frozen-data gaps cannot be judged from one metric alone.

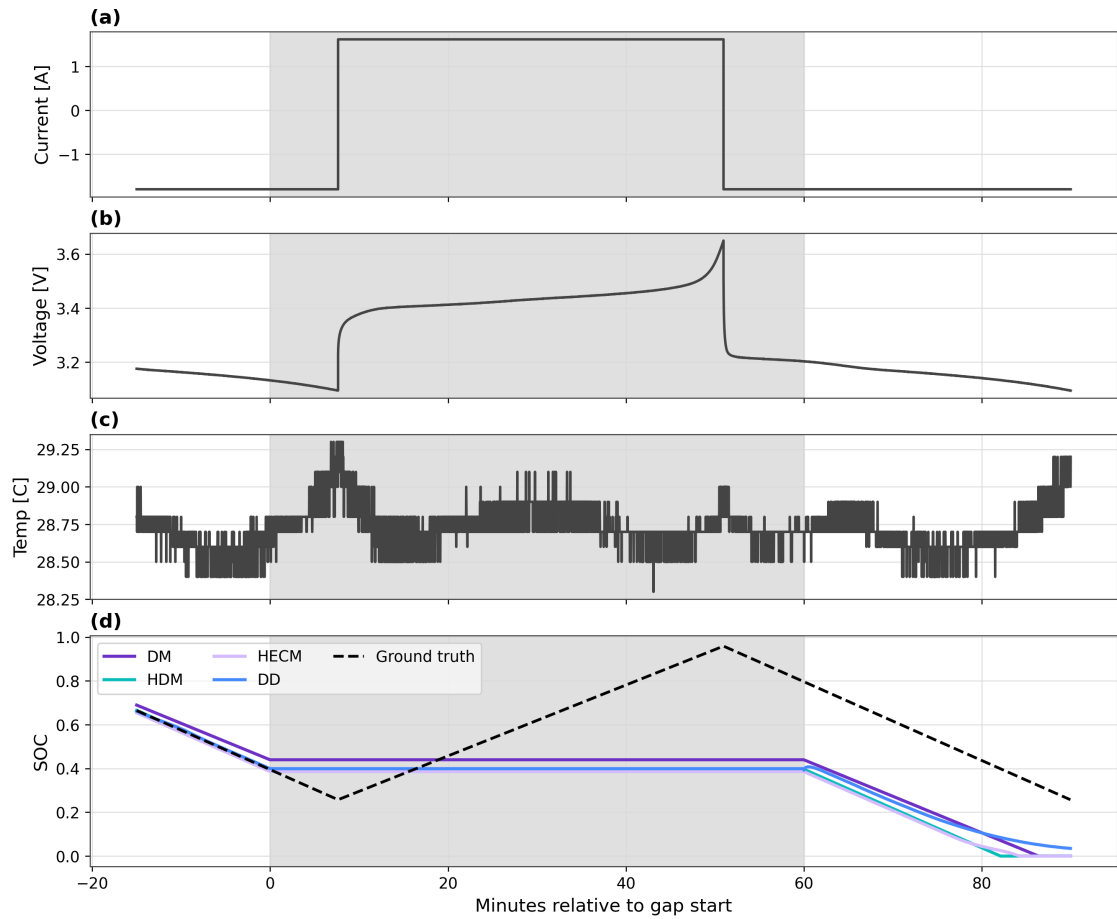


Abbildung 6.9: Local transition into the burst-dropout interval. The shaded region marks the frozen-data window and makes visible how the estimator outputs behave while the input stream is held constant. This view isolates the immediate local response that is hidden by full-run aggregate metrics.

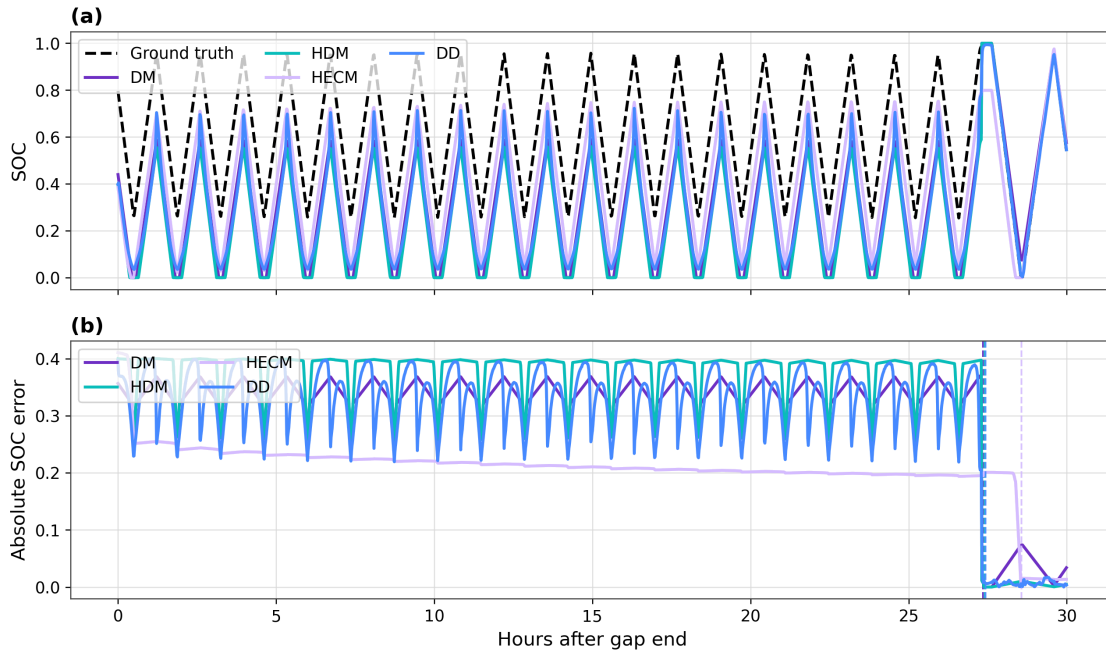


Abbildung 6.10: Post-gap recovery after burst dropout. The figure shows the disturbed SOC trajectories after the frozen interval together with the corresponding absolute SOC error, so that the delayed return toward the baseline bands becomes directly visible. It therefore complements the moderate full-run ΔMAE values with the practically more revealing local recovery behavior.

6.7.6 Spike sensitivity

Single voltage spikes lead to strongly different local reactions even when the full-run MAE remains nearly unchanged. The spike protocol injects one $\pm 0,20$ V single-sample spike every 1000 samples, corresponding to every 1000 s on the nominal 1 s time base. HECM is almost unaffected under this protocol, whereas DD shows the clearest spike sensitivity in both the representative local window and the aligned post-spike excess-error response. The post-spike excess error is evaluated event-wise rather than as a global full-trajectory percentile: for each of the first 40 valid spike events, the SOC-error trajectory is aligned in a window from -60 s to 180 s around the injected spike, the mean absolute error over the final 10 s before the spike defines the local pre-spike baseline, and the largest positive increase above this baseline after the spike is stored as the event severity. The compact severity value in Figure 6.11 is the 95th percentile of these event-wise peak increases. The figure therefore emphasizes local spike behavior instead of relying only on signed

full-run averages; the corresponding global and local metrics are listed in Appendix Tables A.2 and A.3.

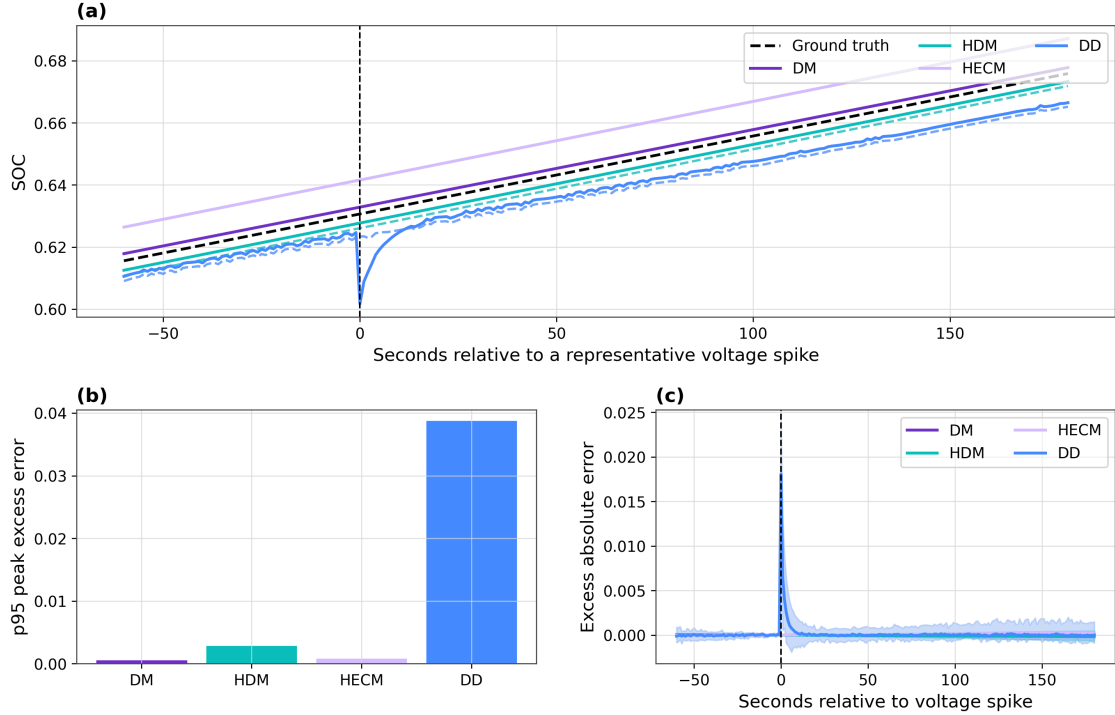


Abbildung 6.11: Voltage-spike sensitivity. The figure combines a representative local SOC response around an injected spike, a compact spike-severity summary based on the 95th percentile of event-wise post-spike peak excess errors, and the aligned post-spike excess-error behavior across repeated spike events. This local perspective is necessary because isolated spike effects can remain understated in signed full-run averages even when the transient response differs strongly across estimators.

6.7.7 Cross-scenario comparison

The cross-scenario comparison confirms that no single estimator dominates all criteria. HDM is the clean-condition leader, HECM is strongest in the scenarios that most clearly change the estimator ranking, especially persistent current bias and voltage noise, and DD is most attractive when missing-sample stability or recovery after initialization mismatch is prioritized. ADC quantization remains comparatively weak in this benchmark and is kept in the scenario space for completeness rather than because it changes the main ranking. Figure 6.12 condenses the disturbed-scenario ΔMAE values into one heatmap, while Appendix Tables A.2,

A.4, and A.5 provide the numerical details. Appendix Figure A.1 adds the dedicated ADC-quantization illustration from the benchmark protocol.

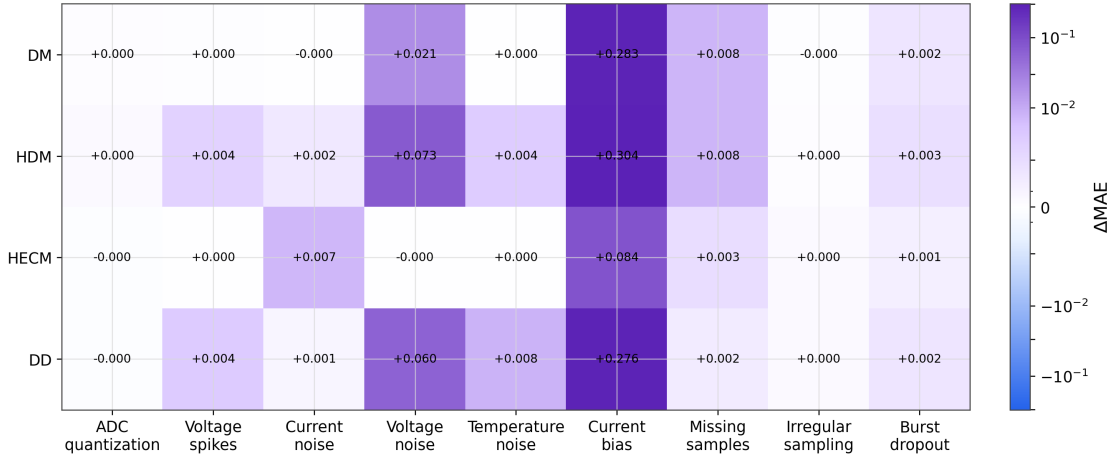


Abbildung 6.12: Cross-scenario error increase relative to the baseline case. The heatmap condenses the disturbed-scenario penalties, including the ADC-quantization case, and makes estimator-specific ranking shifts visible across disturbance classes. It therefore serves as the compact scenario-overview view for identifying where robustness advantages are stable and where they are disturbance-specific. Error values in the figure are SOC fractions on a 0–1 scale; multiply by 100 to obtain percentage points.

6.8 Discussion

6.8.1 Model-specific performance behavior

Before the individual SOC cores are interpreted, it is important to note that the online SOH support used by HDM, HECM, and DD is a shared LSTM-based neural network. The following comparison therefore separates the SOC-estimator families under a common learned health-context signal rather than comparing independently designed SOH models.

The direct-measurement model behaves as the expected drift-prone reference. Persistent current disturbances accumulate directly into the SOC estimate, and missing information cannot be corrected through an additional feedback path. HDM improves nominal calibration by adapting the effective capacity through online SOH support, but it remains structurally exposed to the same current-integration failure mode and additionally inherits the dynamics of the learned SOH assistance.

HECM benefits from interpretable parameterization, explicit voltage-residual correction, and causal observer structure. Combined with the same shared online SOH support used by the other SOH-aware variants, this explains why it is the overall robustness leader under persistent current bias and voltage noise. The same result does not mean that all noise channels are irrelevant for HECM: current noise can still affect the observer locally through current-dependent parameter scheduling and update terms. DD behaves as a compact context-rich estimator with channel-dependent strengths and weaknesses. It is especially strong under missing samples and initialization mismatch, but more exposed to voltage spikes and to propagated noise in derived inputs such as dI/dt , Q_c , EFC, and the SOH context. In the present benchmark, this should be read as a design lesson of the compact deployed instance rather than as a hard class limit, because the data-driven family still retains a larger tuning space through architecture, retraining, feature selection, and state-handling choices than the more fixed-form direct or ECM-based alternatives.

6.8.2 Accuracy–performance trade-offs

A central insight of the chapter is therefore that low clean-condition MAE does not imply strong disturbed-operation performance. The trade-off is not only between accuracy and computational complexity, but also between calibration strength, correction capability, recovery speed, disturbance propagation, and the degree to which a model relies on one dominant trigger variable. The benchmark also shows that estimator comparison should not collapse everything into one leaderboard: a deployment-oriented comparison must separate nominal accuracy, disturbance robustness, recovery after inconsistency, and implementation feasibility.

6.8.3 Implications for BMS deployment

If lowest nominal SOC error under clean sensing is the main design priority, HDM is the recommended class because it achieves the lowest baseline MAE of 0.0024. The corresponding trade-off is strong sensitivity to current bias and the lack of recovery to the baseline bands in the inspected initialization-mismatch window. If the primary priority is best overall robustness under uncertain measurement quality, HECM is the strongest choice because it is most resilient under current bias, remains nearly unchanged under voltage noise, and still preserves competitive baseline accuracy; the trade-off is the larger flash footprint and slower strict-band recovery than DD.

If fastest resynchronization after restart or wrong initial state dominates the design, DD is the strongest option because it shows the fastest strict-band recovery in the local initialization analysis, with the trade-off of greater sensitivity to voltage spikes and weaker performance than HECM under current bias.

If resilience to missing samples and signal freezing dominates the design, DD again becomes attractive because it shows the smallest MAE increase under missing samples and strong post-disturbance stability, while clean-condition accuracy remains below HDM and voltage-driven disturbances require a careful feature-pipeline design. If iterative development headroom and future model refinement dominate the design strategy, DD is also attractive because architecture, features, and training can be updated directly while keeping the deployment target fixed; the present compact instance is still weaker than HECM in the strongest bias- and voltage-driven stress cases, but it retains the clearest path for systematic model iteration. If the main objective is maximum implementation simplicity and the largest resource margin, DM remains the simplest class by far, but it also shows the weakest robustness under biased current measurements and missing-sample stress.

The practical message of the chapter is therefore not that one estimator wins universally. Estimator selection must instead be tied to the disturbance profile, recovery requirements, implementation constraints, and expected model-maintenance strategy of the target BMS. The footprint analysis in Section 6.5.4 further narrows this decision space: all four deployment-prepared variants remain within the common embedded budget, persistent state memory is negligible for all classes, HECM is the tightest flash case, and DM keeps the largest implementation margin.

Tabelle 6.4: Decision-oriented recommendation map derived from the robustness benchmark. The table condenses the scenario-wise findings into dominant deployment priorities, the estimator class most strongly supported by the present evidence, and the corresponding trade-off that must still be considered in design decisions.

Primary deployment priority	Recommended class	Main supporting evidence	Main trade-off to consider
Lowest nominal SOC error under clean sensing	HDM	Lowest baseline MAE on the benchmark trajectory (0.0024)	Strong sensitivity to current bias and no return to the baseline bands in the inspected initialization-mismatch window
Best overall robustness under uncertain measurement quality	HECM	Strongest current-bias robustness, nearly unchanged under voltage noise, and competitive baseline accuracy	Larger flash footprint and slower strict-band recovery after initialization mismatch than DD
Fastest re-synchronization after restart or wrong initial state	DD	Fastest strict-band recovery in the local initialization analysis	More sensitive than HECM to voltage spikes and less robust than HECM under current bias
Highest resilience to missing samples and signal freezing	DD	Smallest MAE increase under missing samples and strong post-disturbance stability	Clean-condition accuracy remains below HDM, and voltage-driven disturbances require careful feature-pipeline design
Largest development headroom for iterative model improvement	DD	Architecture, features, and training can be updated directly while retaining embedded-feasible deployment and strong performance in recovery and missing-sample scenarios	Present compact instance is still weaker than HECM in the strongest bias- and voltage-driven stress cases
Simplest implementation and largest resource margin	DM	Minimal architecture and smallest flash footprint by a wide margin	Weakest robustness under biased current measurements and missing-sample stress

Table 6.4 condenses the benchmark into a deployment-oriented recommendation map so that the reader is not left with a large set of disconnected scenario results. Figure 6.13 complements this table with a compact visual synthesis of the same evidence. The plotted values are relative scores rather than additional physical measurements. For each lower-is-better metric, the four estimator classes are min-max normalized as $(x_{\max} - x)/(x_{\max} - x_{\min})$, so that the best class within the present comparison receives 1 and the weakest class receives 0. The accuracy dimension averages the normalized baseline MAE, RMSE, and P95 error, the robustness

dimension averages the normalized ΔMAE values across the disturbed scenarios, and the recovery dimension uses the strict recovery time after the initial-SOC perturbation, with non-recovery in the inspected window treated as the weakest score.

The three profiles in Figure 6.13 then apply illustrative priority weights of 0.60/0.20/0.20, 0.20/0.60/0.20, and 0.20/0.20/0.60 to accuracy, robustness, and recovery, respectively. These profiles should therefore be read as accuracy-prioritized, robustness-prioritized, and recovery-prioritized compromises rather than as replacements for the raw scenario-wise metrics or for the recommendation map in Table 6.4.

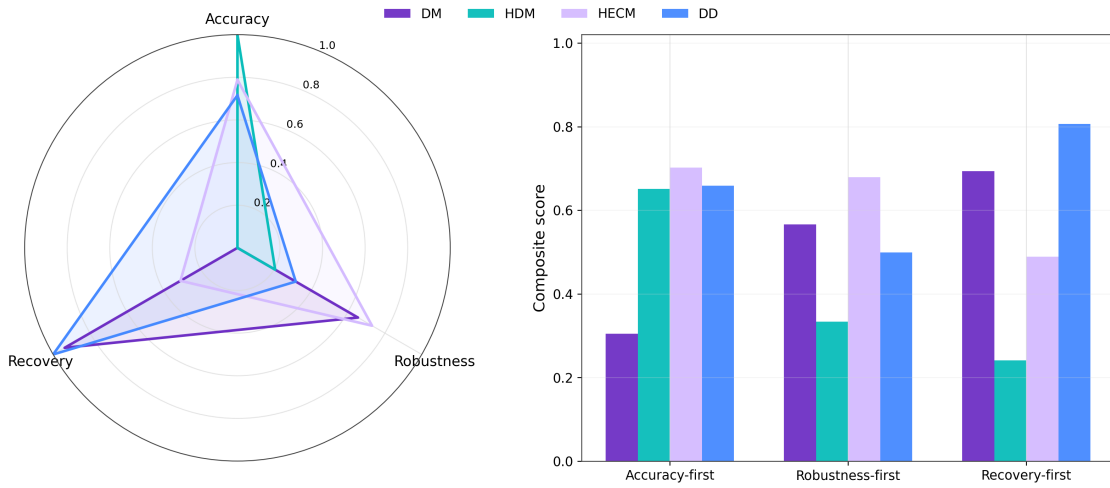


Abbildung 6.13: Decision-oriented synthesis of the robustness benchmark. Panel (a) collapses the benchmark into three relative decision dimensions, namely nominal accuracy, disturbed-scenario robustness, and recovery behavior, using min–max-normalized lower-is-better metrics across the four estimator classes. Panel (b) applies three illustrative deployment-priority profiles and is intended as a visual complement to Table 6.4, not as a replacement for the raw scenario-wise evidence.

6.8.4 Limitations and future extensions

The current benchmark phase is based on one representative dataset trajectory and should later be extended to additional cells, broader ageing states, and cross-cell validation before stronger generalization claims are made. The spike study currently uses isolated spike events; clustered disturbances, longer spike sequences, and hardware-level disturbance injection may reveal additional model differences. The initialization test for DD remains an online-state proxy based on Q_c perturbation rather than an explicit latent-state reset. This is methodologically justified, but it

should still be interpreted as a class-specific approximation. The next benchmark stages should add the remaining extended scenarios, more cells, pack-level experiments, and later hardware-in-the-loop or real embedded runtime measurements while keeping the same performance-oriented evaluation logic.

6.9 Conclusion

Taken together, Chapter 6 shows that SOC-estimator quality for battery-management deployment cannot be reduced to clean-condition accuracy alone. The chapter compares representative estimator families – direct measurement, SOH-supported direct measurement, equivalent-circuit-based hybrid estimation, and data-driven neural estimation – under the same disturbed measurement protocol. The purpose is therefore not to declare one model universally best, but to identify which strengths and weaknesses become visible when different modelling paradigms are exposed to bias, noise, missing information, timing disturbance, spikes, and initialization mismatch.

Under the emphasized disturbance profile, HECM is the strongest overall robustness compromise because its voltage-feedback structure limits several measurement-driven error mechanisms. HDM remains the clean-condition leader, DD shows the strongest recovery and missing-sample behavior, and DM provides the simplest reference while exposing the expected drift vulnerability. This split is the main result: estimator selection must follow the expected disturbance profile, recovery requirement, and implementation context of the target BMS rather than a single ranking.

The broader implication is that practical BMS evaluation needs more than MAE or another average accuracy score. Average error remains necessary, but it must be complemented by local disturbance response, recovery behavior, tail errors, signal-continuity sensitivity, and the way disturbed measurements propagate through derived online features. The next step is therefore not another abstract class comparison, but the concrete transfer of learned state estimators into embedded execution. Chapter 7 builds on this robustness perspective and turns to recurrent neural SOC and SOH estimation on microcontrollers, where causal state handling, compression, quantization, and runtime feasibility become part of the estimator design itself.

Embedded Artificial Intelligence for SOC and SOH Estimation on Microcontrollers

7.1 Embedded Deployment Objective

- Accurate online estimation of state of charge and state of health is central to battery-management operation, yet many recent neural estimators remain evaluated mainly as offline research models rather than as continuously running embedded components.[61, 62, 6, 35]
- Across the broader state-estimation literature, measurement-derived, model-based, and data-driven approaches continue to coexist.[62, 6, 35, 36] Carefully tuned equivalent-circuit and electrochemical observers can reach very low laboratory errors,[43, 42, 44, 45] while recent sequence models report strong SOC accuracy on specialized datasets,[63, 64] and TinyML-style microcontroller deployments typically operate in the low-single-digit percent range.[65, 48, 47]
- This chapter therefore examines how recurrent SOC and SOH estimators can be transferred from floating-point training releases to practical microcontroller execution under explicit flash, SRAM, latency, and energy limits. The focus is not only on predictive accuracy, but on the combined interaction between estimation utility, compression, firmware integration, and runtime cost on the target controller.
- The application context is modular stationary storage in remote microgrids within the MG-FARM programme, where storage modules must operate autonomously and the battery management system must compute SOC and SOH on

an STM32H753ZI-class microcontroller without depending on cloud-side infrastructure.[66, 67, 68, 69]

- Against this background, the chapter addresses an embedded-AI gap in the literature. A small number of TinyML-style battery studies already report microcontroller deployments,[65, 47, 50] but detailed treatment of scaling statistics, hidden-state handling, quantization constants, and long-horizon streaming behaviour on the actual controller remains uncommon.
- Recurrent neural estimators are particularly relevant in this context because they can absorb long temporal dependencies and evolving degradation behaviour without requiring an explicit electrochemical model at runtime.[52, 45] Within the dissertation, this chapter therefore extends the earlier SOH-estimation study from Chapter 5 toward embedded SOC and SOH deployment on one concrete MCU platform.
- The central contribution is therefore the complete deployment path from trained recurrent models to executable MCU firmware: stateful LSTM + MLP estimators, structured pruning, INT8 weight quantization, manual export into transparent C kernels, and on-device benchmark measurements on the STM32 target. In this sense, battery AI is treated explicitly as an embedded-systems problem: an estimator is only practically useful if it remains accurate under long-horizon streaming replay and still fits into the resource and timing budget of the controller.

7.2 Dataset Basis and Experimental Scope

- All experiments rely on the LFP DoE Cycle Ageing Dataset described in Section 4.1.[2, 70] The three-factor DoE design, the representative trajectory structure, and the full SOH spread across cells are documented there (Figures 4.2, 4.3, and 4.4).[71]
- Between ageing loops, dedicated check-ups with capacity tests, open-circuit-voltage sweeps, and hybrid pulse-power characterisation are inserted. In addition, stationary-storage load profiles derived from residential PV home storage are injected so that the data contain both tightly controlled diagnostics and application-oriented operating conditions.[29]

- The raw measurements are available at 1 Hz and contain voltage, current, terminal temperature, and auxiliary channels. This combination is important because it exposes the estimators simultaneously to fast SOC transients, long-duration integration horizons, and the much slower SOH evolution that unfolds across the full ageing campaign.

7.3 Feature Engineering and Reference Labels

- The SOC estimator consumes the six-dimensional feature vector

$$\mathbf{x}_t^{\text{SOC}} = \left[U(t), I(t), T(t), Q_c(t), \frac{dU}{dt}(t), \frac{dI}{dt}(t) \right],$$

where the temporal derivatives are computed by centred finite differences. The SOH estimator uses a slower-varying feature stack,

$$\mathbf{x}_t^{\text{SOH}} = \left[t, U(t), I(t), T(t), \text{EFC}(t), Q_c(t) \right],$$

with cumulative time stamp t , equivalent full cycles $\text{EFC}(t)$, and cumulative charge state $Q_c(t)$ capturing the longer degradation context.

- All channels are robustly normalised so that the exact same scaling statistics can be embedded into the firmware. Instead of mean and variance, the chapter follows the paper in using median m_k and interquartile range r_k ,

$$\tilde{x}_{t,k} = \frac{x_{t,k} - m_k}{r_k + \varepsilon}, \quad \varepsilon = 10^{-6},$$

which reduces sensitivity to outliers and keeps host-side replay and MCU execution numerically aligned.[72, 73]

- The SOC and SOH reference labels are constructed from the Coulomb-counting and capacity-check-up procedures defined in Section 4.1 (Equations 4.2 and 4.3). Both label traces rely on information unavailable during deployment – sustained CV-based full-charge resets and periodic full-capacity discharges – and are used exclusively for training and evaluation. On the STM32, the estimators operate only on streamed measurements and their internal recurrent states.
- The embedded deployment then follows a strictly stateful streaming regime: hidden and cell states are forwarded sample by sample along the full sequence

without sliding input windows. Host-side replay and STM32 replay therefore use identical inference conditions. Figure 7.1 summarizes this shared stateful LSTM + MLP structure that is later specialized to the SOC and SOH tasks.

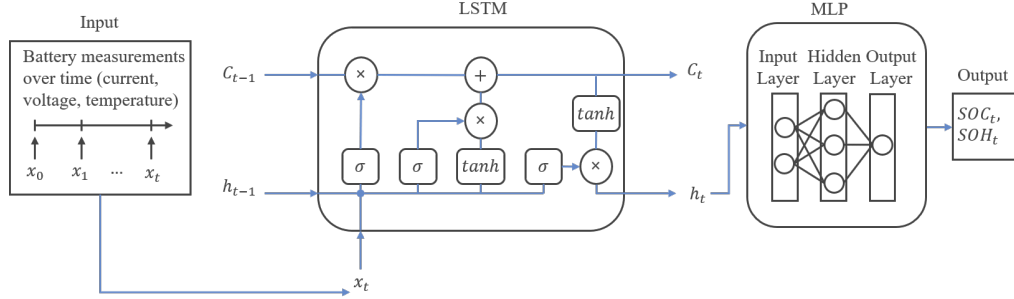


Abbildung 7.1: Stateful LSTM + MLP estimator structure used for both SOC and SOH estimation. Preprocessed feature streams are processed step by step by a single-layer LSTM whose hidden and cell states are propagated across time; a lightweight MLP head then maps the hidden representation to either SOC or SOH. The SOC instance uses hidden size 64 with an MLP width of 64 and a sigmoid output, the SOH instance hidden size 128 with an MLP width of 128 and a linear output. This structure is the basis for the later pruning, quantization, and MCU export steps.

7.4 LSTM Architecture and Training Setup

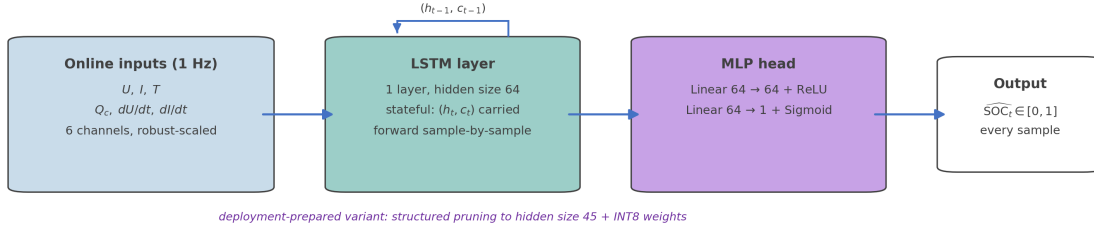
- Both SOC and SOH estimators employ the same recurrent equation set, but with task-specific feature sets and sizes, as summarised in Figure 7.2. The SOC model uses a single-layer LSTM with hidden size $H = 64$ that consumes a six-dimensional feature vector $\{U, I, T, Q_c, dU/dt, dI/dt\}$, followed by an MLP head with one ReLU-activated hidden layer of width 64 and a sigmoid output neuron that constrains the estimate to $[0, 1]$. The SOH model uses the same structure with hidden size $H = 128$ and an MLP hidden width of 128, but consumes a different six-dimensional feature vector $\{t, U, I, T, EFC, Q_c\}$ and ends in a purely linear output neuron because the slowly varying SOH target does not benefit from a saturating output activation. In contrast to the hourly-aggregated SOH estimator of Chapter 6, both models here operate directly on the sample-wise 1 Hz stream. The implementation follows the standard LSTM gate formulation in PyTorch, with propagated hidden and cell states as described in the recurrent-neural-network literature.[74, 75]

- This restriction to a single-layer LSTM + MLP architecture is deliberate. The focus is not an extensive architecture search over GRUs, Transformers, or deeper recurrent stacks, but a transparent and reproducible embedded tool flow for stateful estimators whose hidden-state handling can later be re-implemented faithfully in C. A single-layer LSTM reduces to one compact matrix-vector product per gate step, which can be written as a numerically stable stateful C kernel; sharing the same backbone for both SOC and SOH further minimises the firmware footprint. The deeper SOH estimator of Chapter 6 – a three-layer LSTM with hidden size 192, a feature projection block, and two residual post-processing blocks – achieves a better validation MAE (1.48% vs. 2.62% for this chapter’s model), but structured pruning at any practically useful compression rate collapses that model’s accuracy entirely, making the full training-to-MCU pipeline impossible. The collapse can be attributed to the interaction of depth and residual structure: in a three-layer LSTM, removing even 20 % of the hidden units degrades the representation passed from layer 1 to layer 2, and this degraded signal is amplified again at layer 3, so errors compound multiplicatively across the stack rather than remaining local. The two residual MLP blocks that follow the LSTM exacerbate this further, because they learn fine-grained additive corrections to the full 192-dimensional LSTM output; once the recurrent core has been compressed and its output quality has dropped, the residual corrections can no longer compensate and the prediction head receives a fundamentally degraded input. In a single-layer model with no residual post-processing, the same pruning step removes redundant units without triggering this cross-layer cascade, and the remaining units can be recovered by short finetuning. The architecture choice in this chapter is therefore not a simplification for its own sake but a direct consequence of this pruning-sensitivity finding: a model deep and complex enough to match the accuracy of the Chapter 6 estimator would, by the same token, be too rigid to support the structured compression workflow that the MCU deployment pipeline requires.
- Both SOC and SOH models are trained with mean-squared-error loss against the corresponding reference label at each training sample, while validation additionally tracks MAE, RMSE, and R^2 . The resulting checkpoints, configuration files, and optimizer states are archived so that all later pruning, quantization, and firmware-export steps can be replayed deterministically.
- Training is carried out on NVIDIA A100-class GPU hardware in mixed precision

(FP16). Both tasks use the AdamW optimiser, cosine-annealing warm restarts, gradient clipping, and mini-batch training.[76, 77, 78, 79]

- For SOC, the learning rate is 1.5×10^{-4} , the batch size is 512, and the sequence chunk length is 2024. For SOH, the learning rate is 2.0×10^{-4} , the batch size is 128, and the chunk length is 2048.
- Although deployment is fully stateful and step-wise, training uses truncated backpropagation through time over these chunks. Within each chunk, the hidden and cell states are propagated exactly as in deployment, so the stateful update mechanism is already learned during optimisation while the chunking still enables randomized sampling, faster convergence, and better generalization across the diverse ageing trajectories.
- The base models are first evaluated in streaming mode on the full ageing trajectory of a representative LFP cell and serve as the numerical reference for all subsequent experiments. These same checkpoints are then pruned, quantized, and exported to STM32 firmware, where identical feature streams are replayed so that accuracy, flash, RAM, latency, and energy always refer to the same stateful sequence. Figure 7.3 summarizes this end-to-end path from training to embedded benchmarking.

(a) Embedded SOC estimator: stateful LSTM + MLP



(b) Embedded SOH estimator: stateful LSTM + MLP

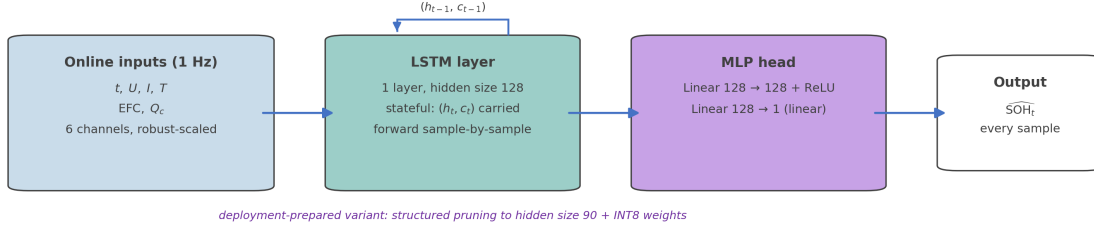


Abbildung 7.2: Block-level architectures of the two stand-alone embedded estimators. (a) The SOC estimator processes a six-channel feature vector $\{U, I, T, Q_c, dU/dt, dI/dt\}$ with a single stateful LSTM layer of hidden size 64, followed by an MLP head with one ReLU-activated hidden layer of width 64 and a sigmoid output. (b) The SOH estimator uses the same structure with hidden size 128 and an MLP width of 128, consumes the alternative feature vector $\{t, U, I, T, \text{EFC}, Q_c\}$, and ends in a linear output neuron. Both models forward their hidden and cell states sample by sample along the 1 Hz stream. The purple annotations give the reduced hidden sizes of the deployment-prepared variants after structured pruning and INT8 weight quantization.

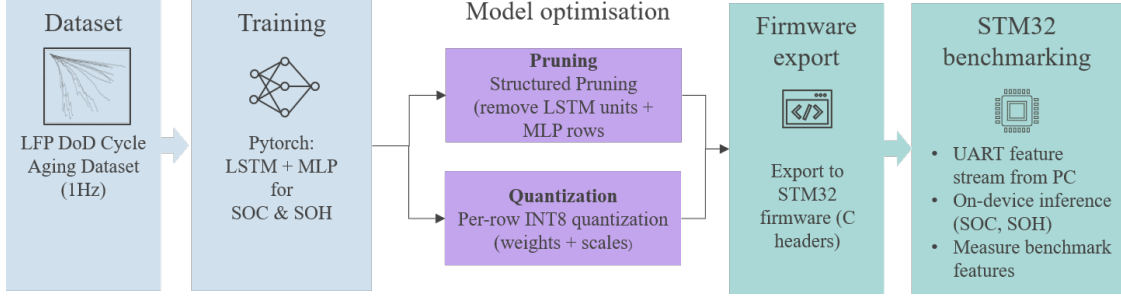


Abbildung 7.3: End-to-end deployment pipeline from the LFP ageing dataset and PyTorch training to pruning, quantization, C-header export, and final STM32 benchmarking. The figure makes explicit that the same trained SOC and SOH checkpoints are reused across the floating-point, pruned, and quantized deployments so that later trade-off comparisons remain numerically consistent.

7.5 Structured Pruning for MCU Deployment

- Pruning is applied as structured magnitude pruning at the level of complete LSTM hidden units, because whole-channel removal is far better suited to efficient C implementations on a microcontroller than unstructured sparse masks.[80, 81]
- The pruning score aggregates the L2 norms of the incoming weights for each hidden unit across all four LSTM gates,

$$s_h = \sum_{g \in \{i, f, o, g\}} \left(\|W_{ih}^{(g)}[h, :]\|_2 + \|W_{hh}^{(g)}[h, :]\|_2 \right),$$

so that hidden units with the smallest aggregated saliency are removed first. This follows the same structured-sparsity intuition used in recurrent-network pruning literature.[82, 83, 84]

- In the final deployment, the SOC estimator is reduced from 64 to 45 hidden units and the SOH estimator from 128 to 90 hidden units. In both cases this corresponds to roughly 30% channel-wise sparsity.
- Rows and columns are removed jointly from the recurrent gate matrices and the downstream MLP weights so that the pruned network remains a valid stateful recurrent mapping.

- After each pruning step the model is fine-tuned again. The chapter therefore treats pruning not as a single static compression event, but as a repeated prune-and-recover cycle that progressively reduces recurrent state size while preserving the learned temporal mapping.
- The engineering purpose of pruning is direct: reduce recurrent state size, shrink flash and RAM demand, and shorten on-device inference time without materially compromising streaming accuracy. Figure 7.4 illustrates this channel-wise removal process at the level of hidden units and associated MLP rows.

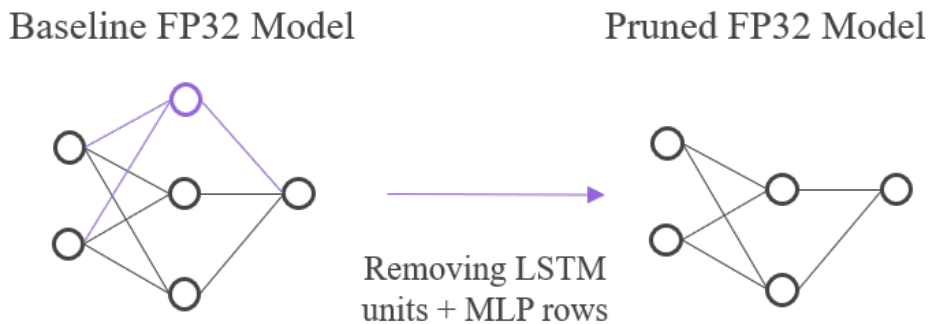


Abbildung 7.4: Structured pruning concept used for the embedded SOC/SOH estimators. The left-hand side highlights a low-saliency hidden channel together with its incident gate connections, while the right-hand side shows the reduced recurrent architecture after removing the corresponding hidden unit and associated downstream weights.

7.6 INT8 Quantization and Manual STM32 Export

- Quantization complements pruning by reducing the numerical precision of the remaining weights while leaving the topology unchanged. In this chapter, the recurrent weights are stored as signed INT8 values with associated floating-point scales so that the firmware can retain a transparent mapping back to the PyTorch reference.[85, 86] This aligns with the broader observation from recent pruning/quantization benchmarks that aggressive compression can preserve useful accuracy when the deployment path is engineered carefully.[87]
- The implementation uses post-training per-row symmetric INT8 quantization for the input-to-gate and hidden-to-gate matrices of the LSTM. For each row,

the scale is chosen as

$$s = \frac{\max_h |w_h|}{127},$$

so that the maximum absolute weight maps close to the signed INT8 limits $\{-128, \dots, 127\}$ while zero remains exactly representable.

- Biases and activations remain in FP32. This design is pragmatic: it preserves numerical stability, keeps the firmware implementation close to the floating-point reference, and still reduces recurrent weight storage by roughly a factor of four relative to FP32 storage.
- During inference, the embedded gate computation reconstructs the effective weights from INT8 codes and FP32 scales. A built-in streaming comparison validates this conversion and typically yields MAE differences below 10^{-4} between exported and reference implementations, which indicates that the recurrent dynamics are preserved numerically.
- The exported model parameters are written as C headers and integrated into manually implemented stateful LSTM + MLP kernels. This manual route is chosen because common export toolchains such as STM32Cube.AI do not natively support the continuous hidden-state handling required for these estimators.[88, 89, 90]
- The result is a fully transparent firmware path: every scaling constant, every recurrent state update, and every arithmetic step can be audited and reproduced outside the proprietary toolchain. Figure 7.5 summarizes the FP32-to-INT8 conversion, scale handling, and STM32 export flow.

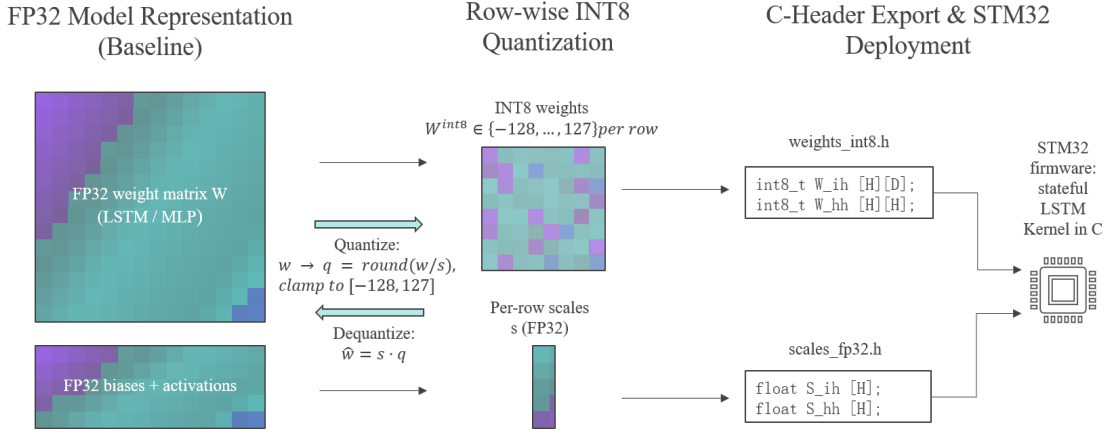


Abbildung 7.5: Per-row INT8 quantization and export concept used for the STM32 deployment. The figure shows how FP32 recurrent weight rows are mapped to the signed INT8 codebook together with floating-point scale factors and then exported as C headers for the stateful STM32 kernel, which reconstructs the effective gate weights during inference.

7.7 Benchmark Methodology and KPI Logic

- The benchmark is executed on an STM32 Nucleo-144 board with an STM32H753ZI microcontroller, providing 2 MB flash and 1 MB SRAM, which together define the available code, weight, and runtime-buffer budget for the deployed estimators.[91, 92]
- Starting from the trained LSTM + MLP checkpoints, three embedded variants are evaluated for each task: the dense FP32 baseline, the structured-pruned FP32 model, and the quantized INT8-weight deployment. For all of them, identical feature streams from the LFP dataset are replayed both on the host and on the STM32 so that accuracy, latency, and resource measurements always refer to the same stateful input sequence.
- The host-side replay operates directly on the reference PyTorch implementation and produces MAE, RMSE, R^2 , error histograms, parity-style comparisons, and long-horizon trajectory views. These simulations serve as the high-precision numerical reference against which the embedded deployments are later judged.
- The embedded benchmark injects the same feature vectors via UART into dedicated SOC and SOH firmware images, measures end-to-end host latency, mea-

sures pure on-device inference time through the DWT cycle counter, and derives energy per inference from the average inference time and an empirically calibrated board power of 0,5 W.

- RAM is computed as the sum of statically allocated ‘.data’ and ‘.bss’ sections plus the maximum observed stack usage during streaming. Flash is taken from the linker output of the STM32CubeIDE projects. Tables 7.1 and 7.2 therefore summarize specifically the STM32 resource and timing indicators, whereas the trajectory and distribution metrics remain in the host-side streaming analysis.
- The chapter distinguishes carefully between communication-dominated system latency and pure on-device inference time. This distinction matters because UART framing and host scheduling can dominate the end-to-end delay, although the actual neural computation remains well within the real-time budget of the target platform.

Tabelle 7.1: STM32 engineering KPIs for the SOC deployments on the STM32H753ZI under full-stream replay. The table separates end-to-end host latency from pure on-device inference time and reports the measured flash footprint, total RAM usage, and energy per inference for the dense, pruned, and quantized variants.

Model	Latency [ms]	Inference [ms]	Flash [KB]	RAM [KB]	Energy [mJ]
Base FP32	12.70	1.40	105.32	4.93	0.70
Pruned FP32	12.20	0.80	62.27	4.03	0.40
Quant INT8	18.48	6.99	52.48	3.96	3.49

Tabelle 7.2: STM32 engineering KPIs for the SOH deployments on the STM32H753ZI under full-stream replay. As in the SOC case, the table distinguishes end-to-end latency from core inference time and reports measured flash, RAM, and energy for the three deployment variants.

Model	Latency [ms]	Inference [ms]	Flash [KB]	RAM [KB]	Energy [mJ]
Base FP32	34.88	22.73	335.00	8.69	11.37
Pruned FP32	24.69	12.72	182.41	6.96	6.36
Quant INT8	41.23	29.21	138.00	6.70	14.60

7.8 Streaming Prediction Accuracy

- The results section starts with full-sequence streaming replay over the complete ageing trajectory. Figures 7.6 and 7.7 therefore provide the central long-horizon view of how the dense, pruned, and quantized deployments behave under identical stateful replay conditions.
- For SOC, the tracking remains tight across the entire dynamic range. The residuals stay centered around zero, and the quantized model overlaps the full-precision baseline so closely that the two traces are often visually almost indistinguishable. This indicates that the per-row symmetric quantization preserves the recurrent dynamics sufficiently well for accurate long-horizon SOC tracking.
- The pruned SOC model also remains highly accurate, but shows slightly stronger smoothing in highly dynamic regions. In the interpretation of the paper, this is the expected consequence of the reduced hidden-state dimensionality after structured channel removal.
- For SOH, the task is structurally harder because the estimator has to infer slow degradation from sparse check-up events and continuous monitoring data. The paper therefore removes the discrete capacity tests from the estimator input and retains only the remaining check-up blocks and operating segments, so that the estimator must infer the degradation trend from causally available evidence only.
- A causal smoothing stage is applied to the raw SOH predictions. Specifically, the chapter follows the paper in using an exponential moving average with smoothing factor $\alpha = 10^{-6}$ together with a downward-rate limiter of 2×10^{-8} per step in order to suppress unphysical jumps while maintaining a monotonic long-term degradation trend.
- After this filtering, the pruned and quantized SOH models still track the dense baseline with similar fidelity and preserve the long-term trend despite visible differences in small-scale variability. Across the embedded variants, the resulting full-stream accuracy remains in the same overall range already noted earlier: about 2,3% bis 2,8% MAE for SOC and about 0,9% bis 1,5% MAE for SOH, both reported relative to full scale.

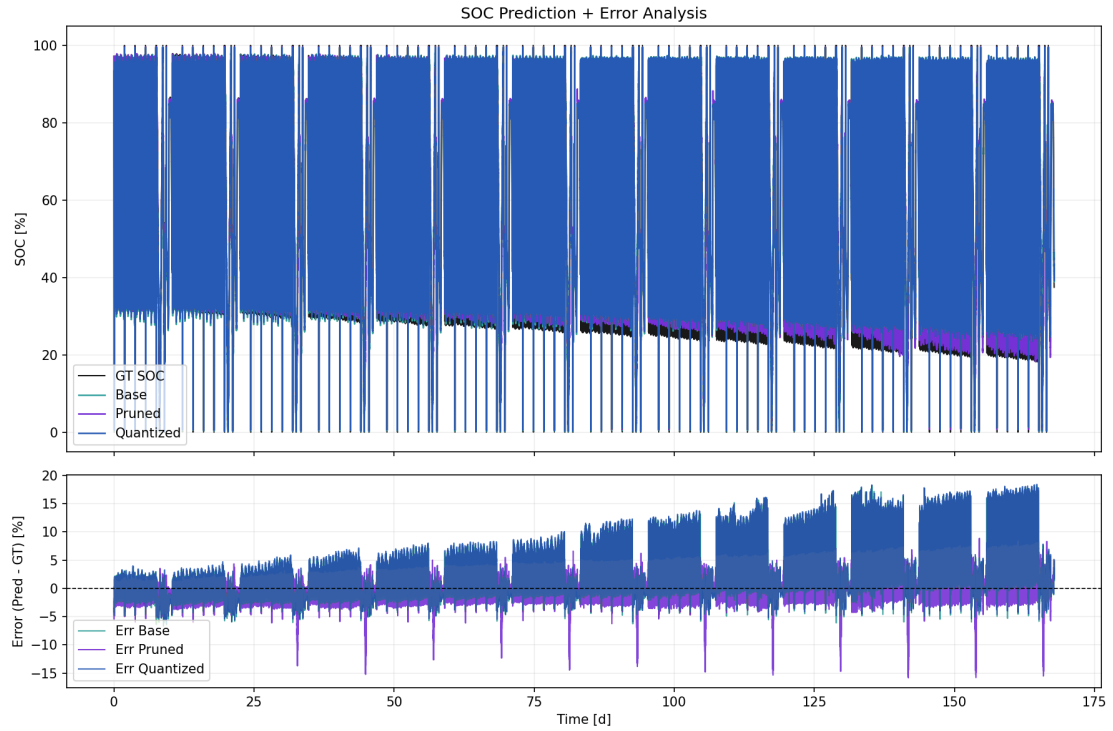


Abbildung 7.6: Streaming SOC dashboard for the base, pruned, and quantized SOC deployments over the full ageing trajectory. The upper panel compares the three stateful predictions with the SOC reference label, while the lower panel shows the instantaneous prediction error. The near-overlap of the dense and quantized traces is one of the central observations of the chapter.

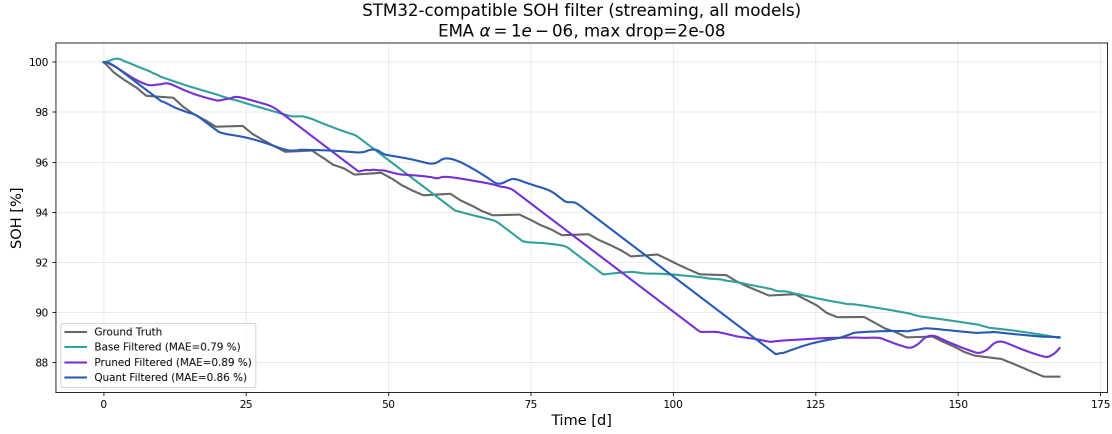


Abbildung 7.7: Streaming SOH dashboard with causal filtering for the three deployment variants. The upper panel shows the reference SOH together with the base, pruned, and quantized predictions after on-line smoothing, while the lower panel displays the corresponding prediction errors. The figure highlights the much slower degradation dynamics and the stronger role of causal post-processing in the SOH task.

7.9 Error Distribution and Local Transient Behaviour

- Figures 7.8 and 7.9 analyse the statistical distribution of the prediction errors. For SOC, the distribution is narrow and symmetric, which indicates the absence of systematic bias across the full streaming replay.
- The SOC boxplots further show that the pruned model achieves the smallest median MAE and the narrowest interquartile range. In other words, pruning does not merely reduce resource demand for the SOC task, but acts in this case like a mild regularizer that slightly improves the dense baseline.
- For SOH, the error distribution is broader, which is consistent with the greater intrinsic difficulty of estimating capacity fade from operational data. Even so, the bulk of the SOH errors remains within a practically usable tolerance band of approximately $\pm 2\%$, which keeps the embedded estimator relevant for onboard diagnostics despite the more difficult regression target.
- Global metrics alone would still hide local failures. The chapter therefore also zooms into a high-dynamic pulse segment between 30k and 40k seconds (Figure 7.10) and into a representative check-up sequence between 657k and 897k

seconds (Figure 7.11).

- In the pulse region, the quantized model follows the reference transients with high fidelity and captures voltage-recovery phases accurately, whereas the pruned model smooths these sharper local features slightly because the reduced hidden-state dimension leaves less capacity for modelling fast electrochemical relaxation.
- In the check-up phase, all models converge well and remain stable over long integration horizons. This is an important result for the embedded formulation because it shows that the strictly stateful streaming execution does not accumulate visible drift even across long full-charge and full-discharge sequences.

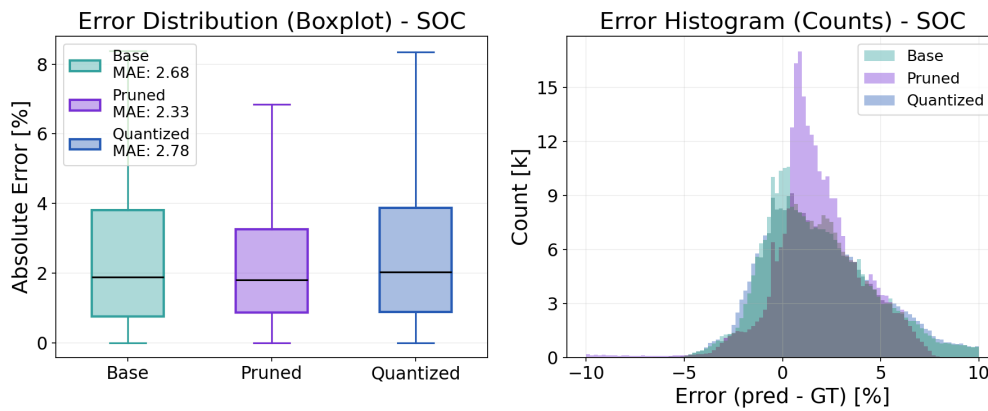


Abbildung 7.8: SOC error distribution for the embedded SOC deployments, including MAE boxplots and an error histogram. The narrow spread around zero confirms unbiased estimation, while the boxplots make visible that the pruned SOC model attains the smallest median error and interquartile range.

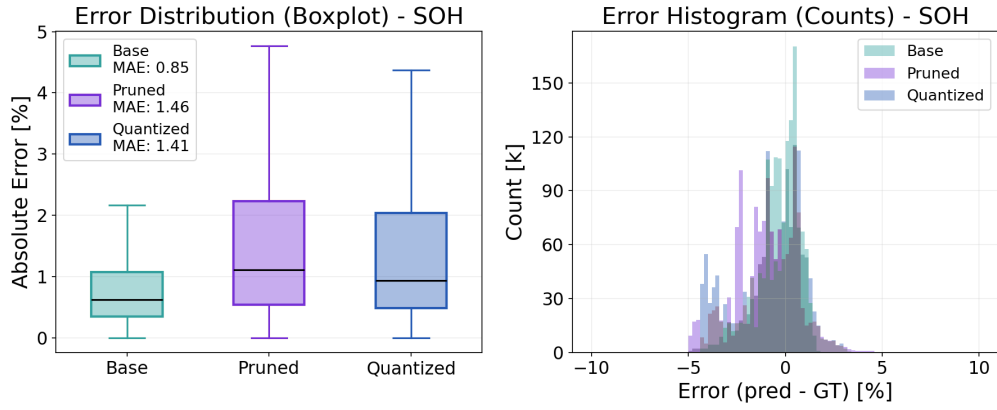


Abbildung 7.9: SOH error distribution for the embedded SOH deployments, including MAE boxplots and an error histogram. The spread is naturally broader than for SOC, but the majority of errors remains within a practically useful tolerance band around the reference trajectory.

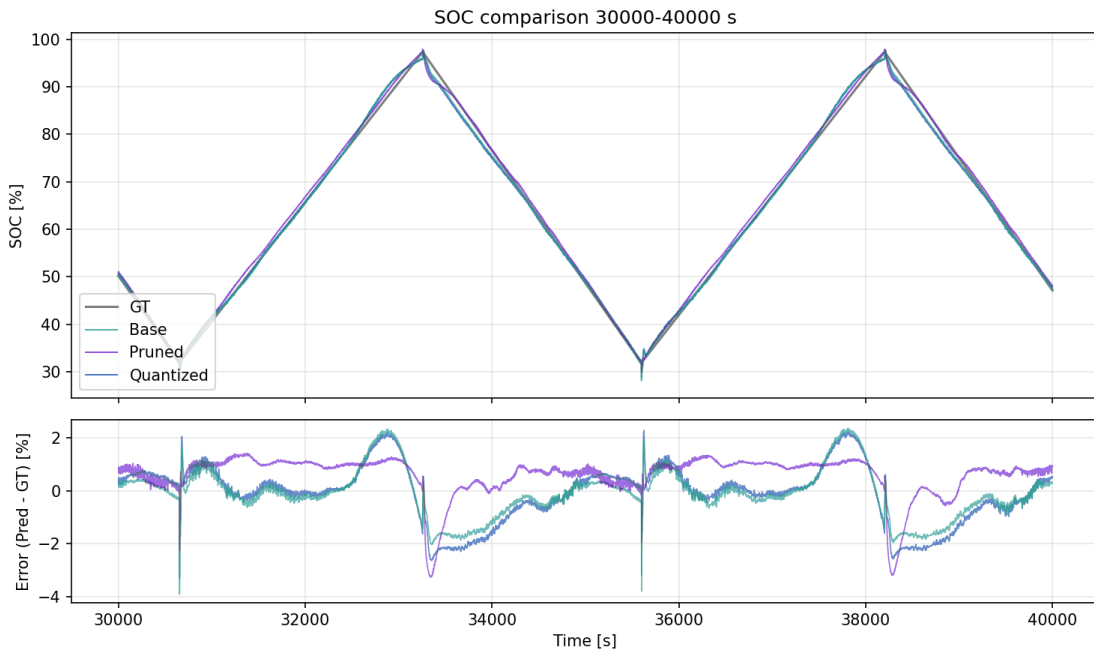


Abbildung 7.10: Detailed pulse-region analysis of embedded SOC streaming replay. The figure focuses on a highly dynamic load segment and shows how the three deployment variants capture fast transients and voltage-recovery behaviour under identical stateful inference conditions.

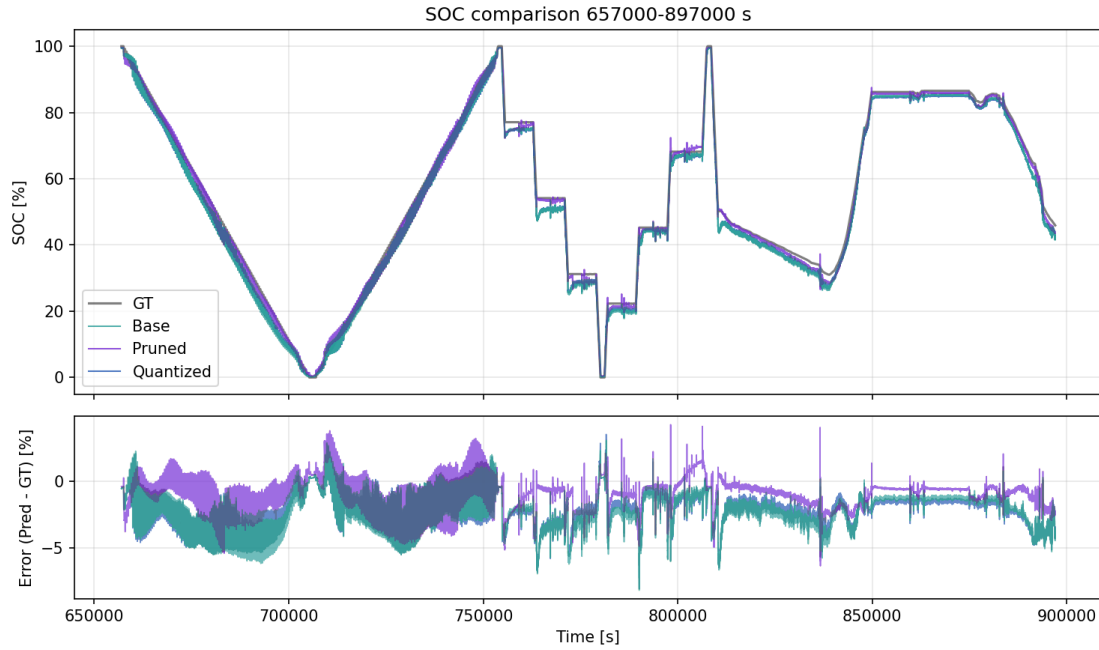


Abbildung 7.11: Detailed check-up-region analysis of embedded SOC streaming replay. The figure covers a representative diagnostic interval with long integration horizons and illustrates the stability of the stateful deployments during extended charge/discharge behaviour.

7.10 Resource Efficiency, Latency, and Trade-Off

Summary

- The KPI tables already show that pruning and quantization affect SOC and SOH differently, but Figures 7.12 and 7.13 condense the same trade-off visually across model size, memory, and timing.
- For each deployment, the model complexity can first be estimated from the parameter count under idealized storage assumptions, namely four bytes per FP32 weight and one byte per INT8 weight. The measured flash footprint is then taken from the STM32 linker output and is systematically larger because it also includes code, constant tables, and runtime support. This gap is particularly visible for the larger SOH firmware.
- The RAM view in Figure 7.12 is based on the measured static data sections plus worst-case stack usage during streaming. It therefore approximates the

true runtime requirement of the deployed kernels more realistically than a pure parameter-memory count.

- For SOC, pruning reduces flash from 105,32 kB to 62,27 kB, reduces inference time from 1,40 ms to 0,80 ms, and reduces energy per inference from 0,70 mJ to 0,40 mJ. This corresponds to gains of roughly 41–43%.
- Quantization reduces the SOC flash footprint further to 52,48 kB, but increases pure inference time to 6,99 ms and energy per inference to 3,49 mJ because the software kernel has to reconstruct FP32 values from INT8 codes and scales.
- For SOH, pruning reduces flash from 335,00 kB to 182,41 kB, reduces inference time from 22,73 ms to 12,72 ms, and reduces energy per inference from 11,37 mJ to 6,36 mJ. Quantization reduces flash further to 138,00 kB, but raises inference time and energy to 29,21 ms and 14,60 mJ.
- The latency distribution in Figure 7.13 reveals the same hierarchy. On-device SOC inference remains in the sub-10 ms regime even for the quantized kernel, whereas SOH updates remain in the approximate range of 12 ms bis 30 ms. Including UART overhead still leaves end-to-end latencies below roughly 20 ms for SOC and below roughly 50 ms for SOH, which is comfortably inside typical stationary-storage BMS timing limits.[5, 6]
- The model-size and latency plots therefore condense a clear pattern: quantization is the strongest lever for non-volatile memory, while pruning is the strongest lever for energy and inference time on the current hand-written STM32 implementation.
- To summarize this multi-objective trade-off more compactly, the chapter uses the composite utility score

$$U_v = \frac{1}{4} \left(\frac{\text{MAE}_v}{\text{MAE}_{\text{Base}}} + \frac{F_v}{F_{\text{Base}}} + \frac{R_v}{R_{\text{Base}}} + \frac{E_v}{E_{\text{Base}}} \right),$$

where MAE denotes the streaming mean absolute error, F the measured flash footprint, R the measured total RAM usage, and E the measured energy per inference. By construction, the dense baseline has $U = 1$, and values below one indicate an average improvement across these four objectives.

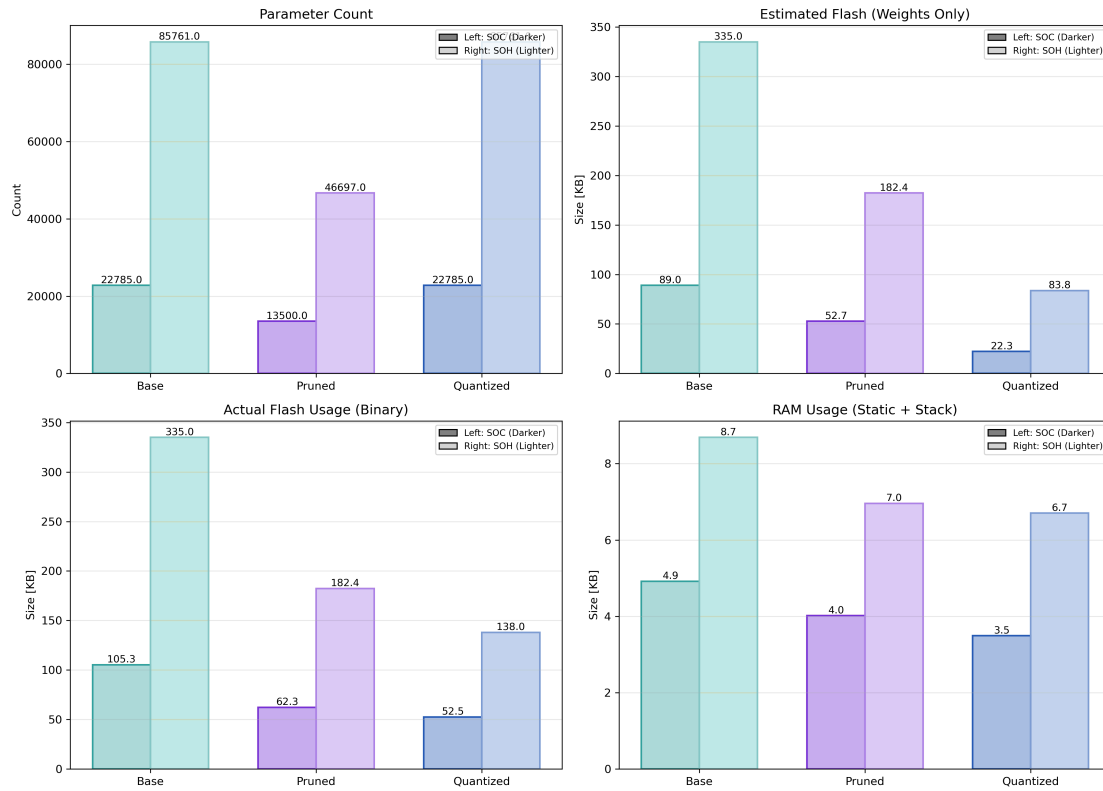


Abbildung 7.12: Resource landscape of the embedded SOC and SOH variants. The figure compares parameter count, idealized flash estimates from weight storage alone, measured flash on the STM32 firmware, and measured RAM usage. The separation between estimated and measured flash clarifies the additional code and runtime overhead of the embedded implementation.

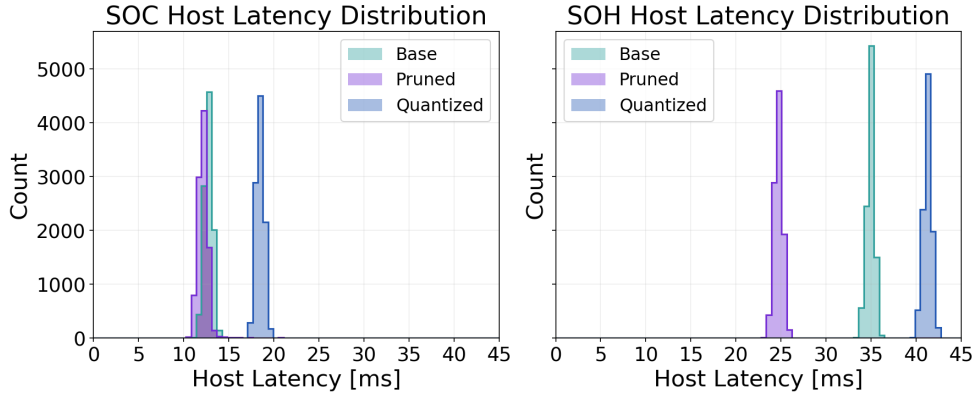


Abbildung 7.13: Latency distributions for SOC and SOH variants on the STM32H753ZI. The histograms summarize repeated on-device measurements and make visible the distinction between faster pruned kernels and slower quantized kernels with additional de-quantization overhead.

Tabelle 7.3: Composite pruning/quantization summary. Values are reported as SOC/SOH. Δ MAE is given in percentage points of full scale, flash/RAM/energy are relative changes with respect to the corresponding dense baseline, and the utility score U follows the normalized average defined in the text, with lower values indicating a more favorable overall trade-off.

Variant	Δ MAE [pp]	Δ Flash [%]	Δ RAM [%]	Δ Energy [%]	U
Pruned	−0.35 / +0.61	−41 / −46	−18 / −20	−43 / −44	0.71 / 0.91
Quantized	+0.10 / +0.56	−50 / −59	−20 / −23	+399 / +29	1.83 / 1.03

7.11 Implications for MCU-Based BMS

- The central engineering outcome of the chapter is that all investigated deployments satisfy the timing constraints of typical stationary-storage BMS applications with comfortable margin.
- Even when communication overhead is included, end-to-end SOC latencies remain below roughly 20 ms and SOH latencies below roughly 50 ms. The pure compute time on the MCU stays below 1,5 ms for dense and pruned SOC and within roughly 12 ms bis 30 ms for the SOH variants.

- The chapter also makes clear that there is no universally best compression strategy. On the current STM32 implementation, pruning provides the most balanced operating point because it reduces flash, RAM, inference time, and energy simultaneously, whereas quantization is attractive mainly when flash capacity is the dominating constraint and additional runtime overhead remains acceptable.
- This conclusion is reflected directly in the utility score: for SOC, the pruned variant reaches $U = 0.71$ and is therefore the most favorable trade-off, whereas the quantized SOC deployment reaches $U = 1.83$ because the INT8 kernel increases energy strongly. For SOH, both compressed variants remain close to the baseline with $U \approx 0.9\text{--}1.0$, which means that flash savings and accuracy degradation roughly balance each other in the aggregate score.
- From the perspective of the dissertation, this chapter provides the concrete bridge between data-driven estimator design and the actual embedded controller. It shows that SOC and SOH estimators can be co-located on one STM32H7-class platform, but that architecture, compression, firmware implementation, and task-specific accuracy requirements must be selected jointly rather than sequentially.
- At the same time, the chapter also exposes the limits of the present study. The experiments are based on one rich LFP campaign, one cell chemistry, and one embedded platform, and they do not yet include systematic measurement-noise, sensor-drift, or fault-injection experiments on the microcontroller itself. Future work should therefore extend the analysis to additional chemistries, temperature ranges, loading regimes, and pack-level BMS integration, and should compare the present LSTM backbone against alternative recurrent or hybrid estimators such as GRUs, Transformer-style models, or neural-augmented equivalent-circuit observers.[74, 93, 42, 45]
- This naturally motivates the next chapter. Once Chapter 7 has established that pruning and quantization can change flash, latency, energy, and even accuracy in architecture-specific ways, Chapter 8 examines these compression effects explicitly as an object of study rather than only as one deployment step in a single embedded pipeline.

Architecture-Dependent Effects of Pruning and Quantization

8.1 Problem Statement

- The embedded deployment study already demonstrates that pruning and quantization can improve the feasibility of MCU-based inference.
- The next scientific step is to determine how strongly these gains depend on the underlying estimator architecture.
- This question is important because compression methods that work well for one model family may degrade another model family disproportionately in terms of accuracy, numerical stability, or execution behaviour.
- A cross-architecture analysis therefore extends the embedded perspective from a single deployment pipeline toward a broader model-compression design framework.

8.2 Comparison Framework

- The comparative study varies both estimator architecture and compression strategy under matched embedded evaluation conditions.
- Relevant architecture classes include feed-forward baselines, recurrent models such as GRU or LSTM, and hybrid models with explicit state-related or physics-aware structure.
- Relevant compression variables include structured and unstructured pruning, INT8 quantization, mixed-precision variants, and architecture-specific compression choices.

- The evaluation combines task-specific predictive metrics with Flash, RAM, latency, and numerical-stability indicators so that compression quality is assessed as a joint property of model and hardware.

Tabelle 8.1: Experiment matrix for analysing architecture-dependent pruning and quantization effects under matched embedded conditions.

Model family	Main task	Compression variables	variables	Scientific contribution
MLP	SOH or static surrogate tasks	INT8, mixed precision, structured pruning		Simple low-cost baseline and reference point for interpretability and resource use.
GRU / LSTM	Streaming SOC or SOH	hidden-size pruning, export, activation range analysis	pruning, INT8	Direct continuation of the current embedded AI work.
Hybrid neural / physics-aware models	SOC correction or joint estimation	selective compression of neural sub-blocks	compression	Useful for understanding whether hybrid structures compress differently from purely neural ones.
Tiny deploy-ment baselines	MCU-only inference	aggressive compression latency-first tuning	compression and tuning	Practical lower-bound reference for real BMS integration.

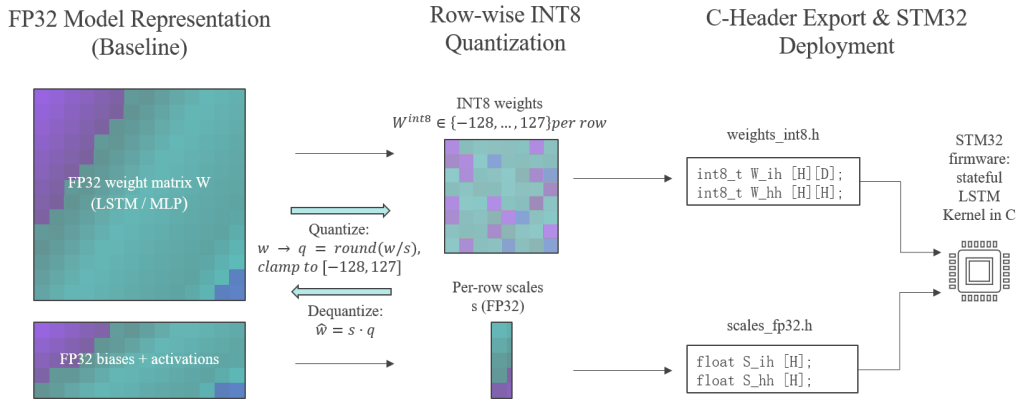


Abbildung 8.1: Quantization workflow used as the methodological starting point for the cross-architecture compression study.

8.3 Scientific Contribution

- The chapter examines which model families retain predictive quality most effectively under aggressive compression and where compression-induced degradation becomes architecture-specific.
- It provides a basis for architecture-dependent design rules that connect estimator structure with quantization and pruning sensitivity.
- In this way, the chapter generalizes the embedded deployment results beyond a single model pipeline and strengthens the hardware-aware interpretation of battery AI.
- The comparative view extends the current embedded work toward transferable deployment guidelines for future BMS integration.

General Discussion and Cross-Paper Synthesis

9.1 Synthesis of the Main Findings

The cumulative findings of the dissertation show that battery-related artificial intelligence cannot be evaluated adequately within a single disciplinary dimension. Model design, disturbance robustness, embedded feasibility, and hardware integration interact directly and must therefore be interpreted together.

- The ageing-estimation chapter shows that relevant battery information can already be captured with a comparatively simple network if the temporal representation and the derived features reflect the underlying electrochemical ageing process.
- The robustness benchmark shows that this representation-oriented view is insufficient on its own, because estimator quality changes once measurement disturbances, timing irregularities, and initialization errors are introduced systematically.
- The embedded-AI chapter shows that the ranking of estimator candidates changes once microcontroller memory, latency, and implementation effort become explicit evaluation criteria in addition to predictive quality.
- The hardware chapter demonstrates that these algorithmic questions are not detached from the physical system. Sensor access, firmware partitioning, estimator exchangeability, and the overall validation chain determine which AI methods can actually be investigated and transferred into BMS-oriented execution environments.

- Taken together, the completed study chapters show that the cumulative scientific gain of the dissertation lies in connecting these layers into one coherent engineering view instead of optimizing only one isolated metric.

9.2 Synthesis Matrix

Tabelle 9.1: Synthesis matrix linking the main contribution chapters to the three central requirement dimensions.

Dissertation contribution	contri-	Performance	Hardware	/	Implementation logic
Neural estimation chapter	ageing-	high relevance for representation quality	indirect relevance	rele-	strong relevance for architecture and feature design
Robustness benchmark chapter	bench-	strongest relevance	medium relevance through sensing realism	rele-	strong relevance for evaluation methodology
Embedded AI chapter	AI chap-	medium relevance	strongest relevance	rele-	strongest relevance for compression and deployment workflow
Architecture-dependent compression chapter	com-	medium relevance	strongest relevance	rele-	strongest relevance for model-hardware co-design
Hardware ecosystem chapter	ecosystem	indirect relevance	strongest relevance	rele-	medium relevance through system integration

9.3 Scientific Overall Contribution

- The dissertation does not merely collect independent studies on battery intelligence. Its core contribution is the combination of representation-oriented modeling, deployment-oriented robustness analysis, embedded implementation, and hardware-grounded validation within one shared research program.
- A second central contribution is the battery-management requirement frame-

work, which makes it possible to compare estimator quality across performance, hardware, and deployment dimensions instead of discussing only one of these layers in isolation.

- A third contribution lies in the methodological connection between offline modelling, disturbance-based benchmarking, and on-device MCU measurements. This link turns the work from a sequence of isolated technical studies into a cumulative framework for deployment-oriented BMS intelligence.
- The overall result is therefore not only a set of chapter-specific findings, but a transferable way of evaluating AI-based battery estimation methods under realistic engineering constraints.

9.4 Methodological Strengths and Limitations Across the Studies

- A key strength of the dissertation is that it spans several layers of evidence: controlled ageing data, structured disturbance benchmarks, measured embedded deployment metrics, and the BMS-STEP hardware platform for practical validation.
- Another strength is the common engineering perspective across otherwise different studies. Ageing estimation, robustness benchmarking, and microcontroller deployment are treated as connected parts of one BMS-related research problem.
- At the same time, the studies rely on different datasets, task definitions, and evaluation settings. Their numerical results are therefore not interchangeable in a strict one-to-one sense, but must be interpreted as complementary evidence for the same overarching design problem.
- The data-driven chapters remain dependent on the breadth and quality of the available datasets. Atypical degradation trajectories, incomplete measurements, and limited coverage of rare operating conditions continue to restrict the generality of the reported results.
- The embedded and hardware-related chapters are likewise shaped by the selected microcontroller targets, firmware environment, and laboratory setup. The

transferability of the exact resource figures to other platforms must therefore be interpreted with appropriate engineering caution.

9.5 Positioning Within the State of Research

- A large part of the existing literature concentrates on one dimension at a time, for example on estimation accuracy, on a particular neural architecture, or on a single deployment toolchain.
- The present dissertation contributes a broader view by showing that battery AI methods must be assessed across model quality, disturbance sensitivity, embedded feasibility, and system integration at the same time.
- In this sense, the work is positioned between algorithm development and embedded systems engineering. It neither treats the battery estimator as a purely offline machine-learning problem nor reduces the hardware side to an implementation appendix.
- This placement is especially important for battery-management applications, because the practical usefulness of an estimator depends on the full chain from measurement acquisition and signal quality to firmware integration and on-device inference behaviour.

9.6 Technical Implications for AI-Based BMS

- The combined results imply that estimator selection for a BMS should begin with the target use case and the expected sensing conditions rather than with benchmark accuracy under ideal signals alone.
- Robustness properties must be treated as a design criterion before and not after embedded compression, because pruning or quantization of a disturbance-sensitive model can further reduce its practical value.
- The hardware chapter additionally shows that an adaptable research platform is not a secondary convenience, but an enabling condition for meaningful embedded-AI experimentation. The ability to exchange estimator methods, separate BMS logic from state-estimation logic, and validate new models in a controlled firmware environment is methodologically important.

- For engineering practice, this means that AI-based battery estimation should be developed together with the execution environment, measurement chain, and integration path instead of being optimized only in an offline data-analysis setting.

9.7 Integrated Design Guidelines

- Estimators should not be selected by clean-condition accuracy alone, because disturbance sensitivity and recovery behaviour strongly affect practical usefulness.
- Robustness analysis should precede aggressive compression, since a highly compressed estimator with poor disturbance tolerance offers limited deployment value.
- Quantization and pruning must be treated as architecture-dependent design choices rather than as universally transferable post-processing steps.
- Dataset quality, estimator quality, and embedded execution quality should be analysed separately so that improvements in one domain are not mistaken for improvements in another.
- Algorithmic claims gain practical relevance only when they are connected to a credible validation chain that includes sensing, firmware execution, and hardware integration.

10.1 Overall Conclusion

- The work shows that deployment-oriented battery AI emerges only when model design, robustness, embedded execution, and hardware integration are treated as one linked problem.
- Neural architectures enable powerful battery state estimation and ageing estimation, but their practical value depends on how they behave under disturbances and how efficiently they can be transferred to embedded hardware.
- Robustness benchmarking reveals that estimator quality changes fundamentally once measurement disturbances, timing irregularities, and initialization mismatch are considered explicitly.
- Embedded implementation demonstrates that compression, latency, and memory demand are not secondary engineering details, but core evaluation dimensions for practical battery-management applications.
- The combination of algorithmic studies and custom hardware integration therefore leads to a unified framework for analysing battery intelligence across performance, hardware, and deployment requirements.

10.2 Boundary Conditions and Remaining Gaps

- The reported findings must be interpreted in the context of the available datasets, the selected estimator classes, and the concrete hardware environments used throughout the work.
- The general design logic of the dissertation is transferable, but the exact quantitative trade-offs between accuracy, robustness, memory demand, and latency

can shift when other cell chemistries, sensing chains, control architectures, or MCU families are considered.

- Further validation across broader datasets, longer ageing horizons, and additional embedded targets is therefore necessary to consolidate the transferability of the derived design rules.

10.3 Outlook

- Future work extends the embedded analysis by comparing pruning and quantization across a wider range of model architectures.
- Further experiments on the BMS-STEP platform can deepen the relation between estimator design, signal quality, and practical system behaviour under laboratory and hardware-in-the-loop conditions.
- Long-duration cycling and additional datasets can refine the calibration and validation of SoC and SoH methods across broader ageing states and operating conditions.
- The longer-term objective is the transfer of deployment-oriented battery AI from controlled laboratory systems toward scalable and reproducible BMS integration in larger energy-storage applications.

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This appendix contains supplementary tables, additional hardware documentation, experiment matrices, and publication-specific material that would interrupt the flow of the main text if it were placed directly in the core chapters.

A.1 Supplementary Robustness-Benchmark Result Tables

The following tables collect the numerical robustness-benchmark values used in Chapter 6. Unless stated otherwise, SOC error metrics are reported on the normalized 0–1 SOC scale; multiplying by 100 gives percentage points.

Tabelle A.1: Baseline performance metrics for the selected robustness-benchmark trajectory. Values are reported on the normalized 0–1 SOC scale.

Class	MAE	RMSE	P95 error	Max error	Bias
DM	0.0311	0.0397	0.0800	1.0000	0.0311
HDM	0.0024	0.0061	0.0094	1.0000	0.0022
HECM	0.0090	0.0127	0.0241	1.0000	-0.0029
DD	0.0100	0.0143	0.0336	0.0686	-0.0062

Tabelle A.2: Cross-scenario global performance metrics used in the Chapter 6 result discussion. Values are reported on the normalized 0–1 SOC scale.

Scenario	Class	MAE	Δ MAE	RMSE	Δ RMSE	P95 error
Current (high)	DM	0.0310	-0.0001	0.0394	-0.0002	0.0795
Current bias	DM	0.3143	0.2833	0.3683	0.3286	0.6670
Initial SOC error	DM	0.0322	0.0011	0.0413	0.0017	0.0833
Irregular sampling	DM	0.0309	-0.0001	0.0396	-0.0001	0.0799
Burst dropout	DM	0.0332	0.0022	0.0487	0.0090	0.0819

Scenario	Class	MAE	Δ MAE	RMSE	Δ RMSE	P95 error
Missing samples	DM	0.0386	0.0076	0.0476	0.0079	0.0926
Voltage spikes	DM	0.0312	0.0001	0.0398	0.0001	0.0801
Temperature noise	DM	0.0311	0.0000	0.0397	0.0000	0.0800
Voltage noise	DM	0.0524	0.0213	0.0648	0.0252	0.1228
Current noise (high)	HDM	0.0044	0.0019	0.0074	0.0013	0.0120
Current bias	HDM	0.3068	0.3043	0.3638	0.3577	0.6668
Initial SOC error	HDM	0.0036	0.0011	0.0136	0.0075	0.0101
Irregular sampling	HDM	0.0027	0.0003	0.0061	0.0001	0.0095
Burst dropout	HDM	0.0051	0.0027	0.0325	0.0265	0.0098
Missing samples	HDM	0.0103	0.0079	0.0129	0.0069	0.0221
Voltage spikes	HDM	0.0062	0.0038	0.0140	0.0079	0.0328
Temperature noise	HDM	0.0067	0.0043	0.0110	0.0049	0.0210
Voltage noise	HDM	0.0751	0.0727	0.0908	0.0847	0.1578
Current noise (high)	HECM	0.0165	0.0075	0.0280	0.0154	0.0701
Current bias	HECM	0.0932	0.0842	0.1303	0.1176	0.2954
Initial SOC error	HECM	0.0091	0.0001	0.0143	0.0016	0.0241
Irregular sampling	HECM	0.0095	0.0004	0.0131	0.0004	0.0248
Burst dropout	HECM	0.0104	0.0014	0.0228	0.0101	0.0246
Missing samples	HECM	0.0119	0.0029	0.0163	0.0036	0.0329
Voltage spikes	HECM	0.0090	0.0000	0.0127	0.0000	0.0241
Temperature noise	HECM	0.0090	0.0000	0.0127	0.0000	0.0241
Voltage noise	HECM	0.0090	0.0000	0.0127	0.0000	0.0241
Current noise (high)	DD	0.0109	0.0009	0.0151	0.0008	0.0345
Current bias	DD	0.2859	0.2759	0.3426	0.3283	0.6367
Initial SOC error	DD	0.0110	0.0010	0.0175	0.0032	0.0355
Irregular sampling	DD	0.0105	0.0005	0.0144	0.0001	0.0330
Burst dropout	DD	0.0122	0.0022	0.0316	0.0173	0.0348
Missing samples	DD	0.0117	0.0016	0.0149	0.0006	0.0302
Voltage spikes	DD	0.0144	0.0044	0.0213	0.0070	0.0474
Temperature noise	DD	0.0181	0.0081	0.0223	0.0080	0.0451
Voltage noise	DD	0.0698	0.0597	0.0839	0.0696	0.1501

Tabelle A.3: Local behavior and recovery metrics used in the Chapter 6 result discussion. Thresholds are reported on the normalized 0–1 SOC scale where applicable.

Class	Scenario	Local metric	Value	Threshold
DM	Initial SOC error	Recovery time to strict band [h]	1.1658	0.0800
DM	Initial SOC error	Recovery time to fair band [h]	0.2500	0.1599
HDM	Initial SOC error	Recovery time to strict band [h]	–	0.0094
HDM	Initial SOC error	Recovery time to fair band [h]	–	0.0200
HECM	Initial SOC error	Recovery time to strict band [h]	2.3693	0.0241
HECM	Initial SOC error	Recovery time to fair band [h]	1.1564	0.0482
DD	Initial SOC error	Recovery time to strict band [h]	1.0464	0.0336
DD	Initial SOC error	Recovery time to fair band [h]	1.0014	0.0672
DM	Burst dropout	Recovery time after burst dropout [h]	27.3511	0.0216

Class	Scenario	Local metric	Value	Threshold
HDM	Burst dropout	Recovery time after burst dropout [h]	27.4093	0.0008
HECM	Burst dropout	Recovery time after burst dropout [h]	28.5611	0.0095
DD	Burst dropout	Recovery time after burst dropout [h]	27.4351	0.0070
DM	Voltage spikes	Median spike recovery time [s]	0.0000	0.0800
HDM	Voltage spikes	Median spike recovery time [s]	0.0000	0.0094
HECM	Voltage spikes	Median spike recovery time [s]	0.0000	0.0241
DD	Voltage spikes	Median spike recovery time [s]	0.0000	0.0336
DM	Current noise (high)	Late minus early excess rolling MAE	-0.0013	–
DM	Current noise (high)	Mean excess rolling MAE	-0.0001	–
HDM	Current noise (high)	Late minus early excess rolling MAE	-0.0011	–
HDM	Current noise (high)	Mean excess rolling MAE	0.0019	–
HECM	Current noise (high)	Late minus early excess rolling MAE	0.0038	–
HECM	Current noise (high)	Mean excess rolling MAE	0.0073	–
DD	Current noise (high)	Late minus early excess rolling MAE	-0.0002	–
DD	Current noise (high)	Mean excess rolling MAE	0.0009	–

Tabelle A.4: Scenario overview linking disturbance configuration, primary evidence type, and main Chapter 6 figure. This table clarifies which scenarios are interpreted predominantly through global metrics and which require additional local evidence.

Scenario	Configuration	Primary evidence	Main figure(s)
ADC quantization	Steps of 0,01 A, 0,005 V, and 0,5 °C	Global Δ MAE; detailed values in Appendix Table A.5	Figs. 6.12, A.1
Current bias	Constant current offset in the compact matrix, plus dedicated 0.5%/1.5%/3.0% sweep of $ I _{\max}$	Global MAE degradation across the sweep	Fig. 6.5
Current noise	Compact matrix with $\sigma_I = 0,10$ A; local detail sweep up to $\sigma_I = 0,20$ A	Global Δ MAE plus local output-variability and rolling-MAE evidence	Fig. 6.6
Voltage noise	Additive Gaussian noise with $\sigma_U = 0,01$ V	Global Δ MAE	Fig. 6.12
Temperature noise	Additive Gaussian noise with $\sigma_T = 1,0$ °C	Global Δ MAE	Fig. 6.12
Initial SOC error	Additive initial mismatch of -0.10 SOC, corresponding to 10 percentage points; DD via online Q_c offset	Local recovery to strict and fair baseline bands; global MAE is secondary	Fig. 6.7
Missing samples	Every 50th sample frozen on the nominal 1 s grid	Global Δ MAE	Figs. 6.8, 6.12

Scenario	Configuration	Primary evidence	Main figure(s)
Irregular sampling	Uniform jitter of $\pm 0,1$ s around the nominal 1 s grid	Global Δ MAE	Figs. 6.8, 6.12
Burst dropout	One central frozen gap of 3600 s	Global Δ MAE plus local transition and long-horizon recovery	Figs. 6.8, 6.9, 6.10
Voltage spikes	One $\pm 0,20$ V single-sample spike every 1000 samples	Global Δ MAE plus event-wise P95 of post-spike peak excess error	Figs. 6.11, 6.12

Tabelle A.5: ADC-quantization results for the four estimator classes. Values are reported on the normalized 0–1 SOC scale.

Class	Baseline MAE	ADC MAE	Δ MAE
DM	0.0311	0.0313	+0.0002
HDM	0.0024	0.0029	+0.0005
HECM	0.0090	0.0087	-0.0003
DD	0.0100	0.0098	-0.0002

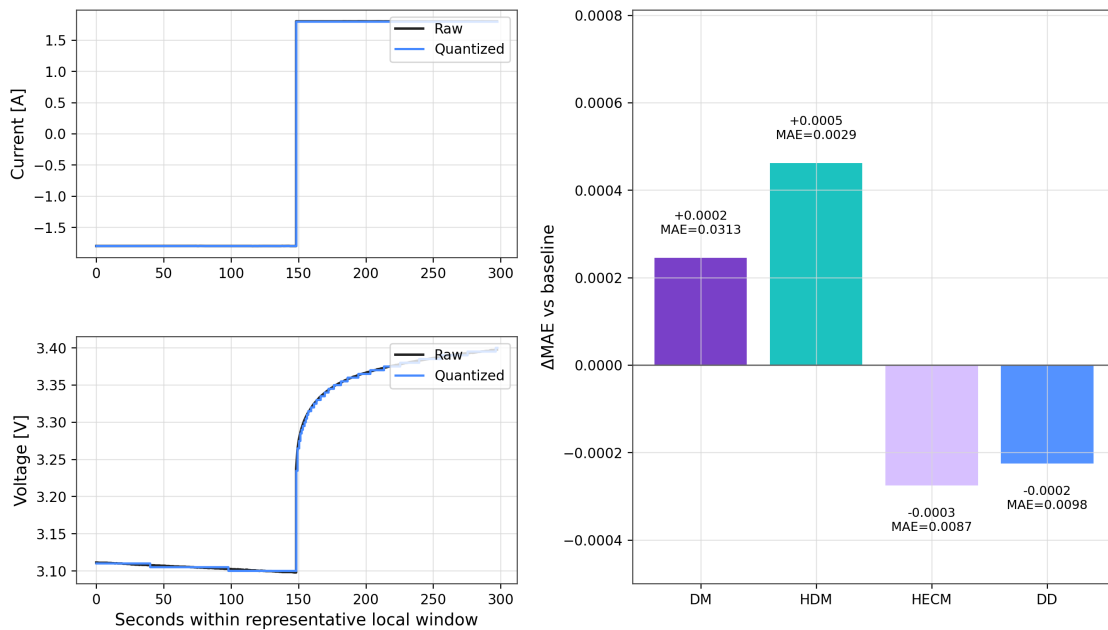


Abbildung A.1: ADC-quantization results for the robustness benchmark. The left panels illustrate representative current and voltage quantization in a local window, and the right panel summarizes the resulting global Δ MAE relative to the baseline case.